ARTIFICIAL INTELLIGENCE

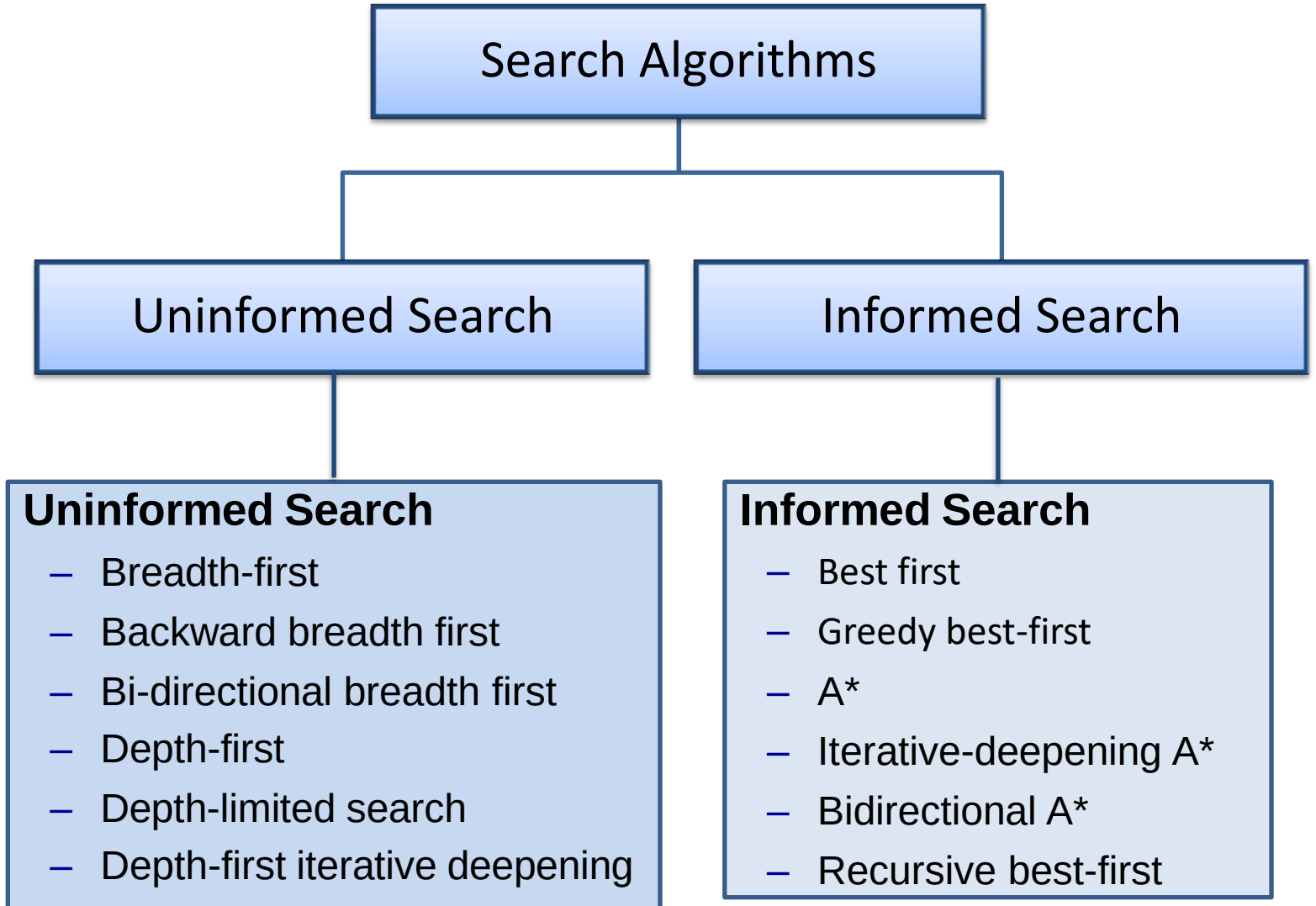
Lecture Set 04
Uninformed Search

Dr. Maria Siddiqua
Assistant Professor
Department of AI & DS
FAST-NUCES Karachi
maria.siddiqua@nu.edu.pk

Course Contents

1. Introduction to AI
2. Agents and Environments
3. Problem Solving by Searching
4. **Uninformed Search**
5. Informed Search
6. Local Search
7. Constraint Satisfaction Problems
8. Adversarial Search
9. Logical Agents
10. First-order Logic
11. Uncertain Knowledge and Reasoning
12. Supervised, Unsupervised, and Reinforcement Learning
13. Recent Trends in AI and Applications of AI Algorithms

Table of Contents



(blind search)

- number of steps, path cost unknown
- agent knows when it reaches a goal

(heuristic search)

- agent has background information about the problem

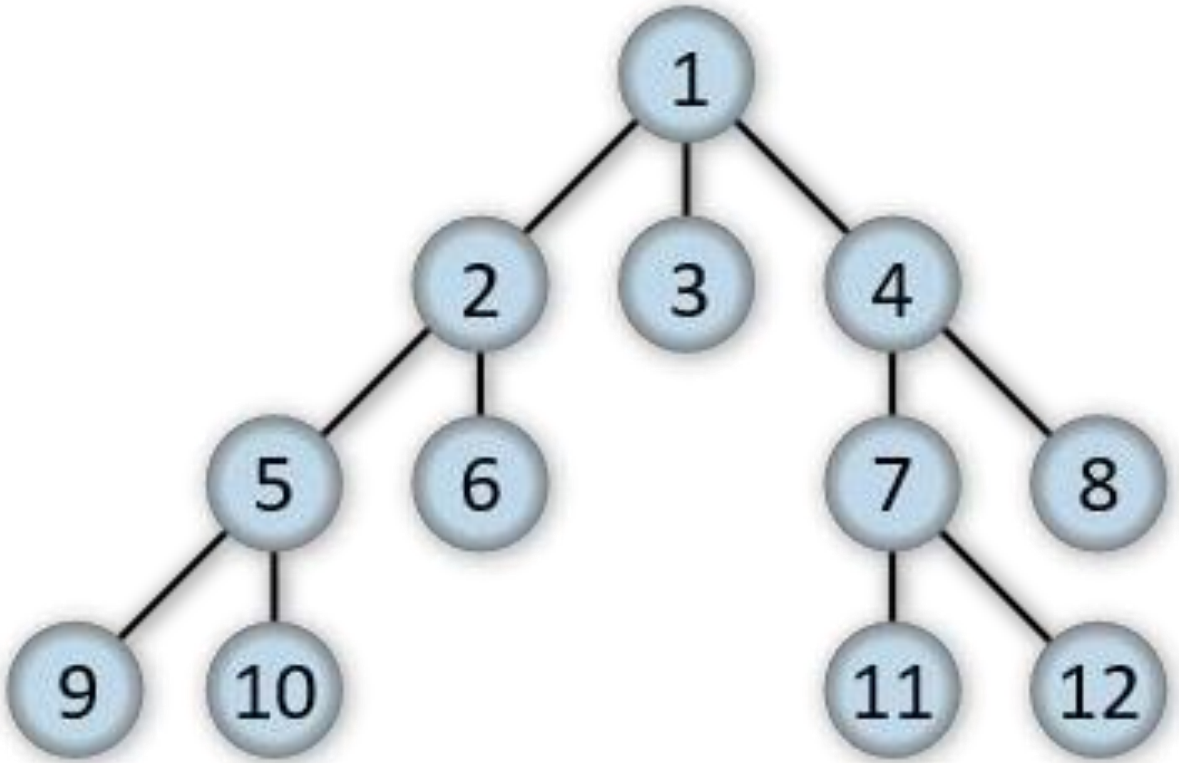
Uninformed Search

- Also called as Blind Search or Brute Force Search.
- A blind search is a search that has no information about its domain. The only thing that a blind search can do is to distinguish a non-goal state from a goal state.
  - *Breadth-First search*
  - *Backward Breadth-First search*
  - *Bi-directional Breadth-First search*
  - *Depth-First search*
  - *Depth Limited search*
  - *Depth-First Iterative Deepening search*
  - *Uniform-Cost search*

Breadth-First Search

- Breadth-first search (BFS) is an algorithm for traversing or searching tree or graph data structures.
- It starts at the tree root and explores the neighbor nodes first, before moving to the next level neighbors.
- Breadth First Search explores the state space in a level by level fashion. Only when there are no more states to be explored at a given level then the algorithm move onto the next level.

Breadth-First Search

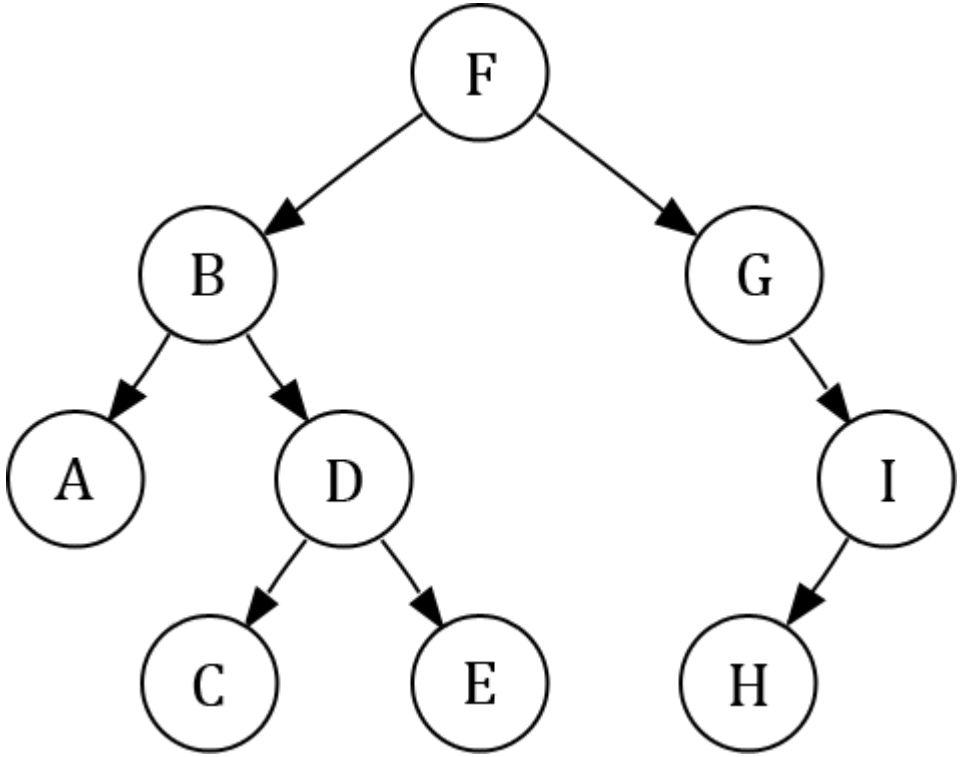


Breadth-First Search

Algorithm:

- Step 01: Choose the starting node, mark it as visited, make it current node.
- Step 02: Find adjacent unvisited node of current node, place it in Queue and mark it visited.
- Step 03: If no unvisited adjacent node found, dequeue node from Queue and make it current node.
- Step 04: Repeat Step 02 to Step 03 until Queue is empty.

Breadth-First Search (Problem 01)



BFS Traversal: **F B G A D I C E H**

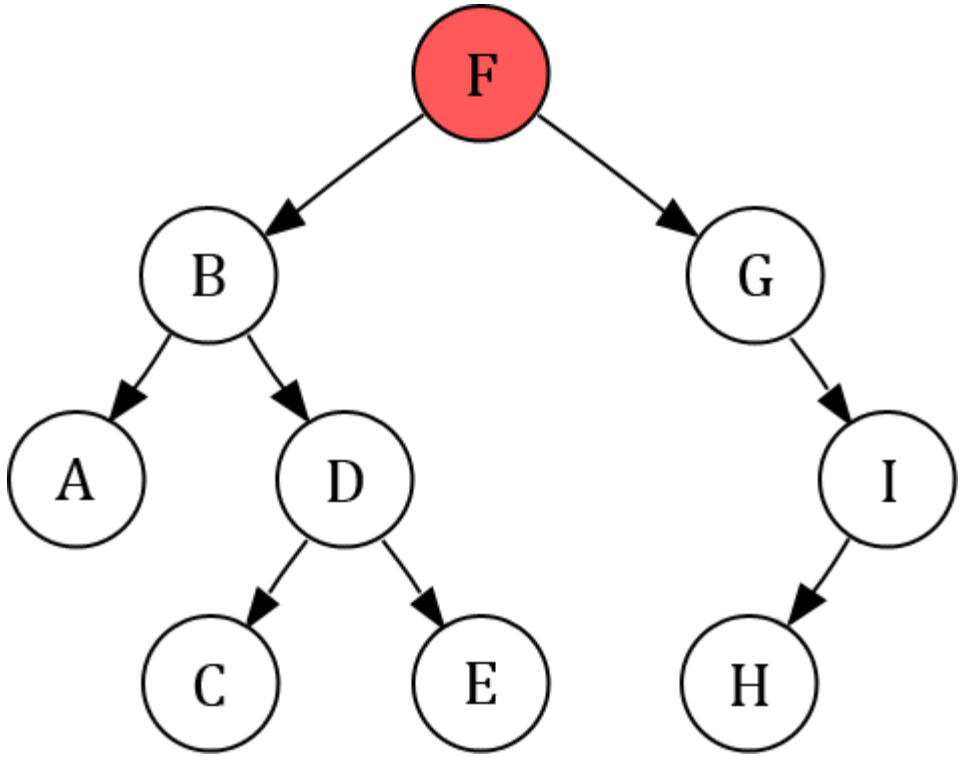
Breadth-First Search (Problem 01)

Current Node = F

Queue:

Visited:

F									



Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 01)

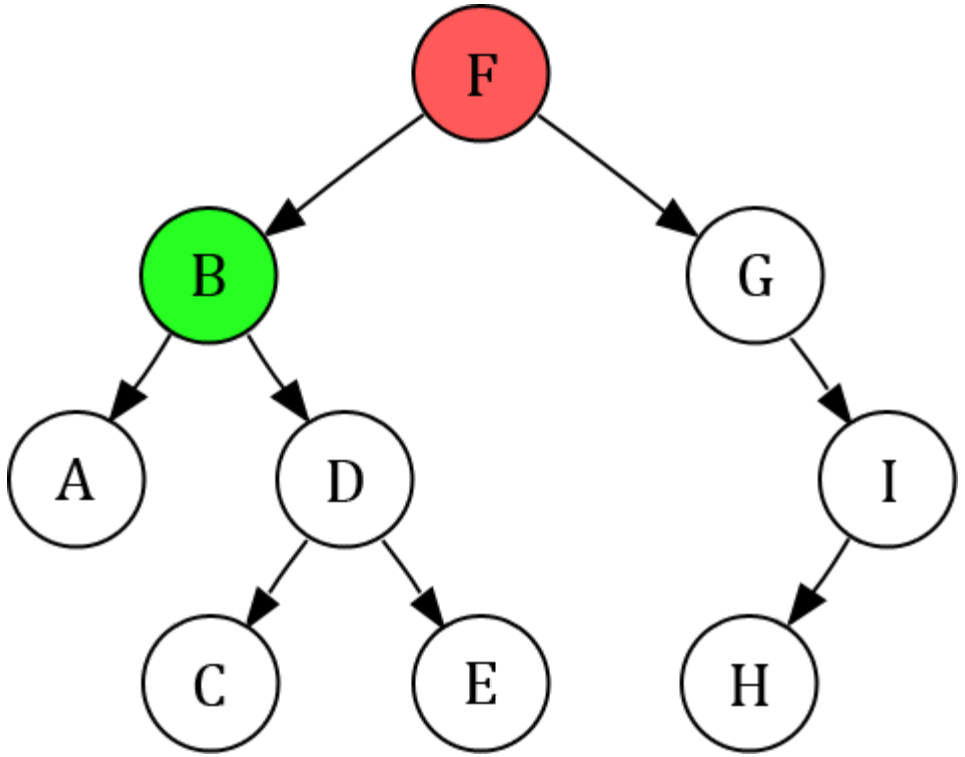
Current Node = F

Queue:

B									

Visited:

F	B								



Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 01)

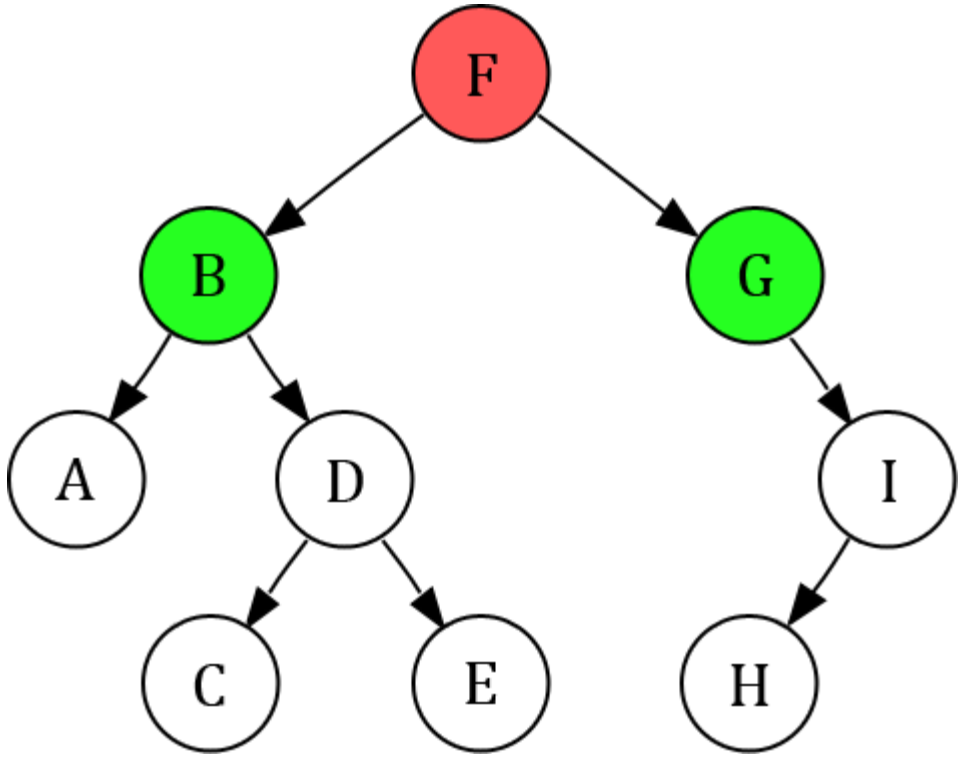
Current Node = F

Queue:

G	B								

Visited:

F	B	G							



Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 01)

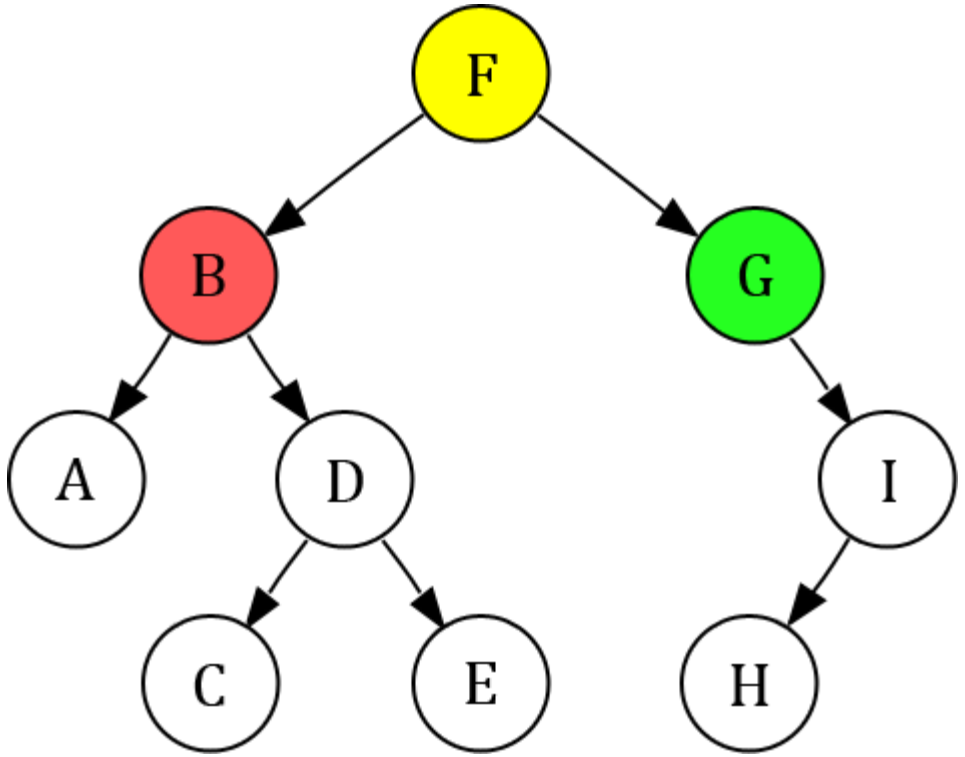
Current Node = B

Queue:

G									

Visited:

F	B	G							



Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 01)

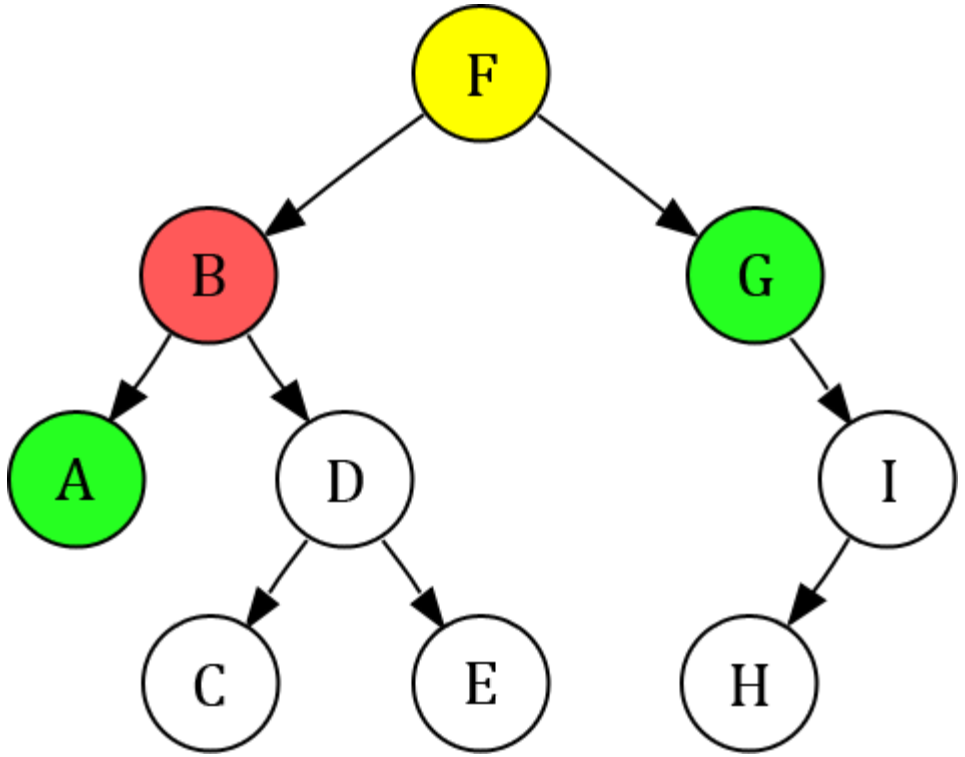
Current Node = B

Queue:

A	G								

Visited:

F	B	G	A						



Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 01)

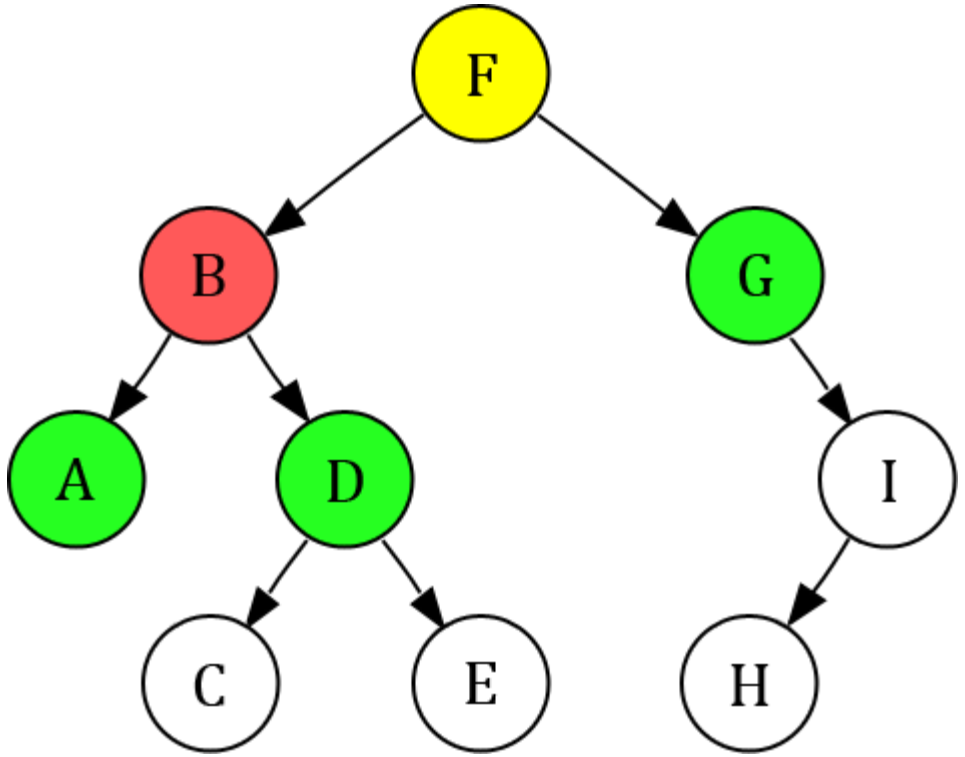
Current Node = B

Queue:

D	A	G							

Visited:

F	B	G	A	D					



Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 01)

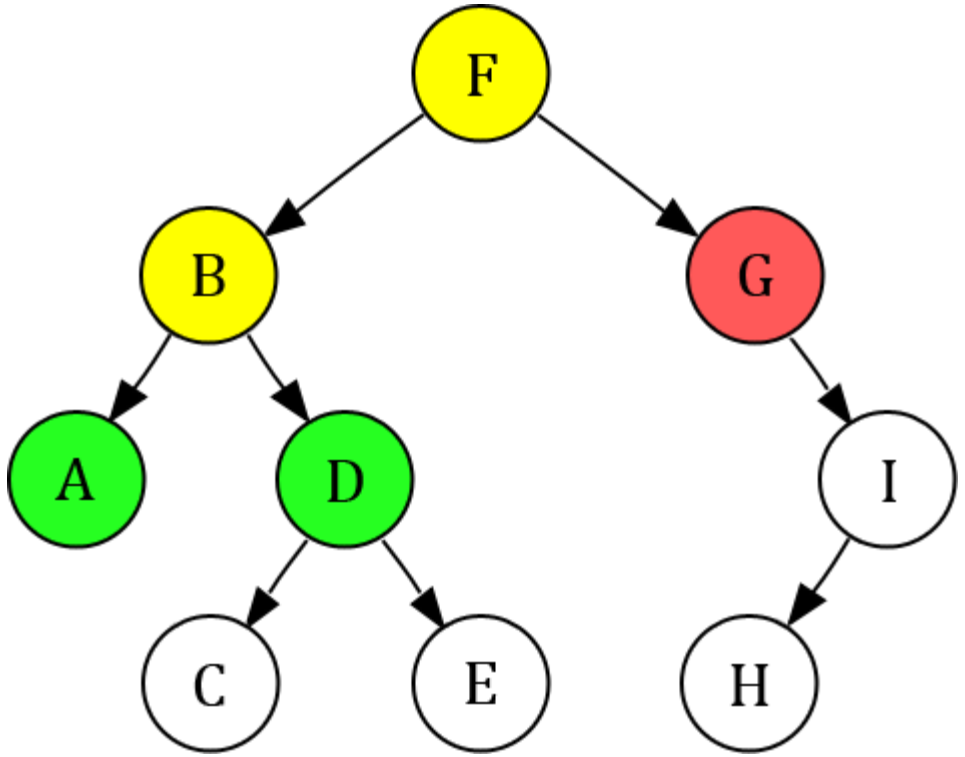
Current Node = G

Queue:

D	A								

Visited:

F	B	G	A	D					



Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 01)

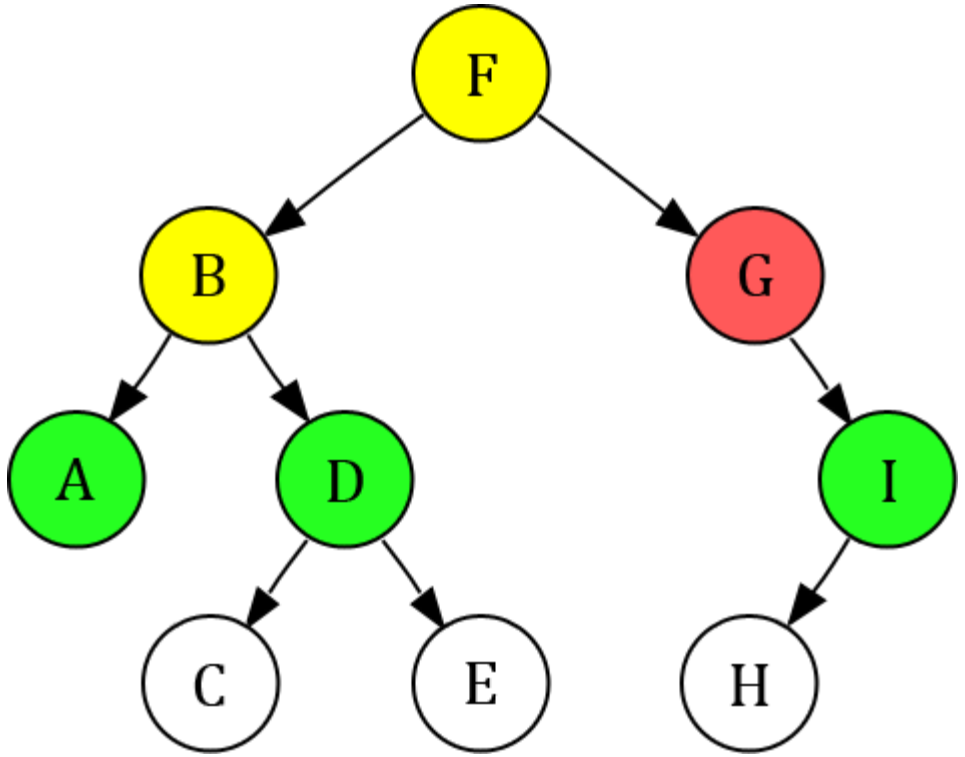
Current Node = G

Queue:

I	D	A							

Visited:

F	B	G	A	D	I				



Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 01)

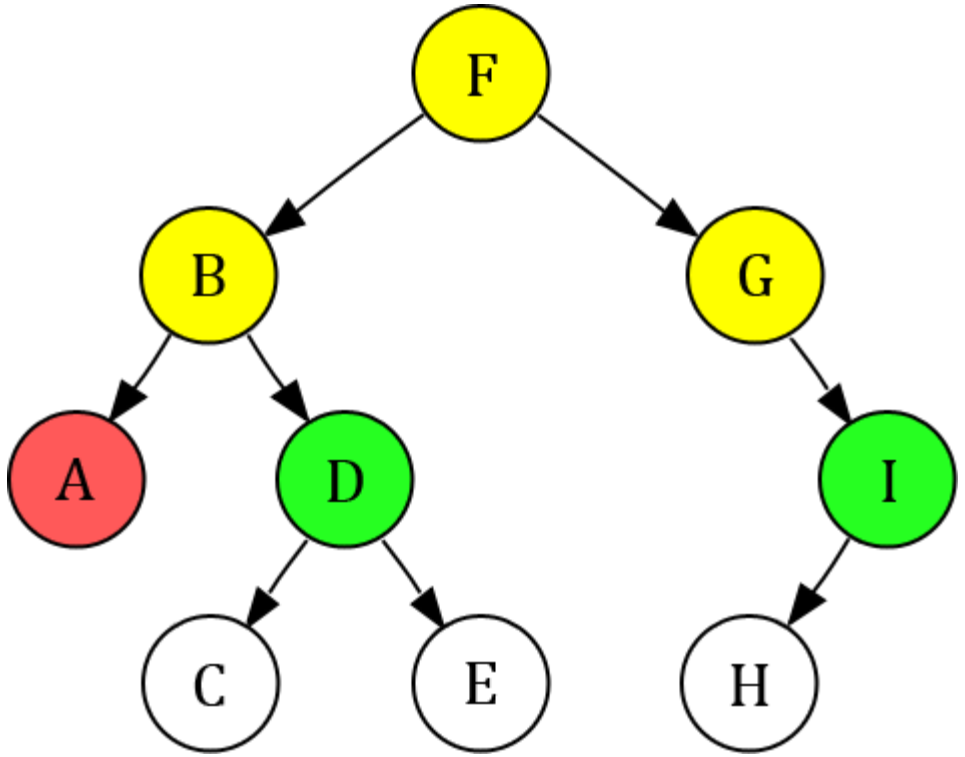
Current Node = A

Queue:

I	D								

Visited:

F	B	G	A	D	I				



Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 01)

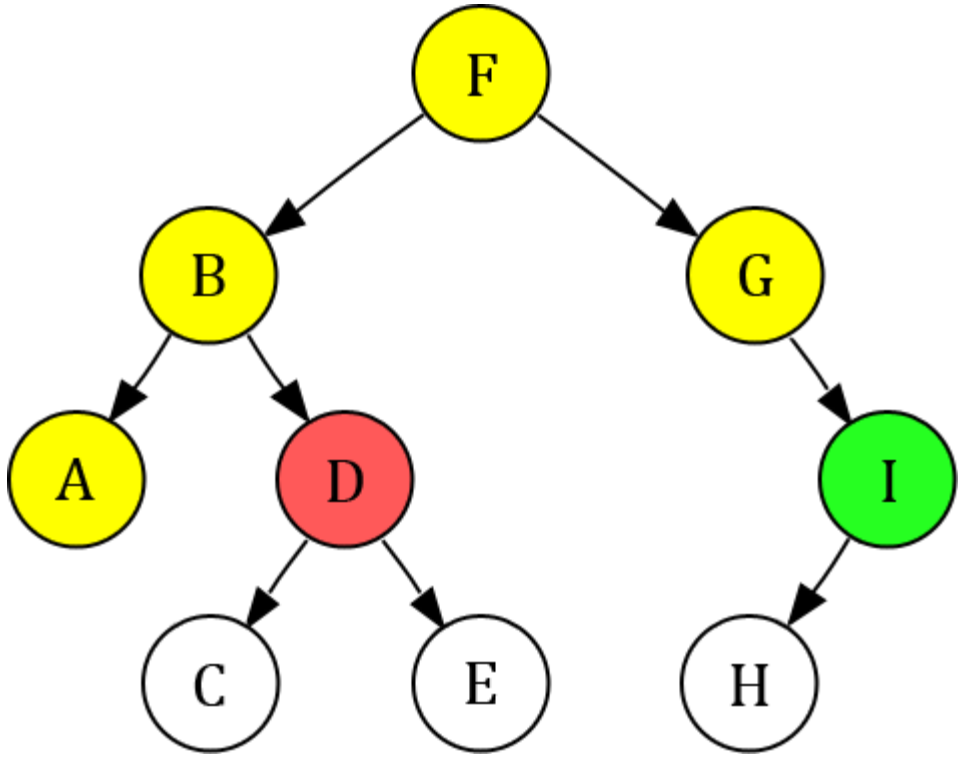
Current Node = D

Queue:

I									

Visited:

F	B	G	A	D	I				



Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 01)

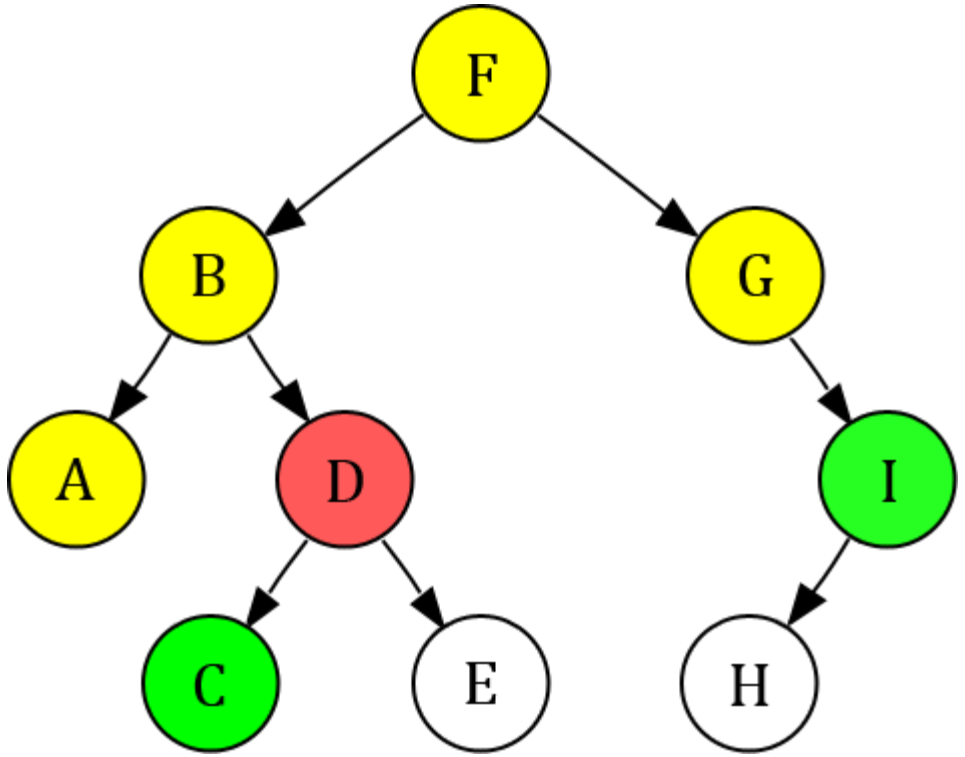
Current Node = D

Queue:

C	I								

Visited:

F	B	G	A	D	I	C			



Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 01)

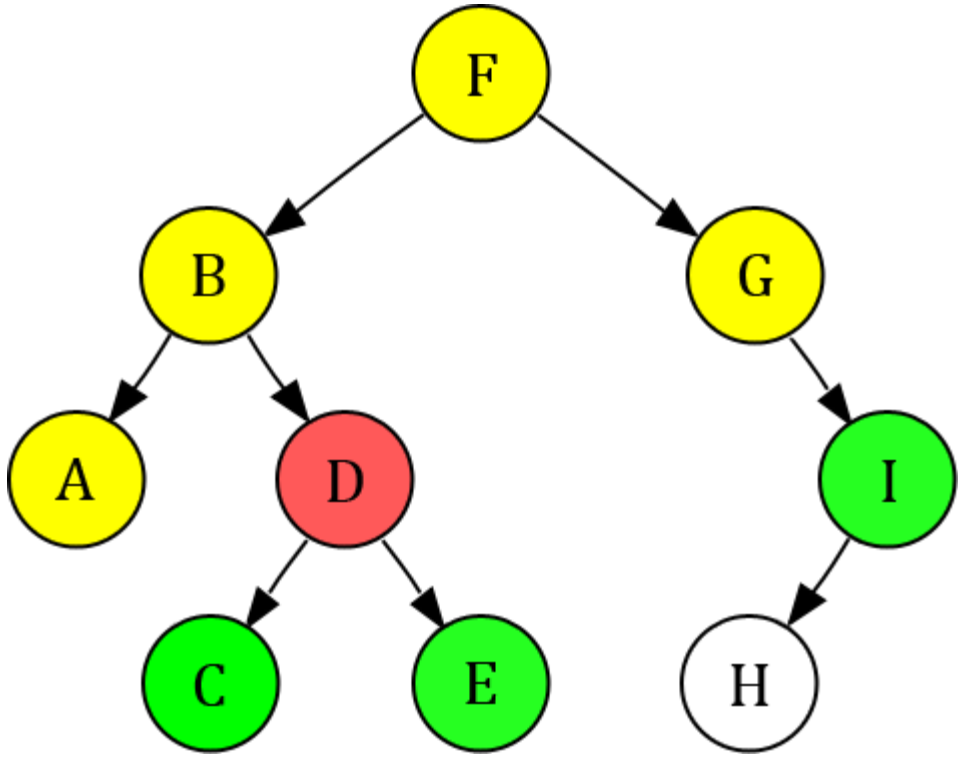
Current Node = D

Queue:

E	**C**	**I**							

Visited:

F	**B**	**G**	**A**	**D**	**I**	**C**	**E**		



Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 01)

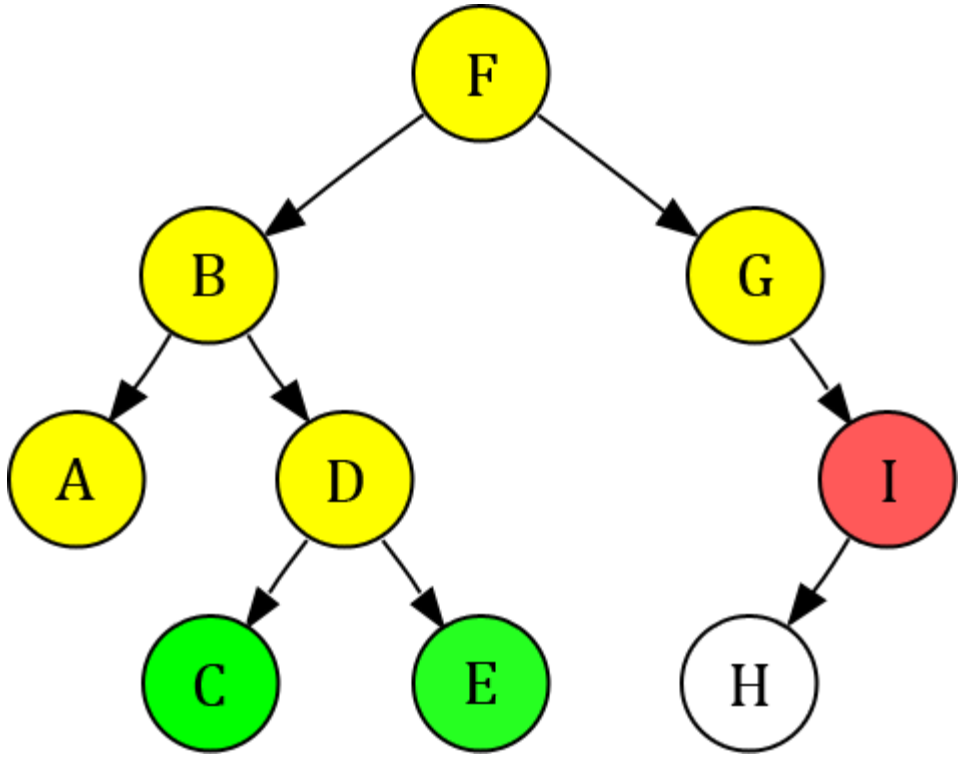
Current Node = I

Queue:

E	**C**								

Visited:

F	**B**	**G**	**A**	**D**	**I**	**C**	**E**		



Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 01)

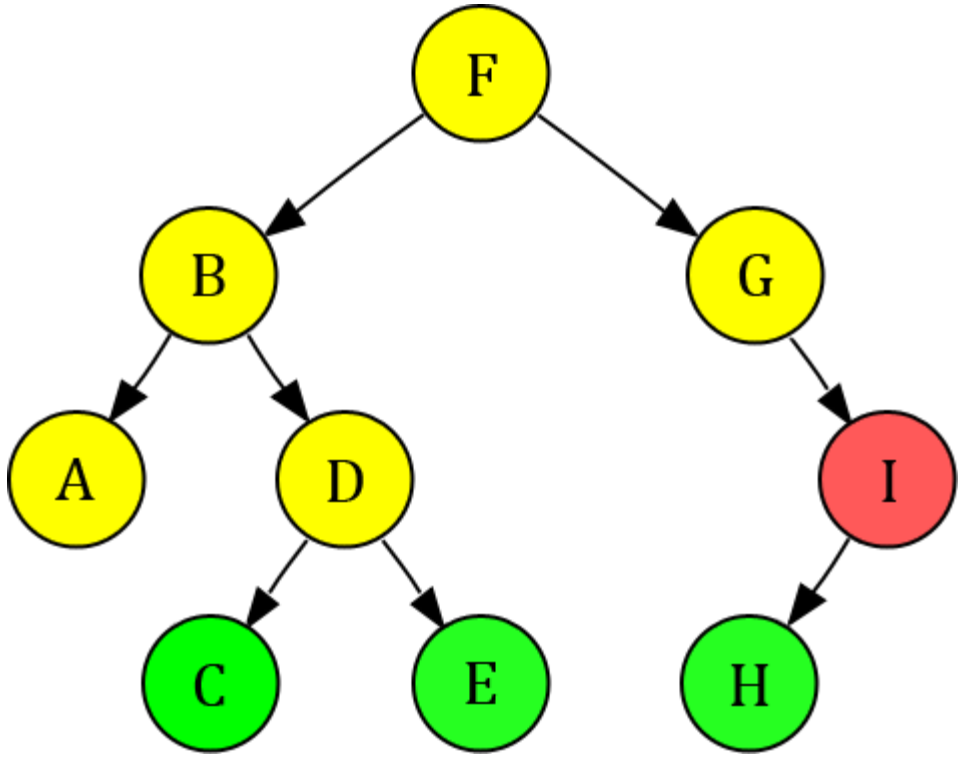
Current Node = I

Queue:

H	E	C							

Visited:

F	B	G	A	D	I	C	E	H	



Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 01)

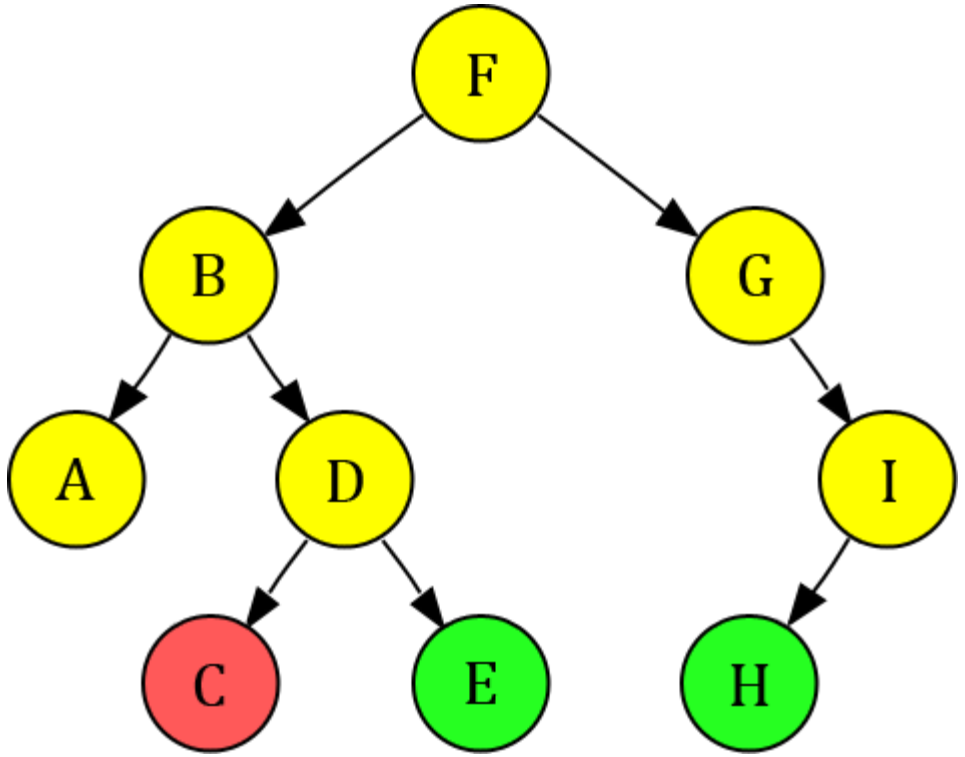
Current Node = C

Queue:

H	E								

Visited:

F	B	G	A	D	I	C	E	H	



Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 01)

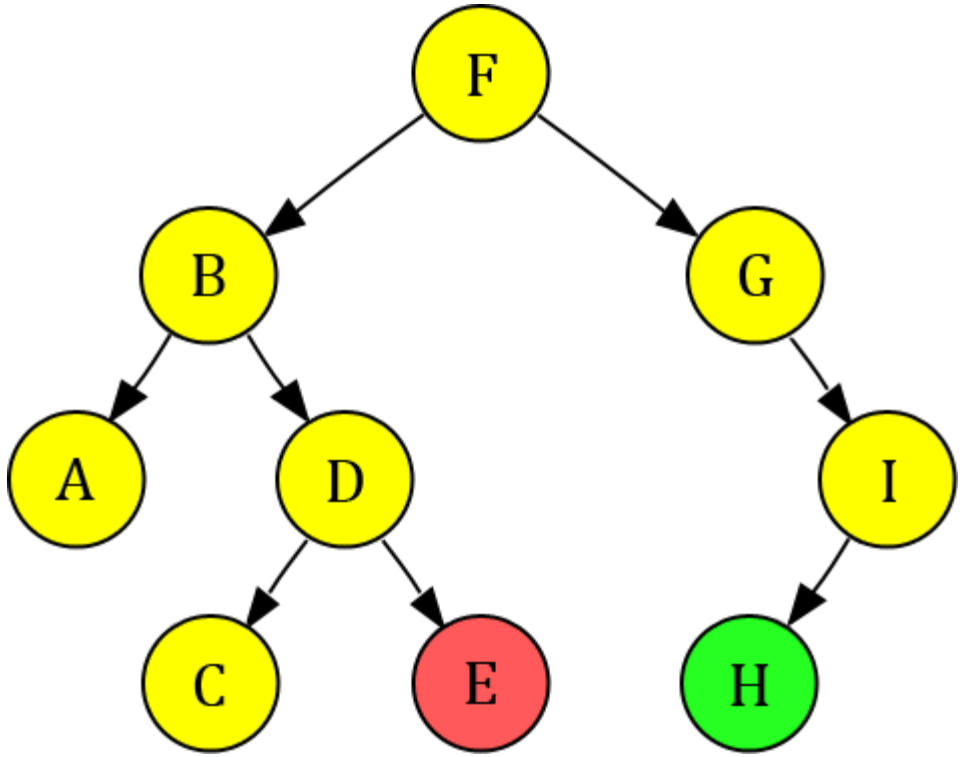
Current Node = E

Queue:

H									

Visited:

F	B	G	A	D	I	C	E	H	



Current Node **Opened Node** **Closed Node**

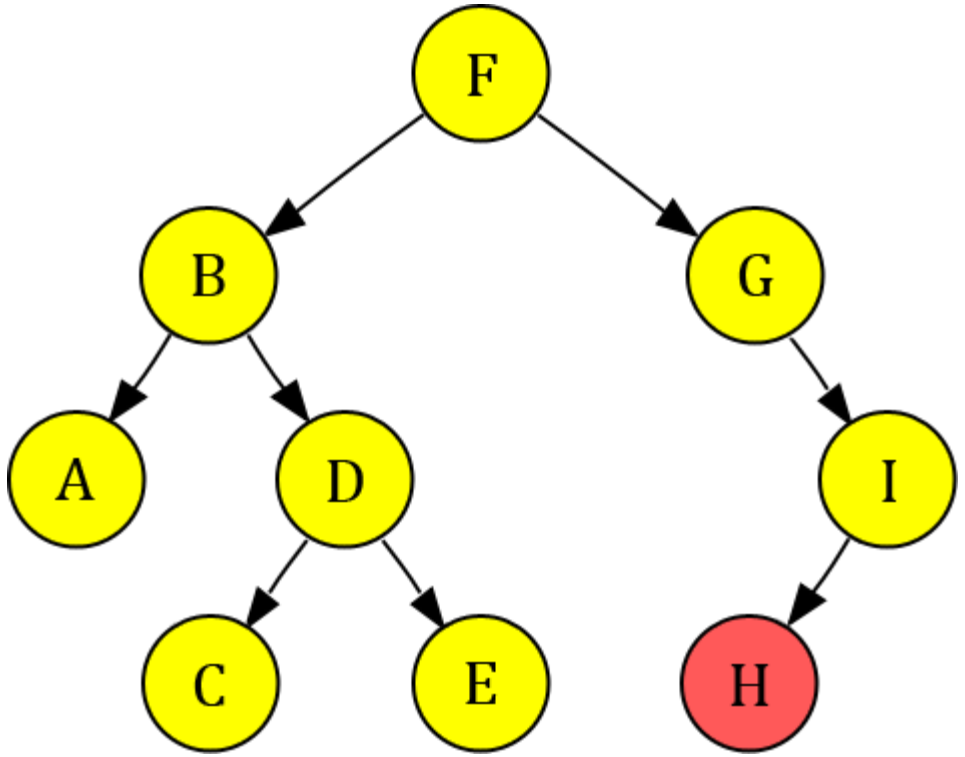
Breadth-First Search (Problem 01)

Current Node = H

Queue:

Visited:

F	**B**	**G**	**A**	**D**	**I**	**C**	**E**	**H**	



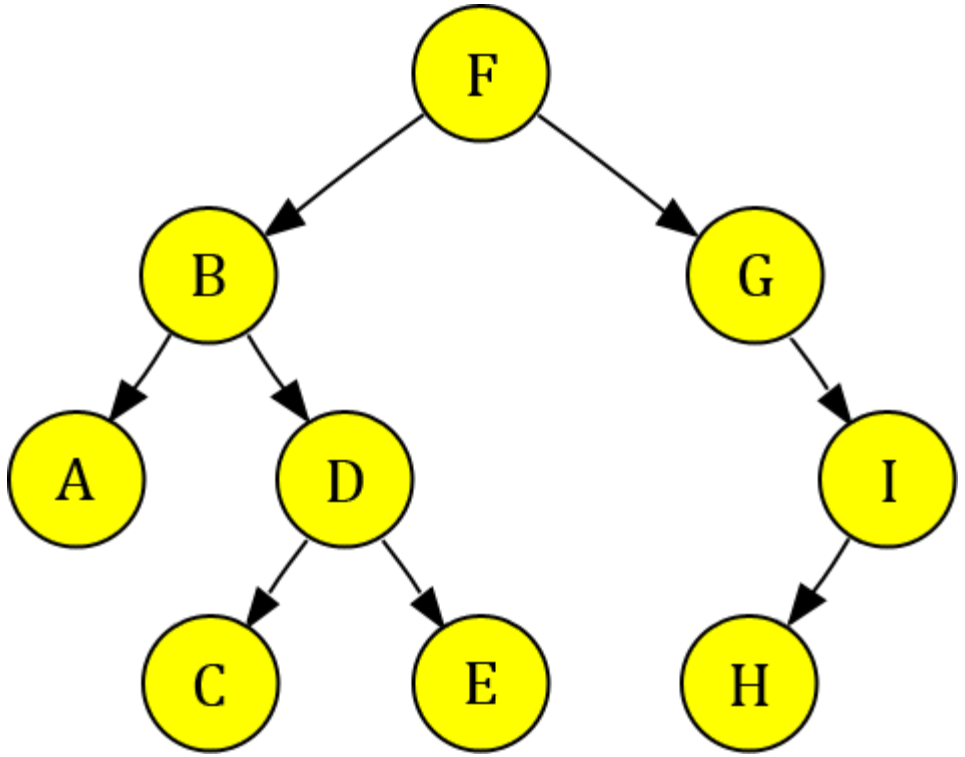
Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 01)

Queue:

Visited:

F	**B**	**G**	**A**	**D**	**I**	**C**	**E**	**H**	

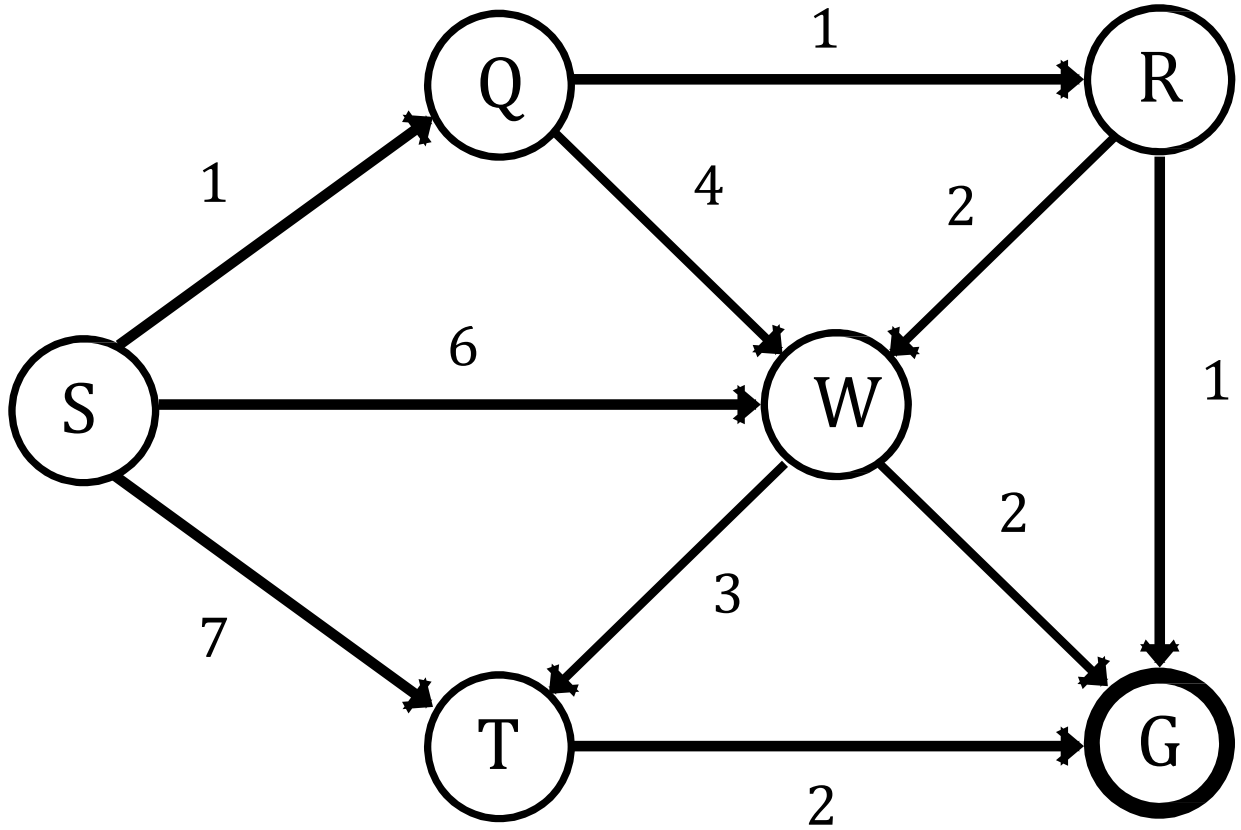


BFS Traversal: **F B G A D I C E H**

Breadth-First Search (Problem 02)

Queue:

Visited:



Current Node **Opened Node** **Closed Node**

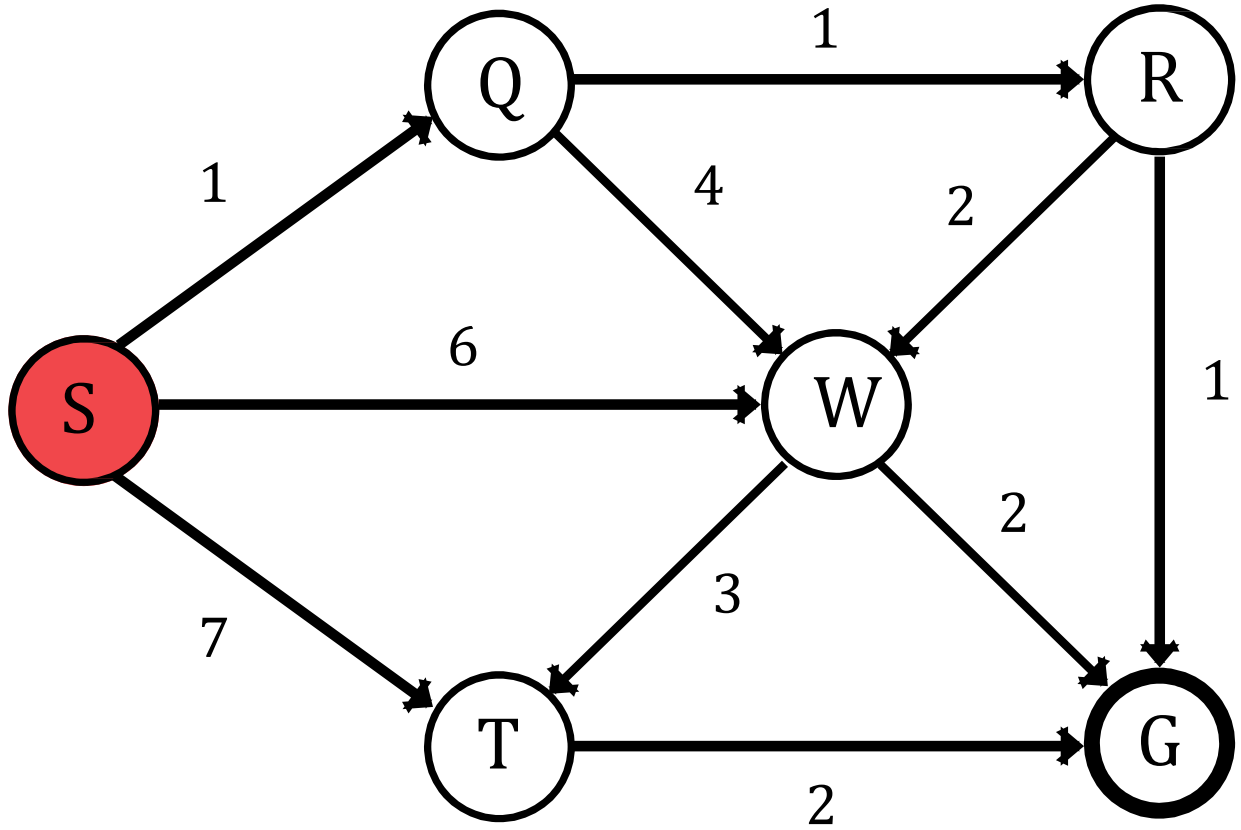
Breadth-First Search (Problem 02)

Current Node = S

Queue:

Visited:

S					



Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 02)

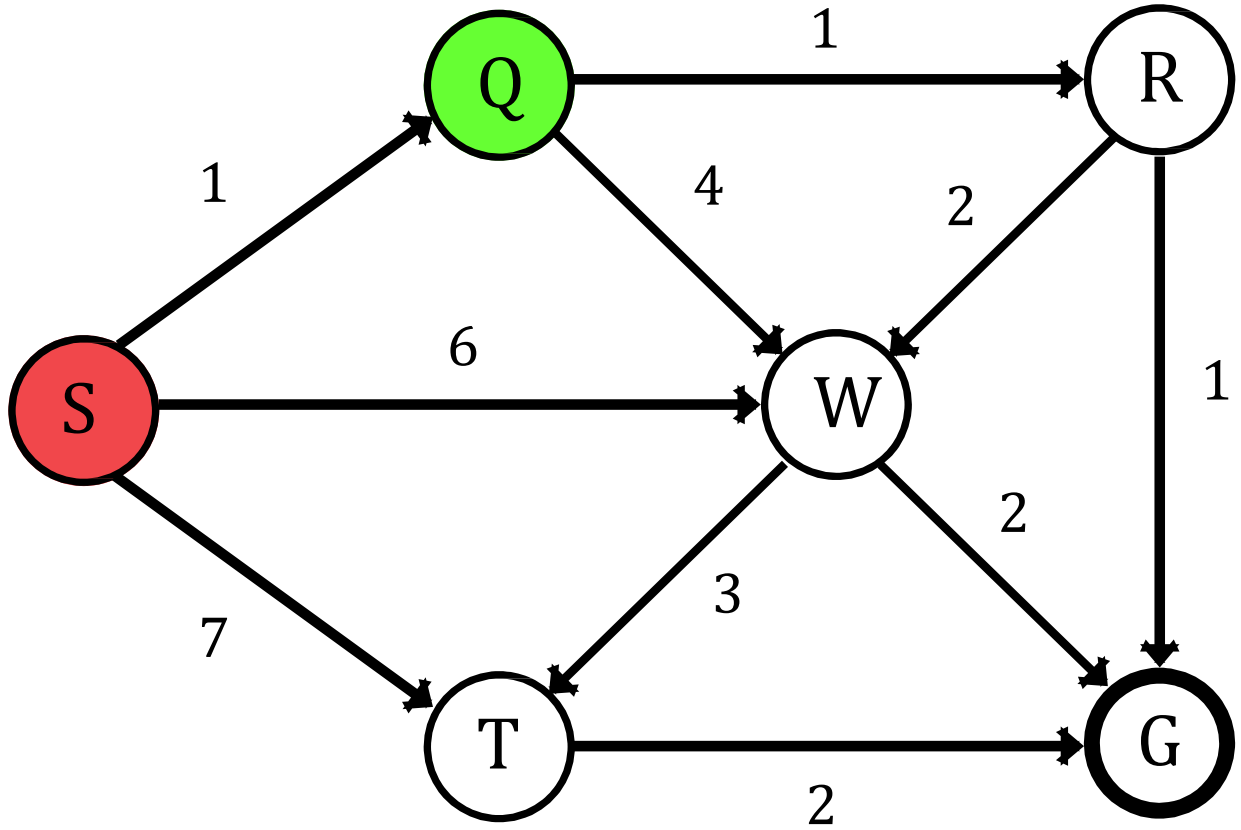
Current Node = S

Queue:

Q					

Visited:

S	Q				



Current Node Opened Node Closed Node

Breadth-First Search (Problem 02)

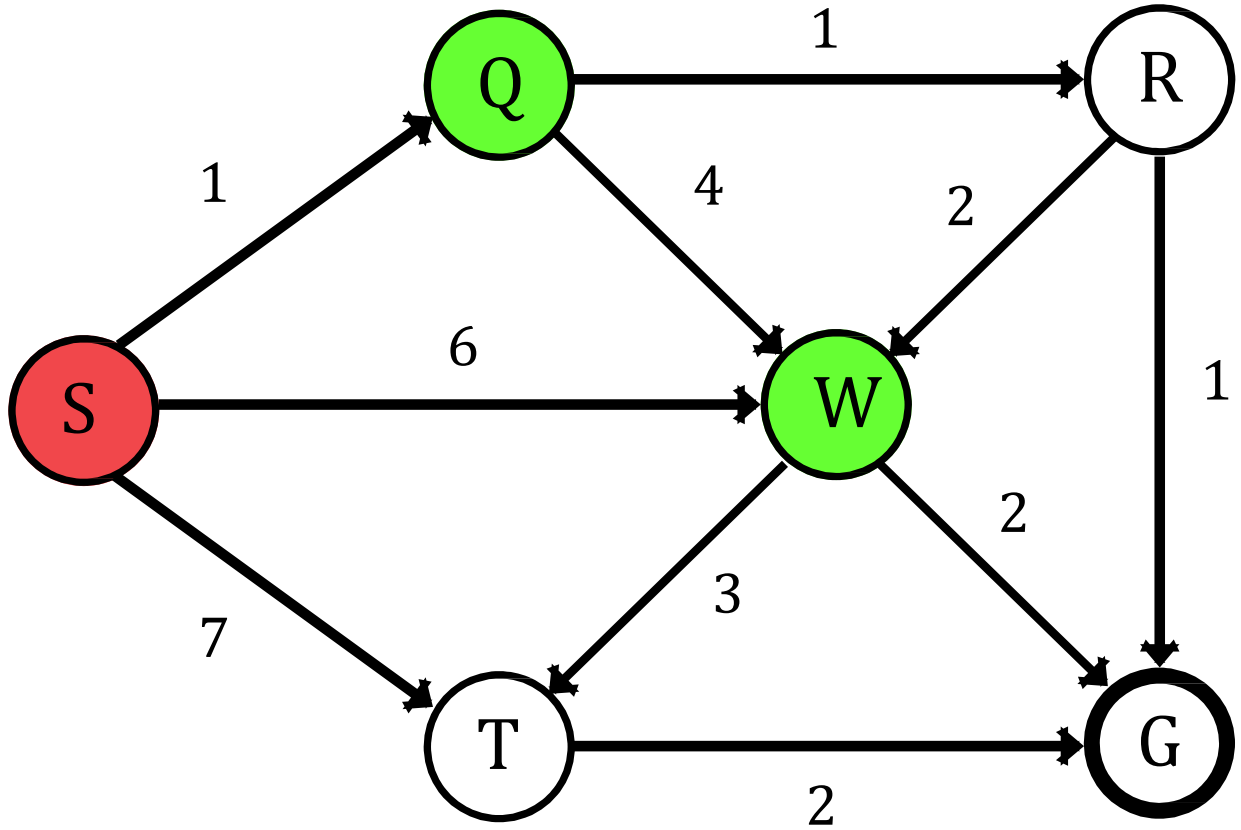
Current Node = S

Queue:

W	Q				

Visited:

S	Q	W			



Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 02)

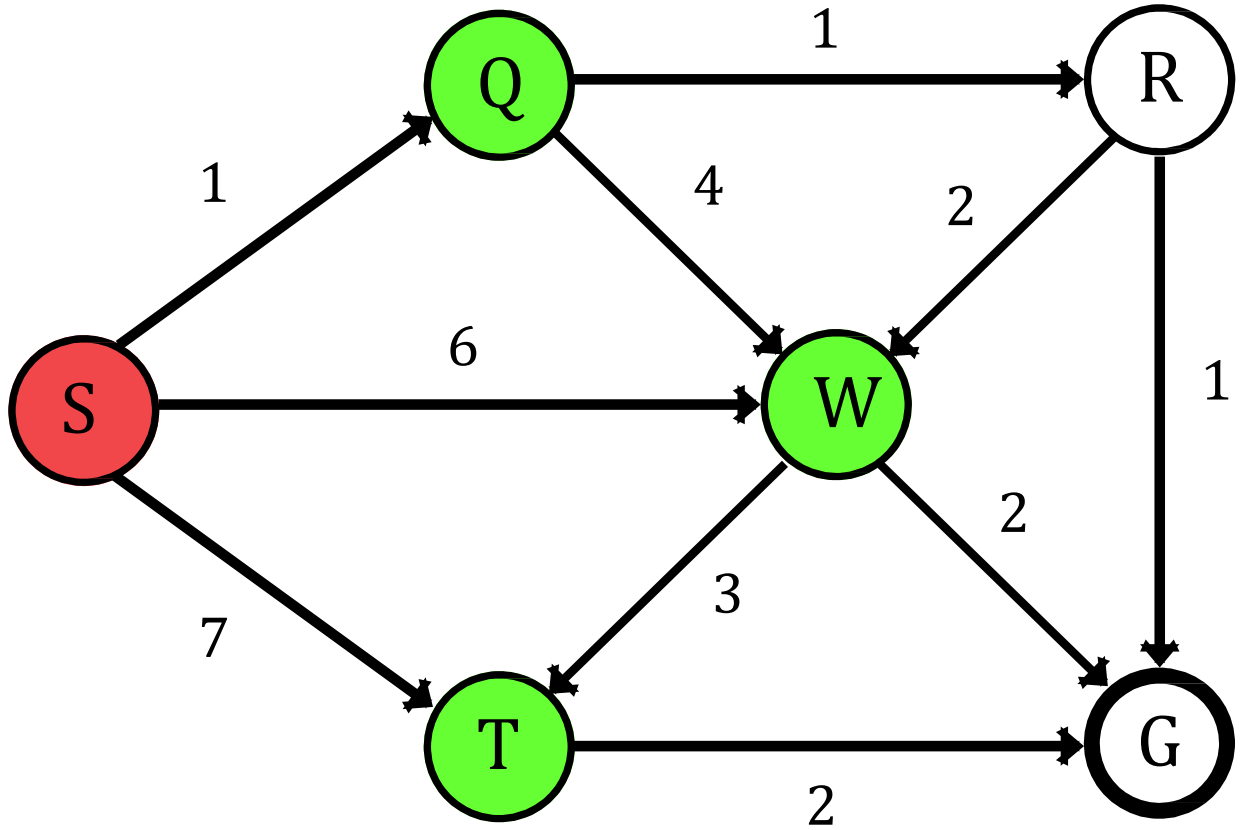
Current Node = S

Queue:

T	**W**	**Q**			

Visited:

S	**Q**	**W**	**T**		



Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 02)

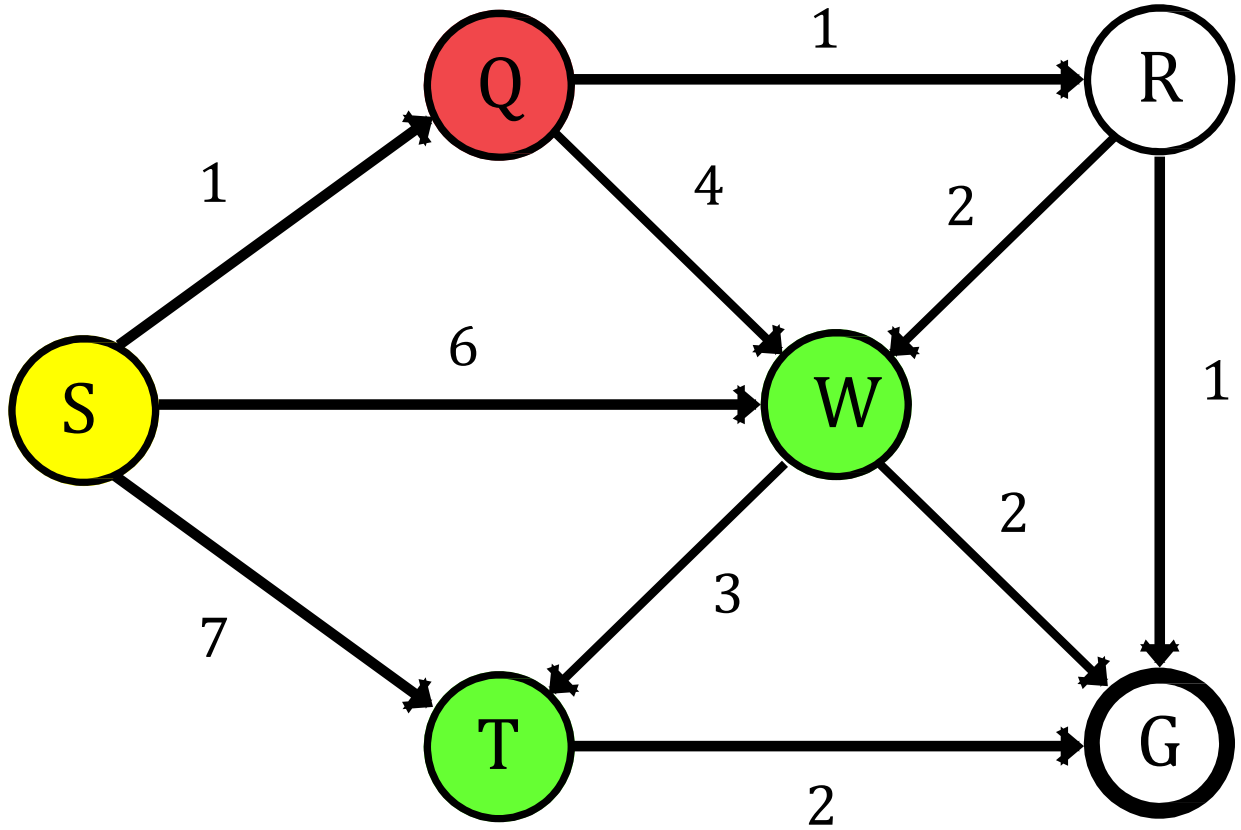
Current Node = Q

Queue:

T	**W**				

Visited:

S	**Q**	**W**	**T**		



Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 02)

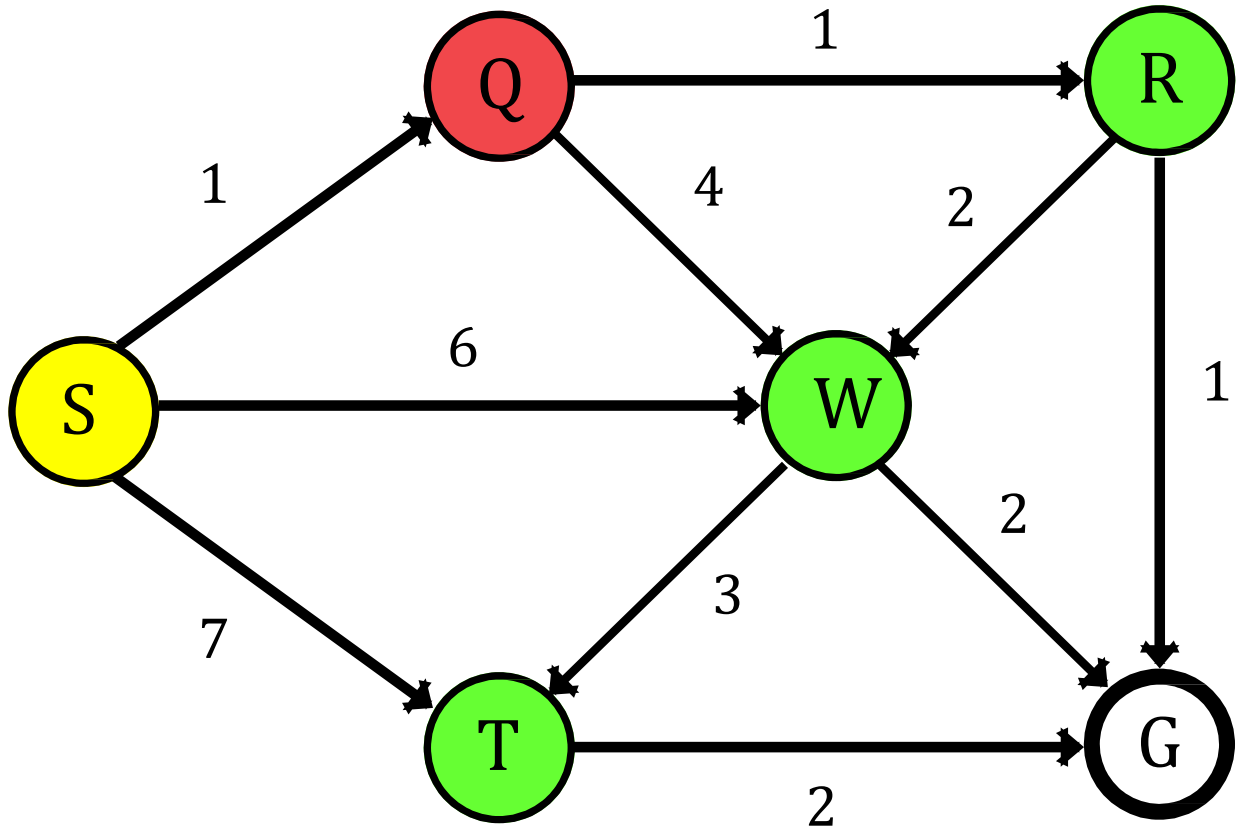
Current Node = Q

Queue:

R	**T**	**W**			

Visited:

S	**Q**	**W**	**T**	**R**	



Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 02)

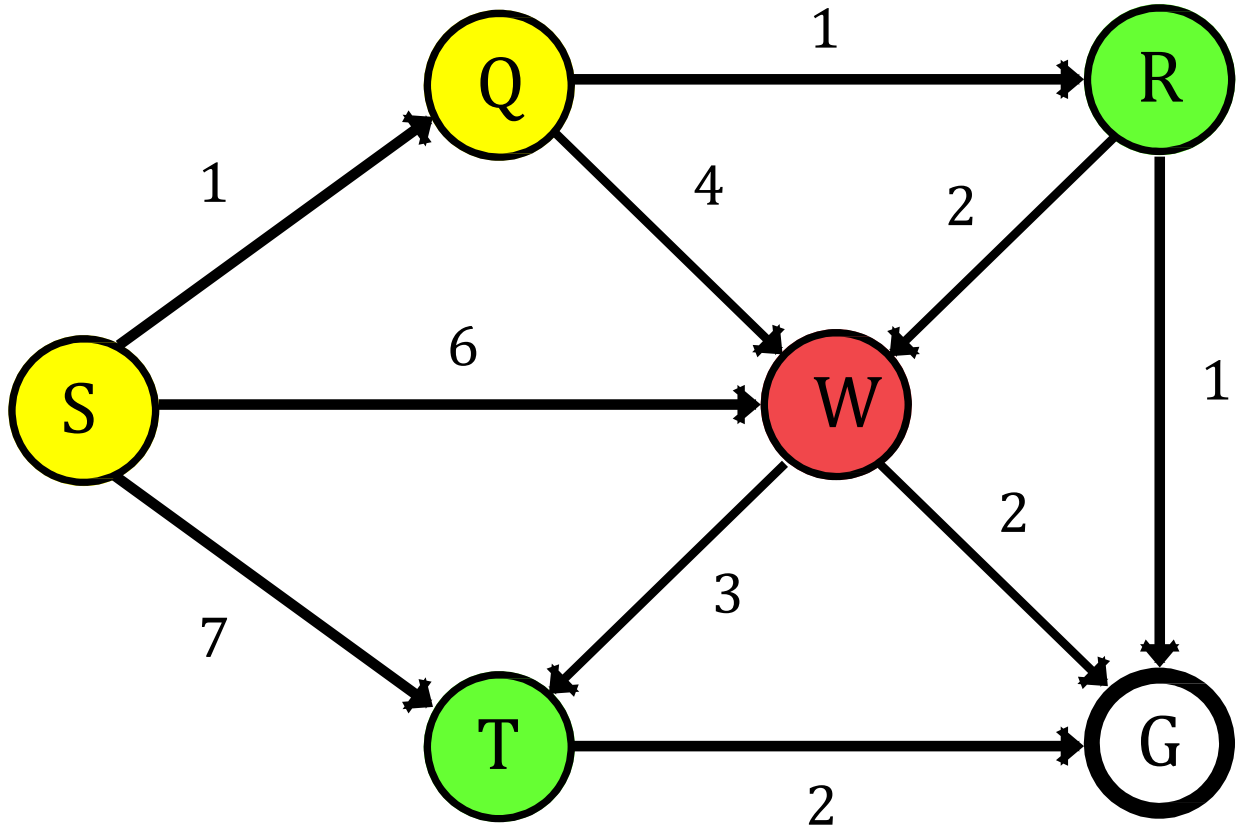
Current Node = W

Queue:

R	T				

Visited:

S	Q	W	T	R	



Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 02)

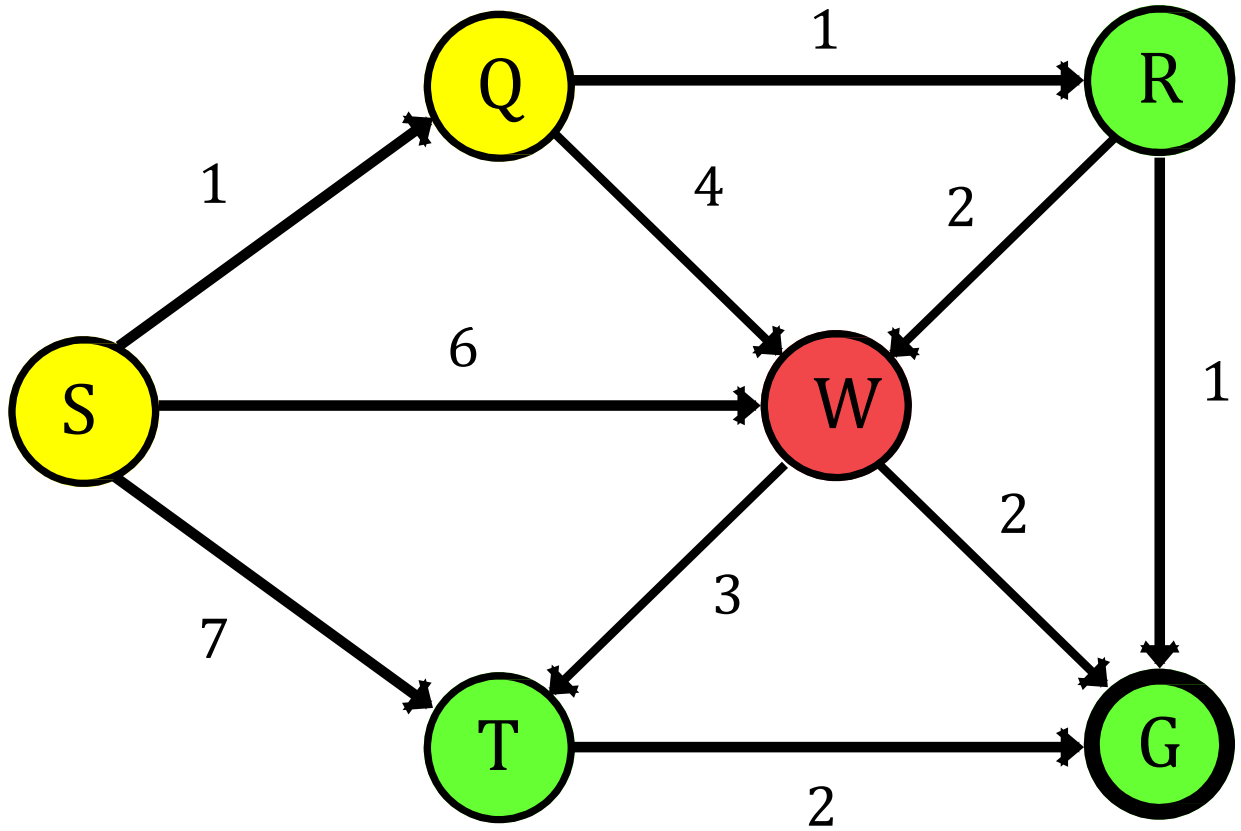
Current Node = W

Queue:

G	R	T			

Visited:

S	Q	W	T	R	G



Current Node Opened Node Closed Node

Breadth-First Search (Problem 02)

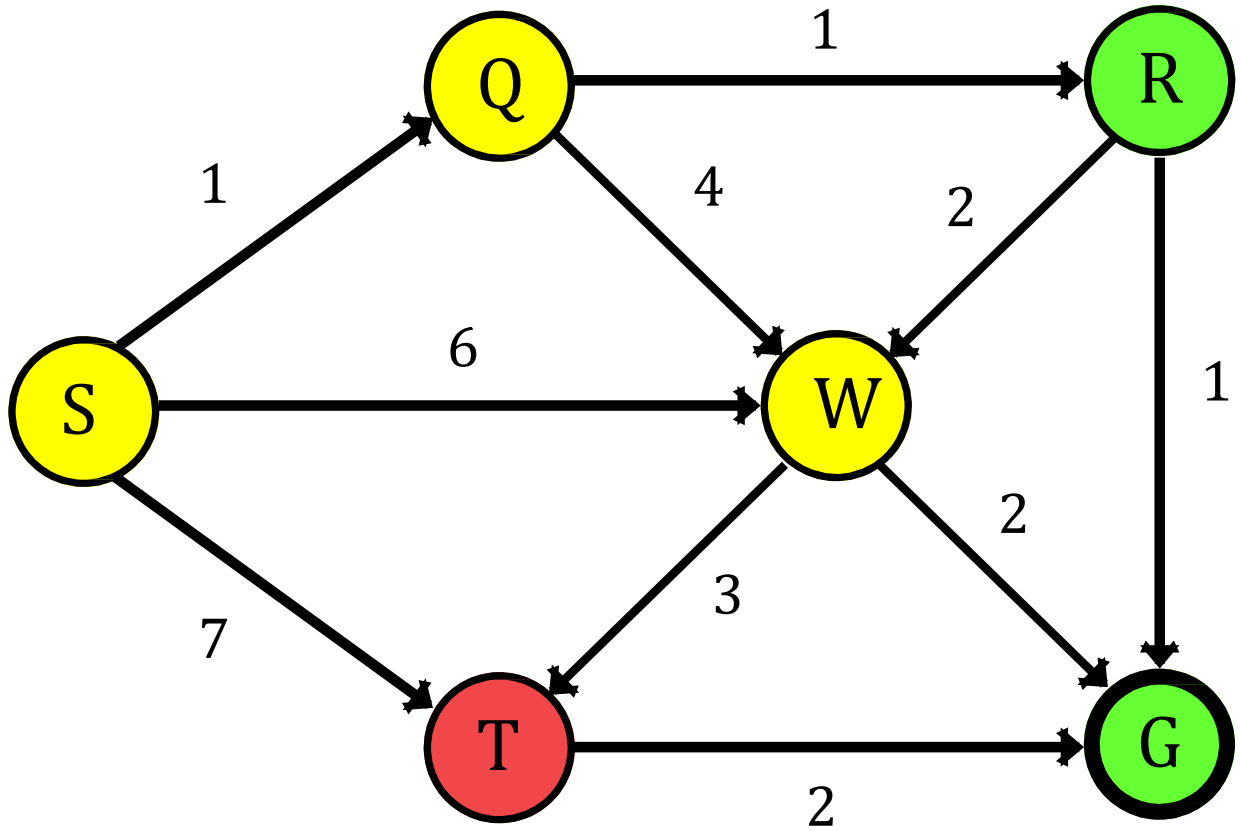
Current Node = T

Queue:

G	R				

Visited:

S	Q	W	T	R	G



Current Node Opened Node Closed Node

Breadth-First Search (Problem 02)

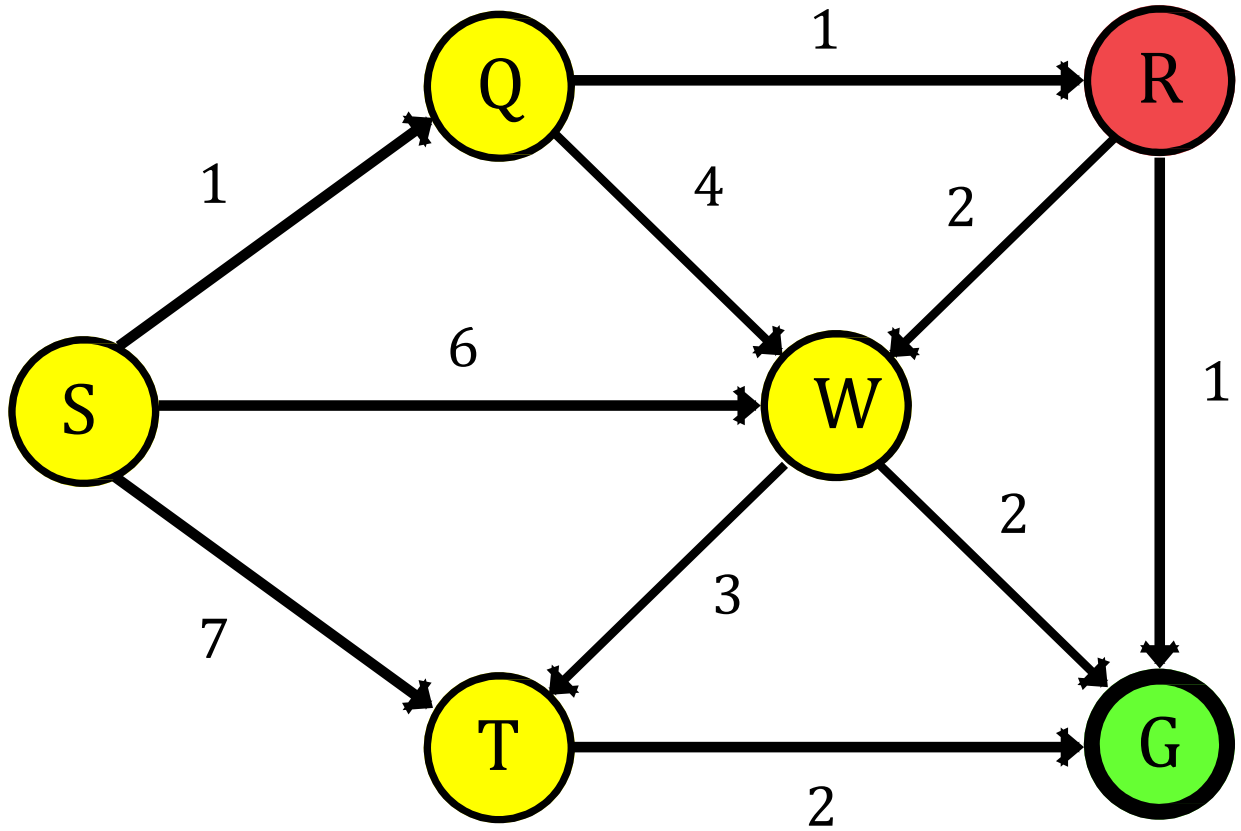
Current Node = R

Queue:

G					

Visited:

S	Q	W	T	R	G



Current Node Opened Node Closed Node

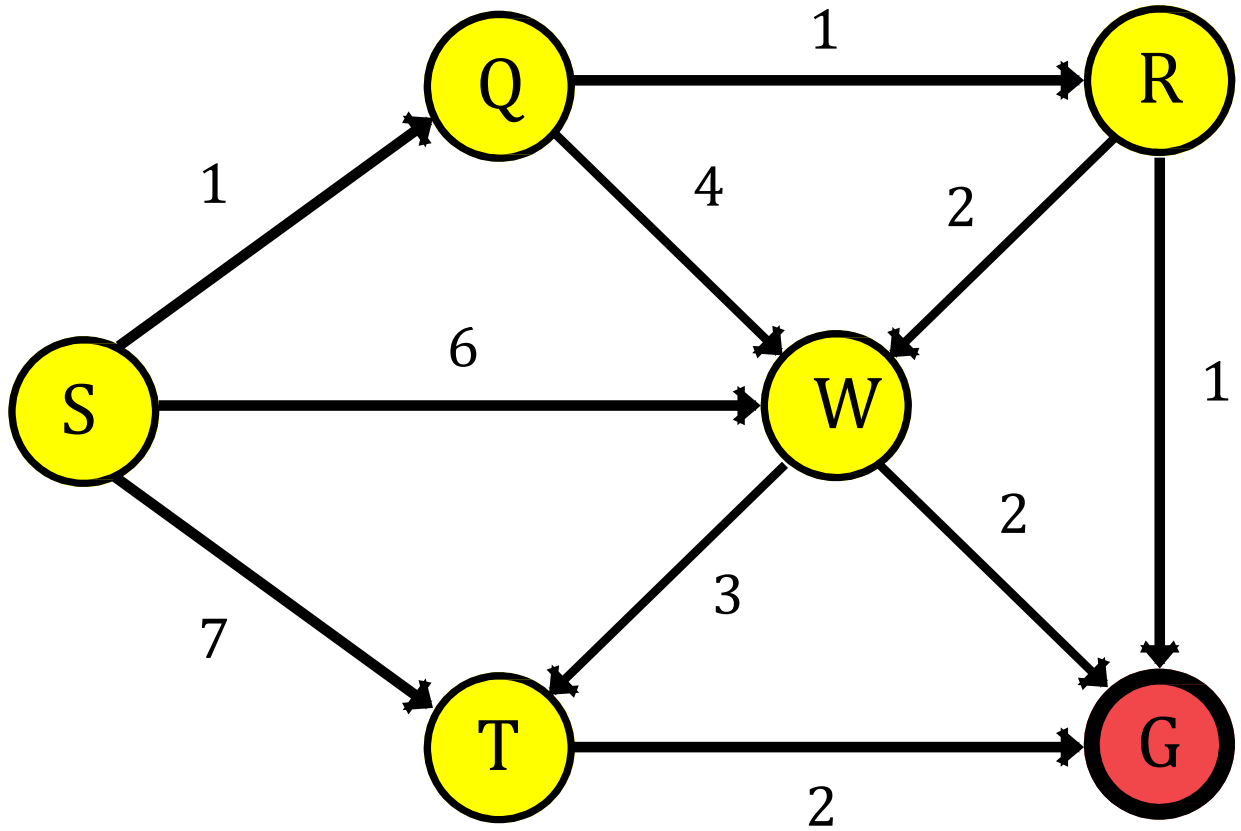
Breadth-First Search (Problem 02)

Current Node = G

Queue:

Visited:

S	**Q**	**W**	**T**	**R**	**G**



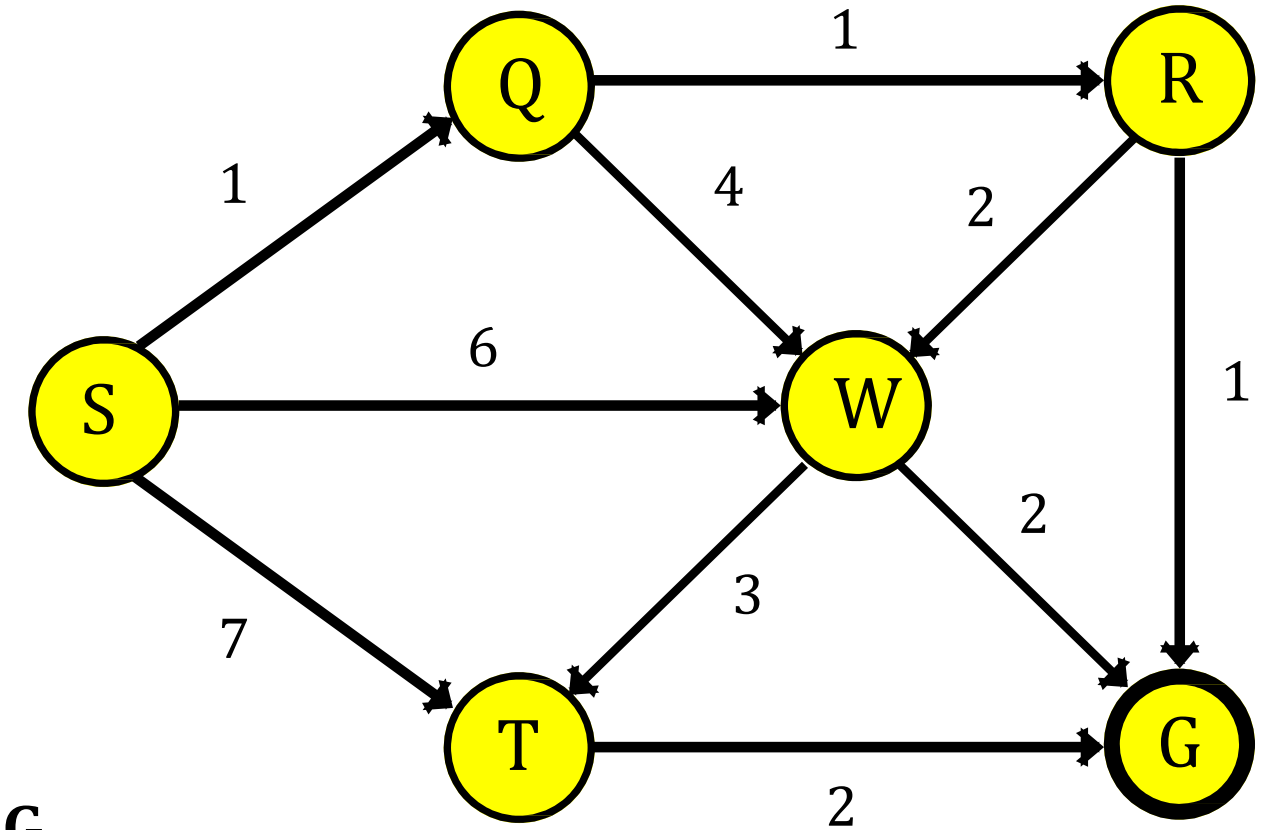
Current Node **Opened Node** **Closed Node**

Breadth-First Search (Problem 02)

Queue:

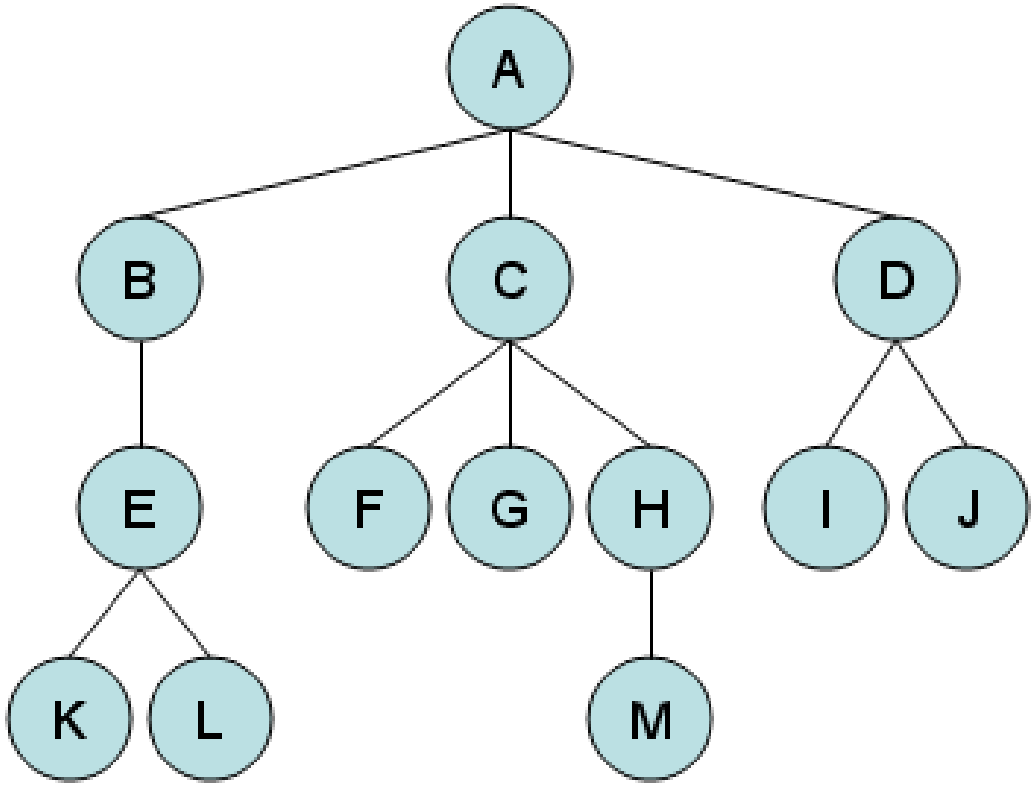
Visited:

S	**Q**	**W**	**T**	**R**	**G**



BFS Traversal: **S Q W T R G**

Breadth-First Search (Problem 03)



BFS Traversal: **A B C D E F G H I J K L M**

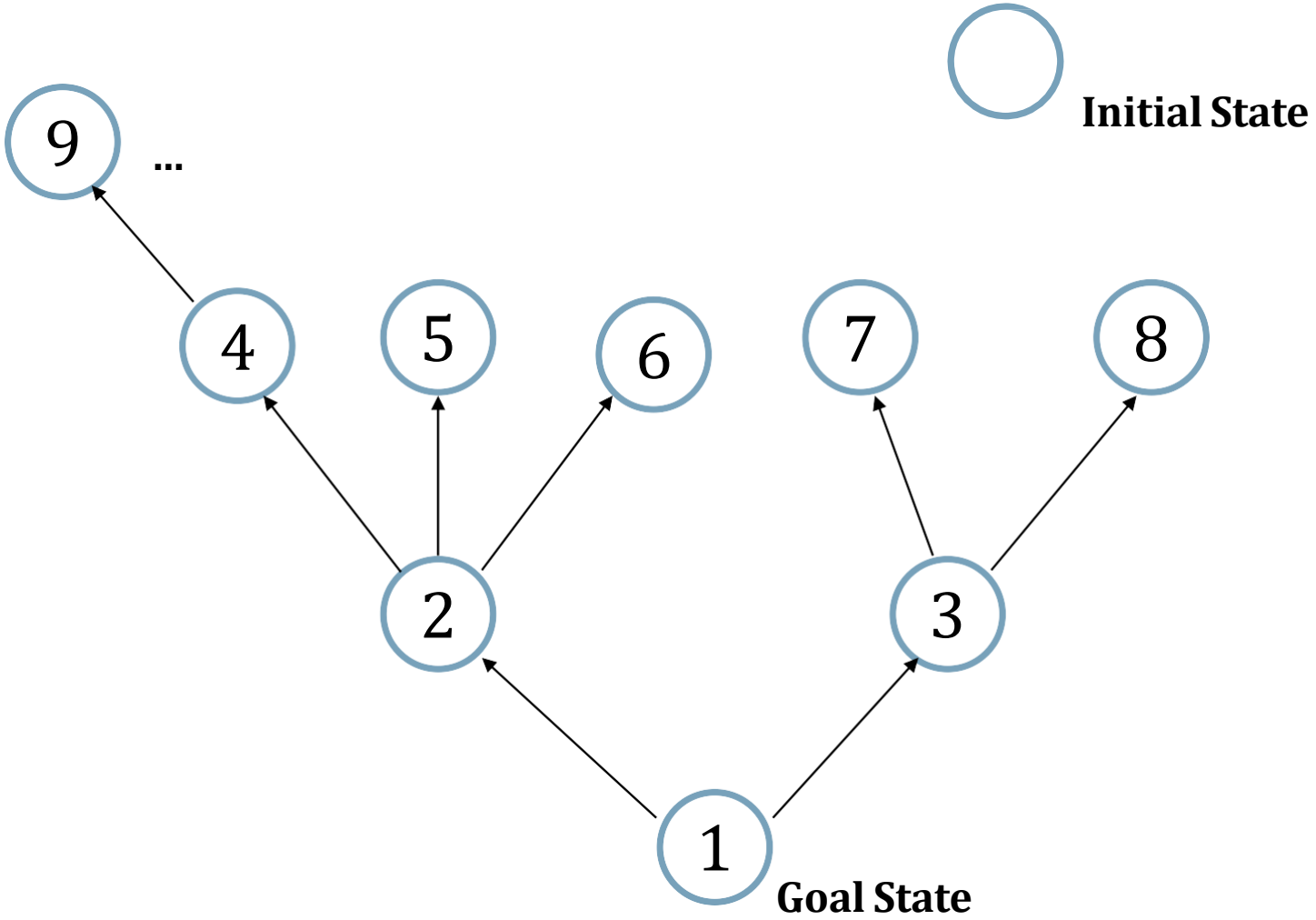
Properties of Breadth-First Search

Property	Description
Completeness	BFS is complete, meaning that it is guaranteed to find a solution if one exists, provided that the branching factor is finite and the search space is finite.
Optimality	BFS is optimal for finding the shortest path in an unweighted graph, ensuring that the first solution found is the shortest path.
Time Complexity	The time complexity of BFS is $O(b^d)$, where b is the branching factor and d is the depth of the shallowest solution.
Space Complexity	BFS has a space complexity of $O(b^d)$, where b is the branching factor and d is the depth of the shallowest solution.
Frontier	BFS maintains a queue as its frontier, ensuring that nodes are explored in the order they were discovered, level by level.

Backward Breadth-First Search

- Searches backwards from a goal state to a start state.
- Obviously this works best when the actions are reversible, and the set of goal states is small.

Backward Breadth-First Search



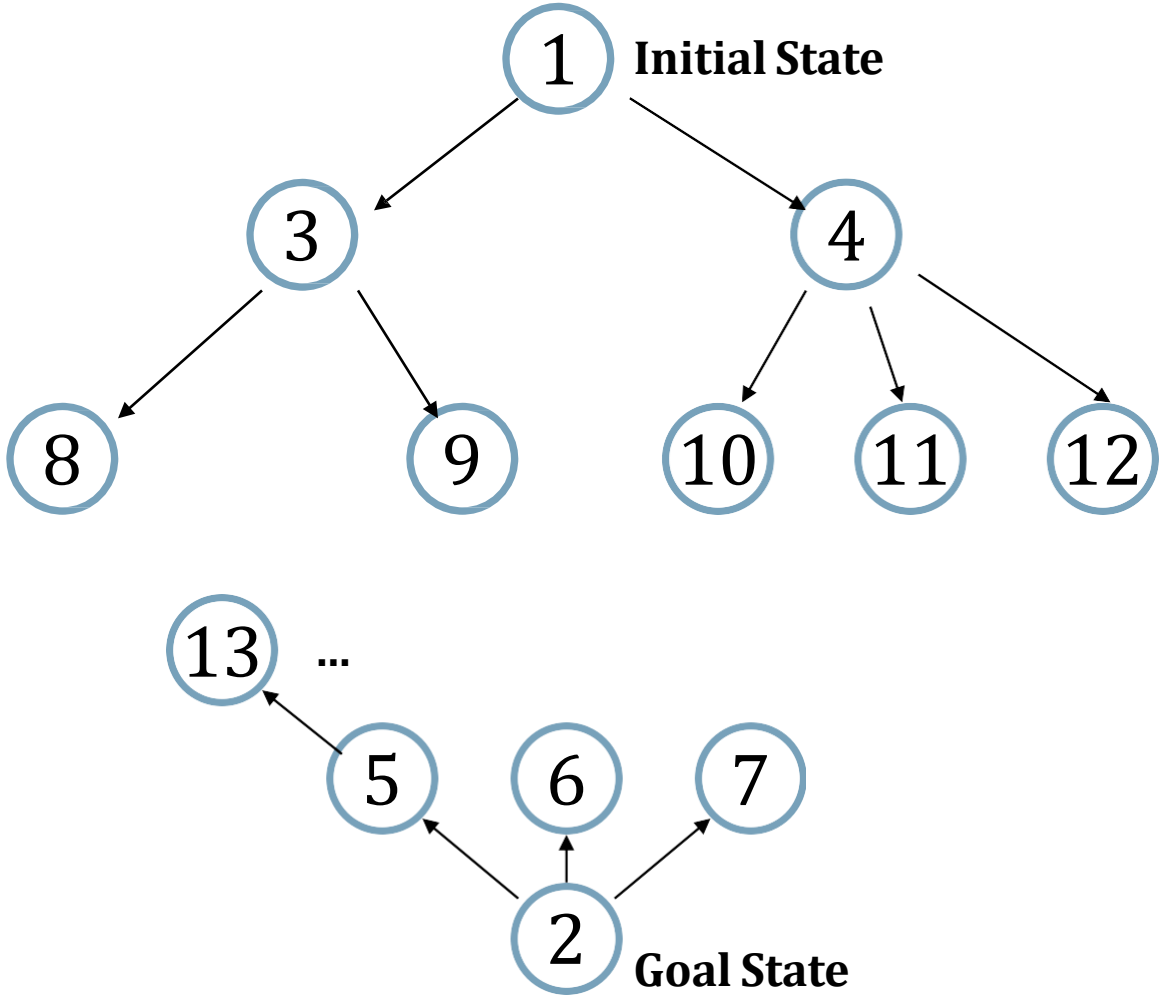
Properties of Backward Breadth-First Search

Property	Description
Completeness	Backward BFS is complete, meaning that it is guaranteed to find a solution if one exists, provided that the branching factor is finite and the search space is finite.
Optimality	Backward BFS is optimal for finding the shortest path in an unweighted graph, ensuring that the first solution found is the shortest path.
Time Complexity	The time complexity of Backward BFS is $O(b^d)$, where b is the branching factor and d is the depth of the shallowest solution.
Space Complexity	Backward BFS has a space complexity of $O(b^d)$, where b is the branching factor and d is the depth of the shallowest solution.
Frontier	Backward BFS maintains a queue as its frontier, ensuring that nodes are explored in the order they were discovered, level by level.

Bi-Directional Breadth-First Search

- The idea behind bidirectional search is to simultaneously search both forward from the initial state j and backward from the goal, and stop when the two searches meet in the middle.

Bi-Directional Breadth-First Search



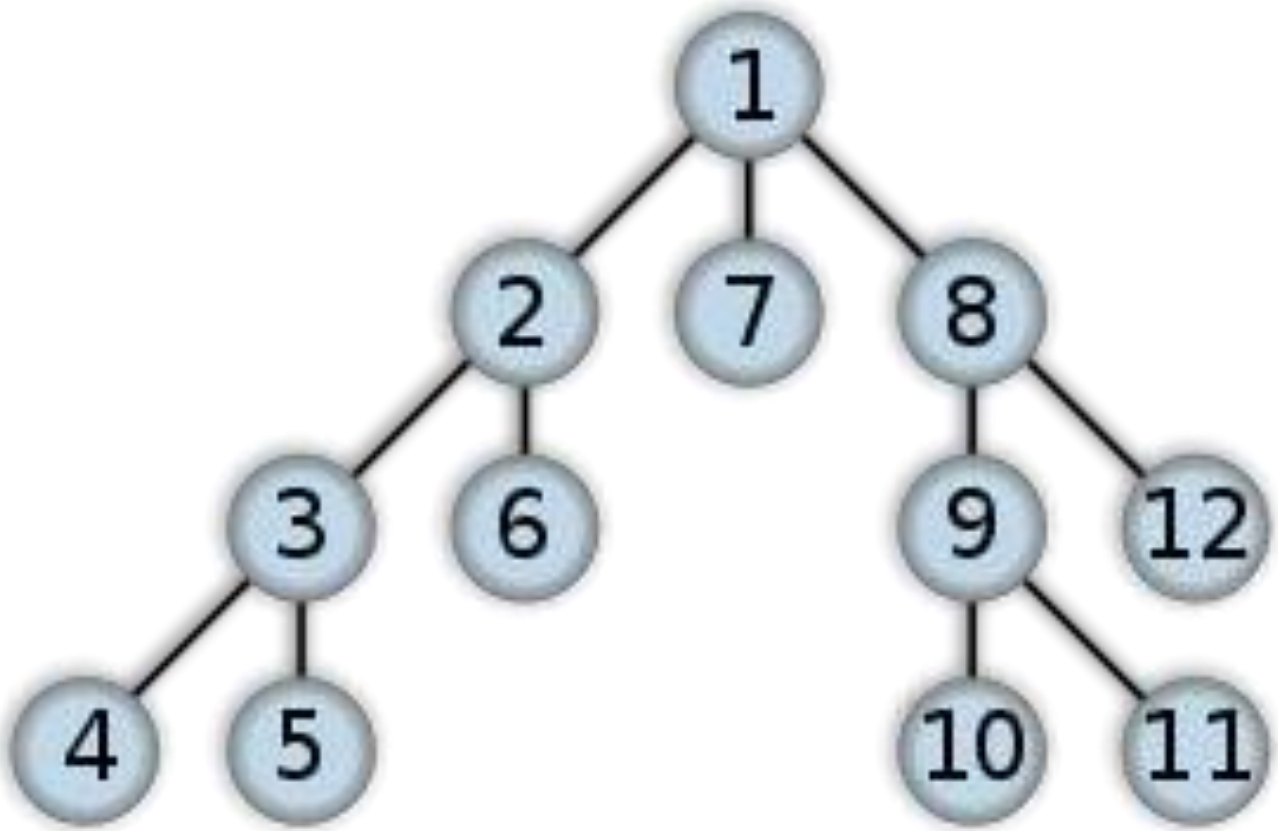
Properties of Bi-Directional Breadth-First Search

Property	Description
Completeness	Bi-BFS is complete, meaning that it is guaranteed to find a solution if one exists, provided that the branching factor is finite and the search space is finite.
Optimality	Bi-BFS is optimal for finding the shortest path in an unweighted graph, ensuring that the first solution found is the shortest path.
Time Complexity	The time complexity of Bi-BFS is $O(b^{d/2})$, where b is the branching factor and d is the shortest path length.
Space Complexity	Bi-BFS has a space complexity of $O(b^{d/2})$, where b is the branching factor and d is the shortest path length.
Frontier	Bi-BFS maintains two queues (forward and backward) as its frontier, exploring nodes from both the start and goal nodes simultaneously.

Depth-First Search

- Depth-first search (DFS) is an algorithm for traversing or searching tree or graph data structures. One starts at the root and explores as far as possible along each branch before backtracking.
- A depth first search begins at the root node (i.e. Initial node) and works downward to successively deeper levels.
- An operator is applied to the node to generate the next deeper node in sequence. This process continues until a solution is found or backtracking is forced by reaching a dead end.

Depth-First Search

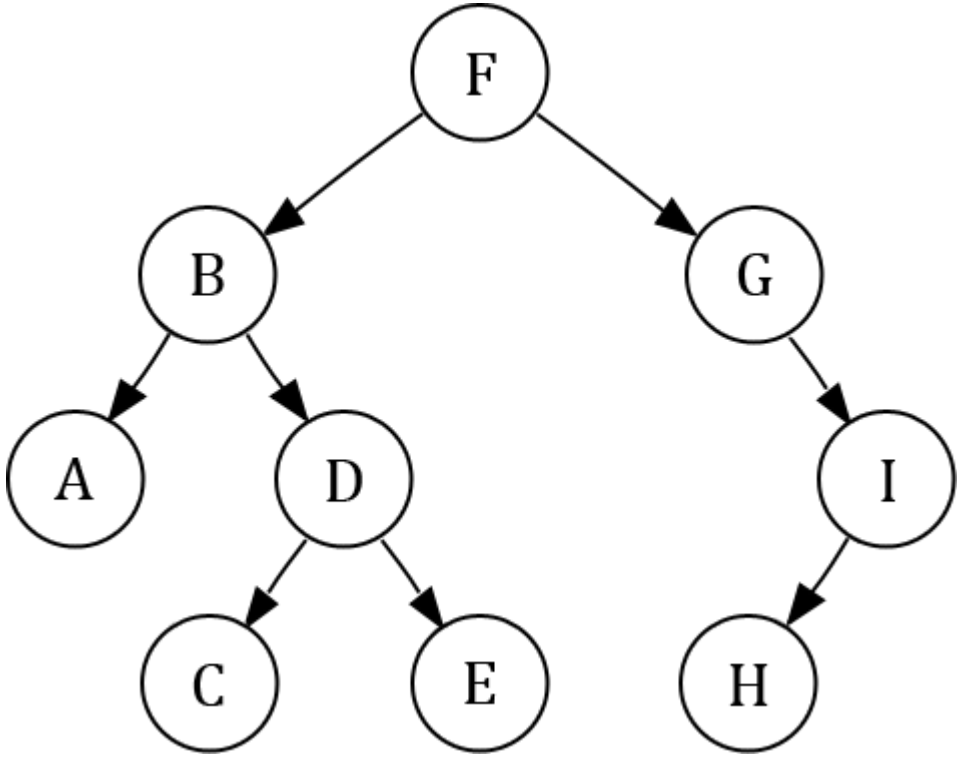


Depth-First Search

Algorithm:

- Step 01: Choose the starting node, mark it as visited, push it on Stack and make it current node.
- Step 02: Find adjacent unvisited node of current node, push it on Stack, mark it visited and make it current node.
- Step 03: If no unvisited adjacent node found, pop out node from Stack and make top of the stack as current node.
- Step 04: Repeat Step 02 to Step 03 until Stack is empty.

Depth-First Search (Problem 01)



DFS Traversal: **F B A D C E G I H**

Depth-First Search (Problem 01)

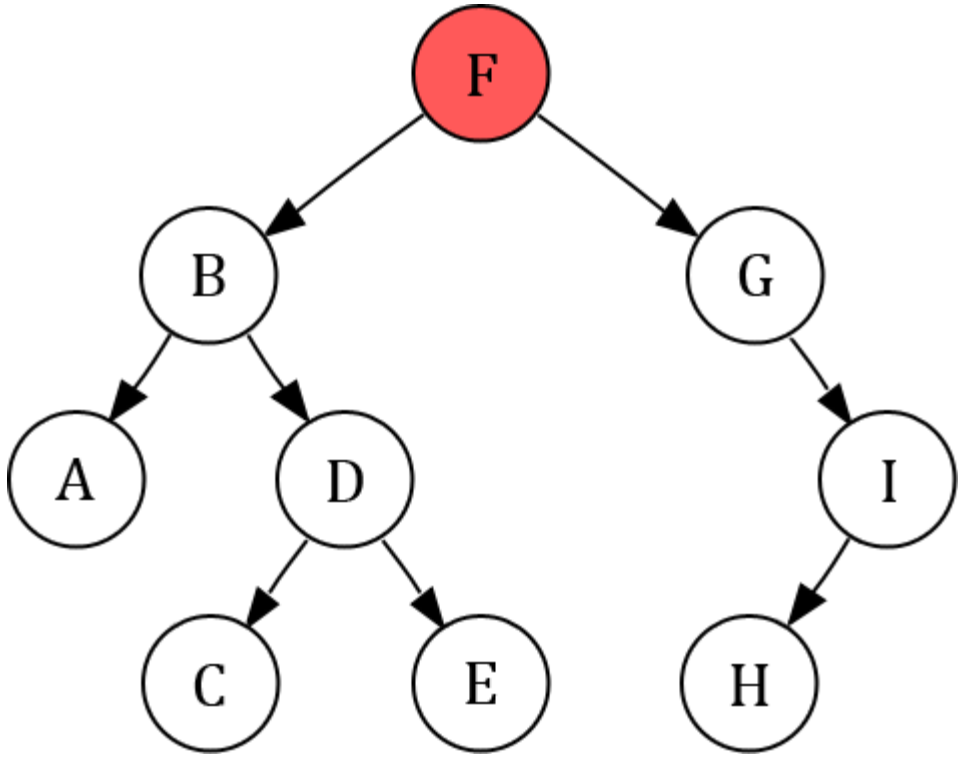
Current Node = F

Stack:

F									

Visited:

F									



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 01)

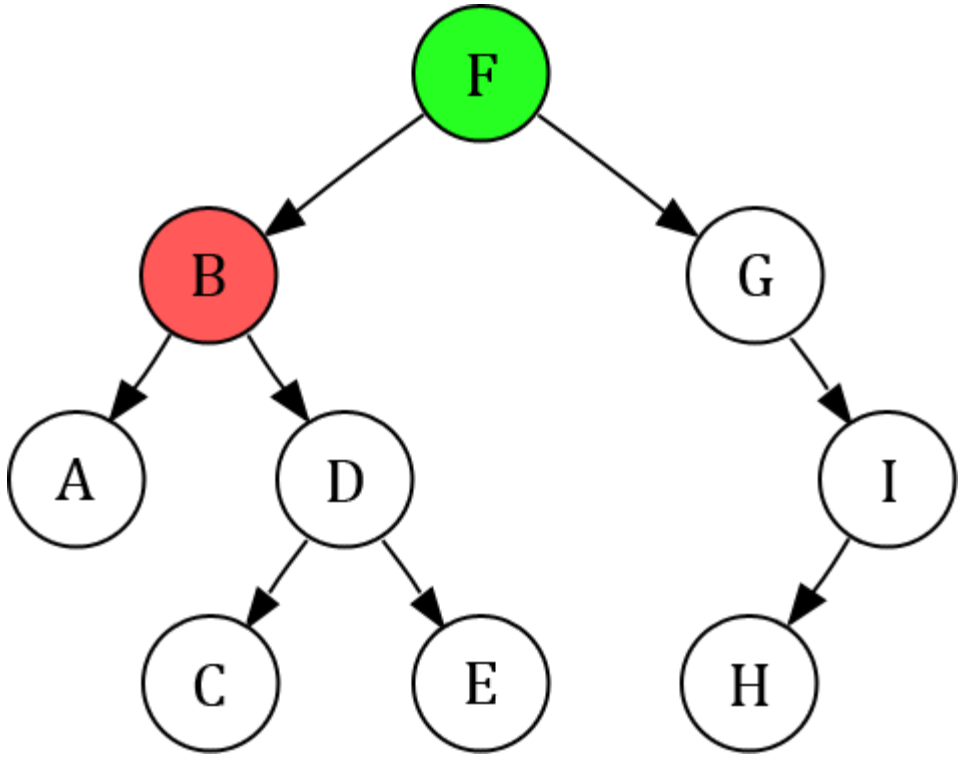
Current Node = B

Stack:

F	B								

Visited:

F	B								



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 01)

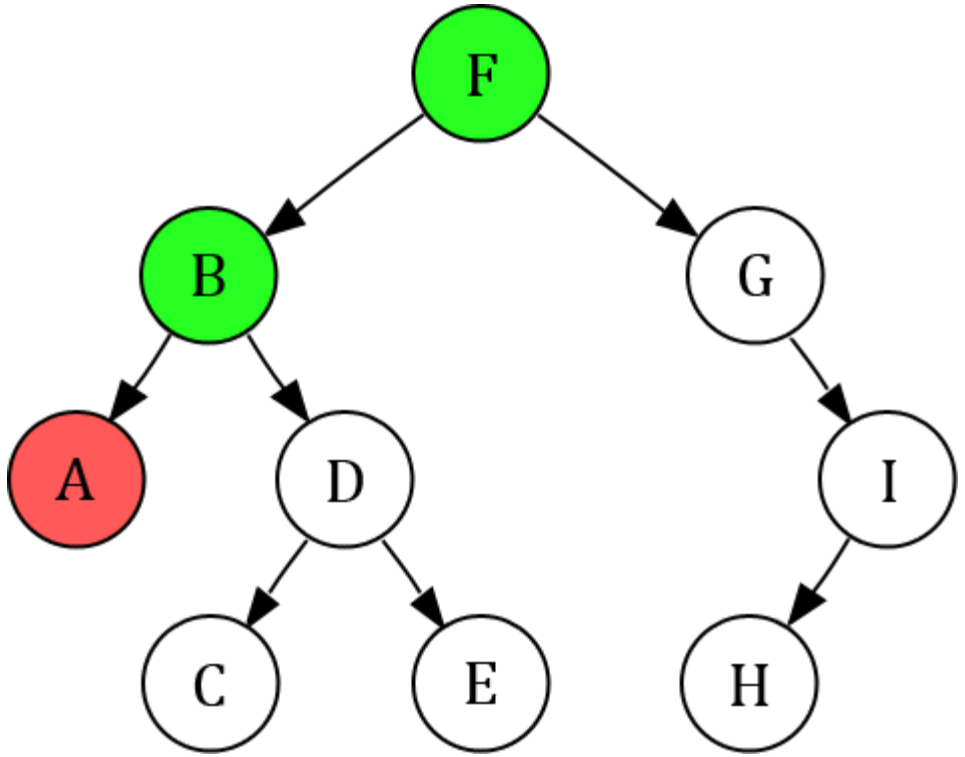
Current Node = A

Stack:

F	B	A							

Visited:

F	B	A							



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 01)

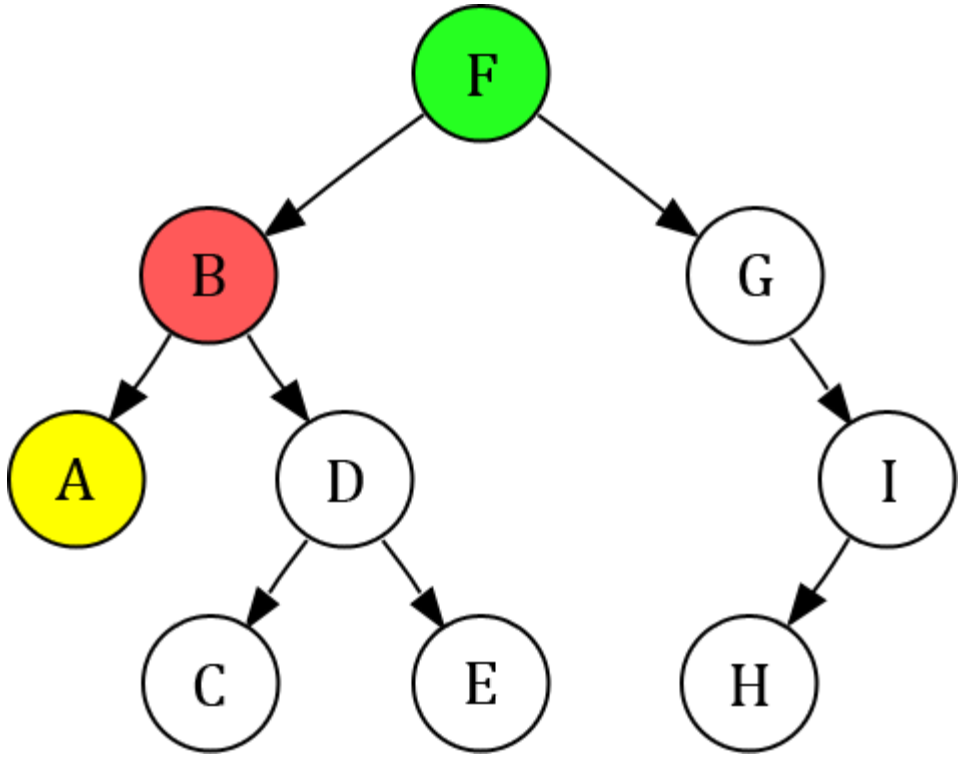
Current Node = B

Stack:

F	B								

Visited:

F	B	A							



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 01)

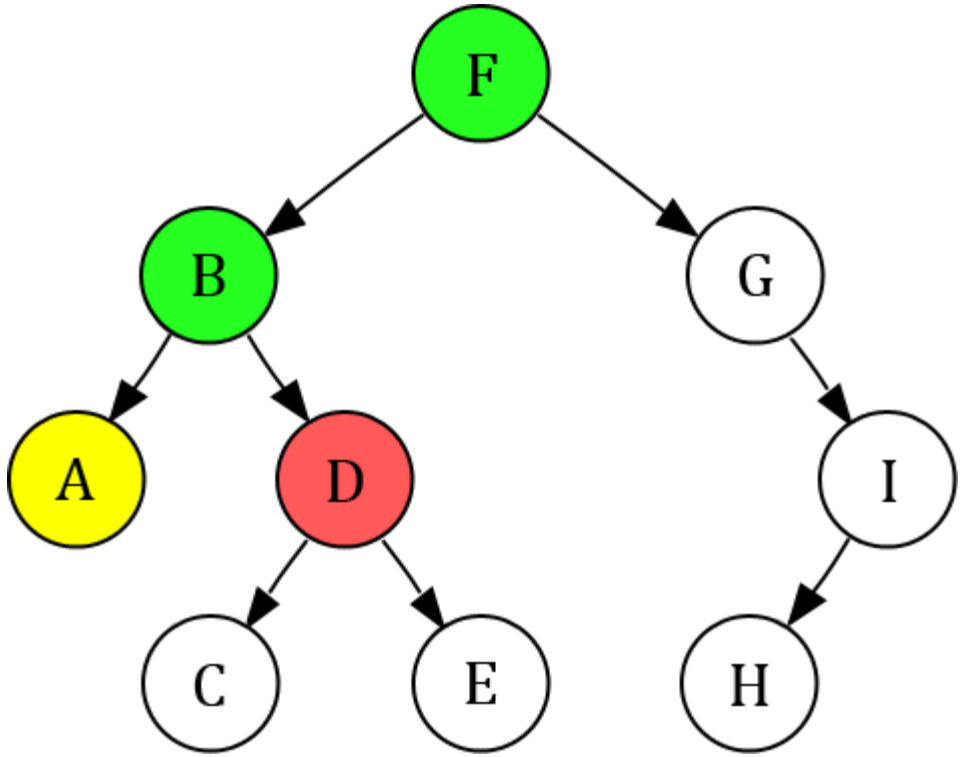
Current Node = D

Stack:

F	B	D							

Visited:

F	B	A	D						



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 01)

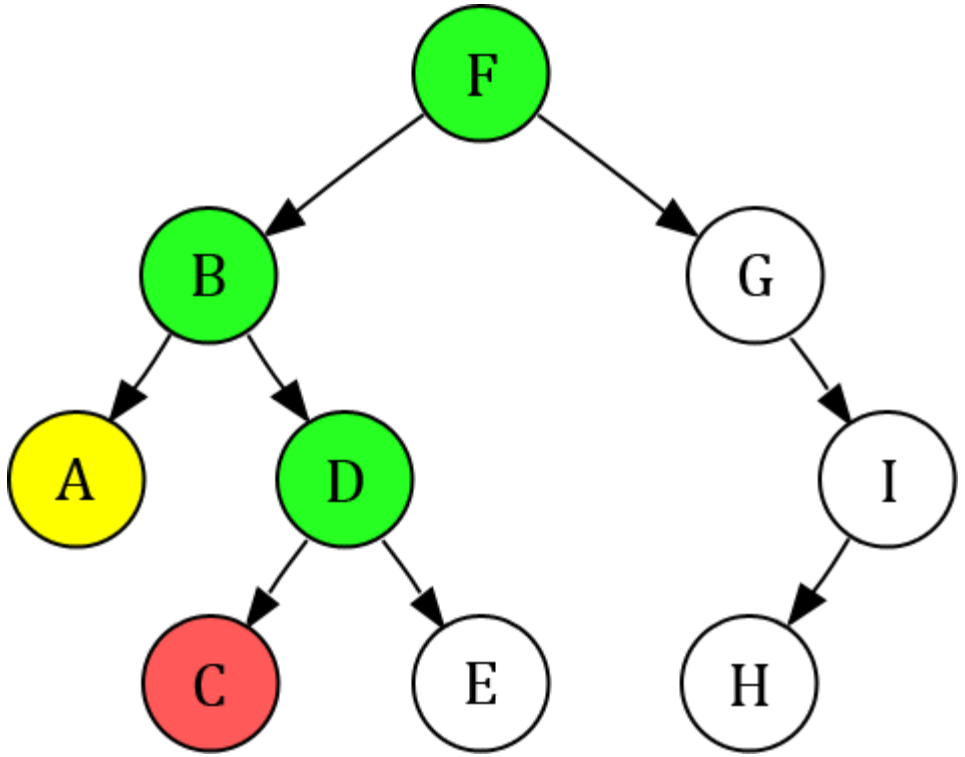
Current Node = C

Stack:

F	B	D	C						

Visited:

F	B	A	D	C					



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 01)

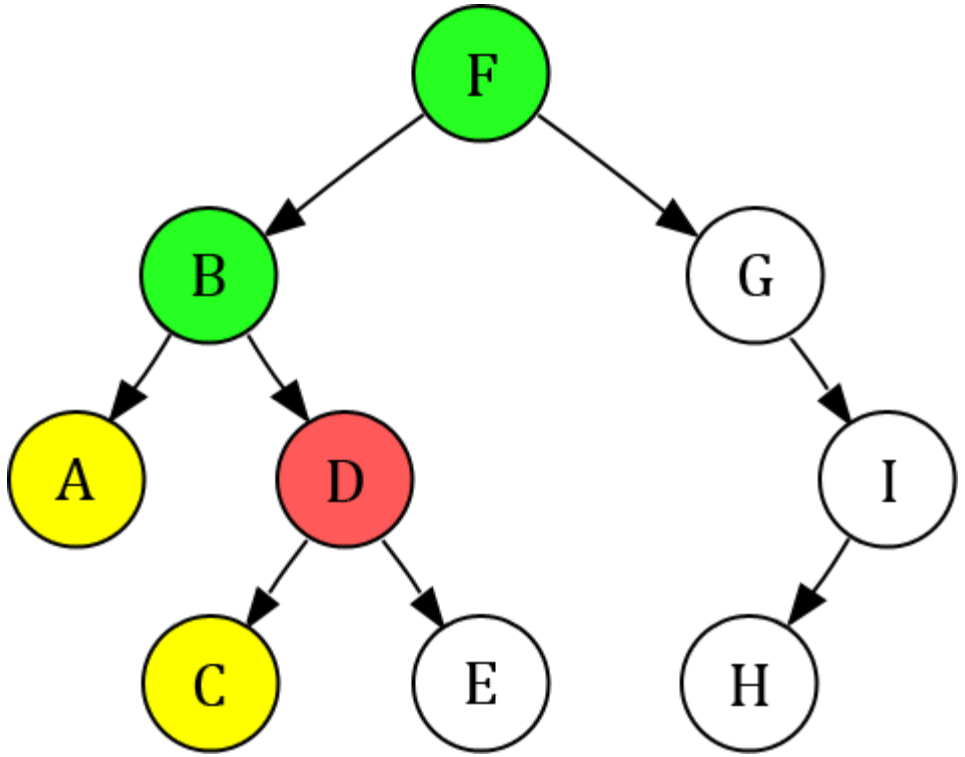
Current Node = D

Stack:

F	B	D							

Visited:

F	B	A	D	C					



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 01)

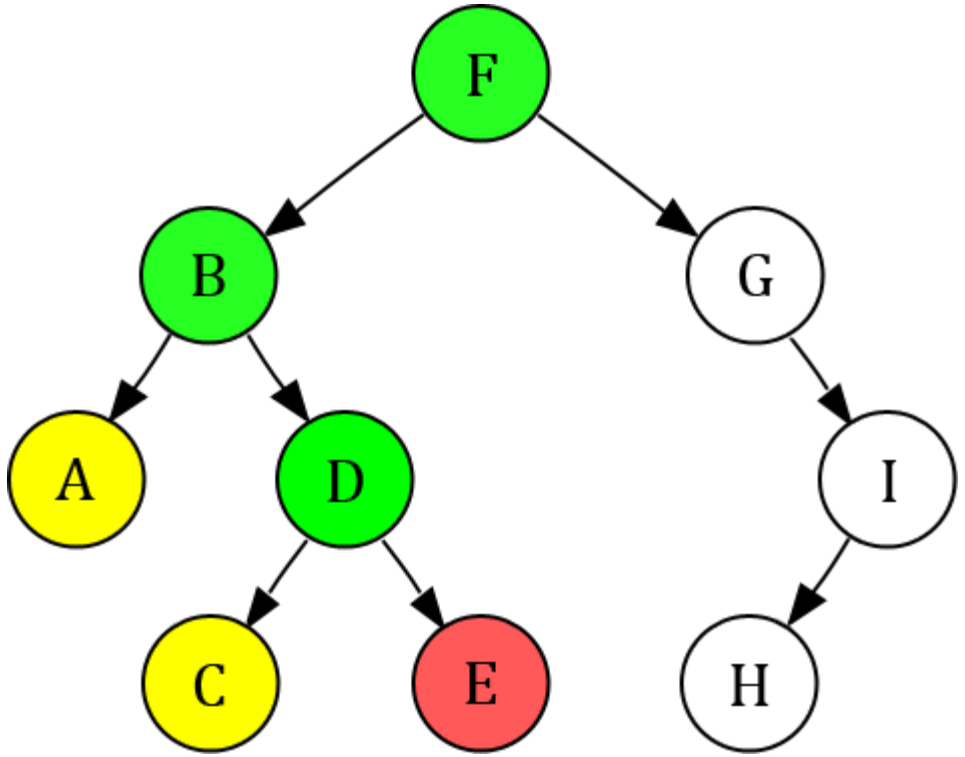
Current Node = E

Stack:

F	B	D	E						

Visited:

F	B	A	D	C	E				



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 01)

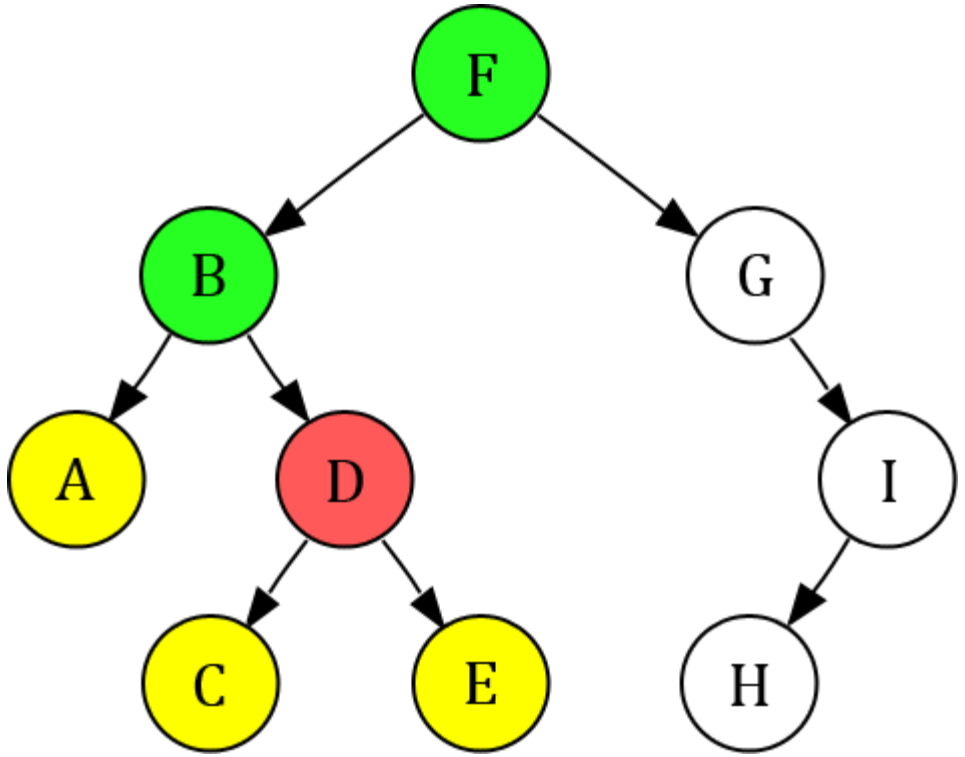
Current Node = D

Stack:

F	B	D							

Visited:

F	B	A	D	C	E				



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 01)

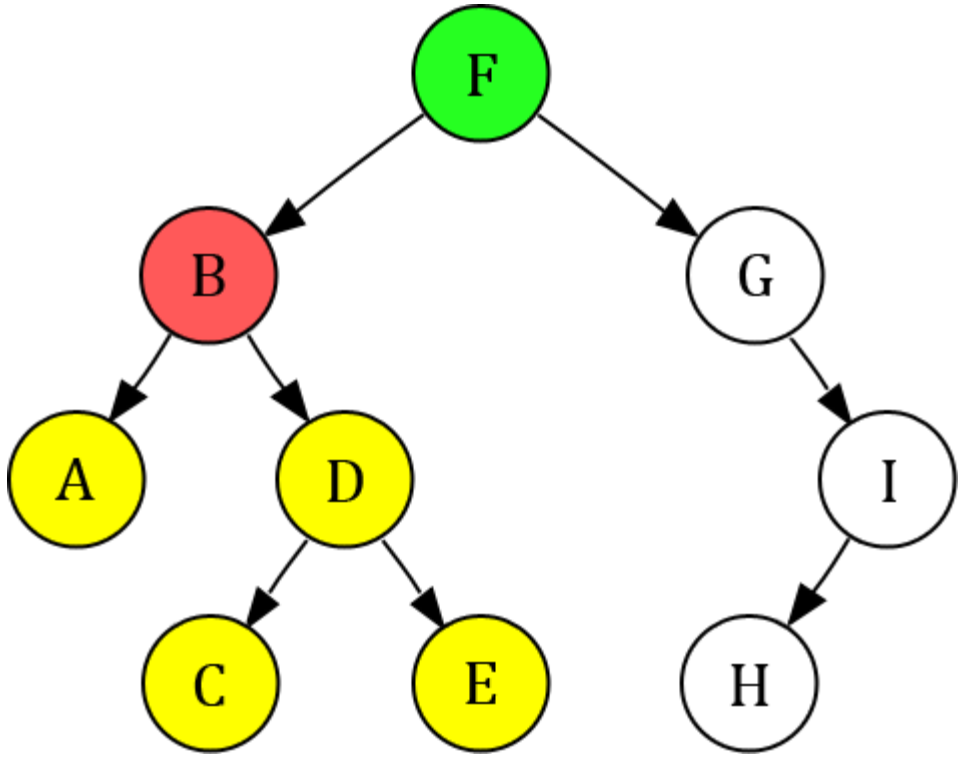
Current Node = B

Stack:

F	B								

Visited:

F	B	A	D	C	E				



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 01)

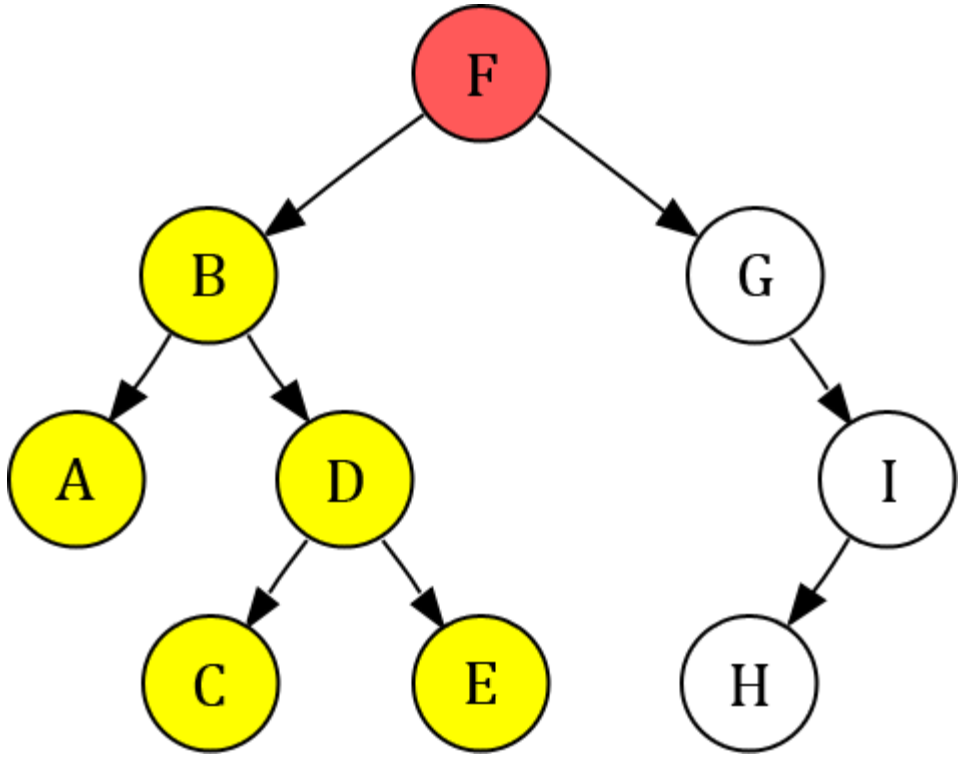
Current Node = F

Stack:

F									

Visited:

F	**B**	**A**	**D**	**C**	**E**				



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 01)

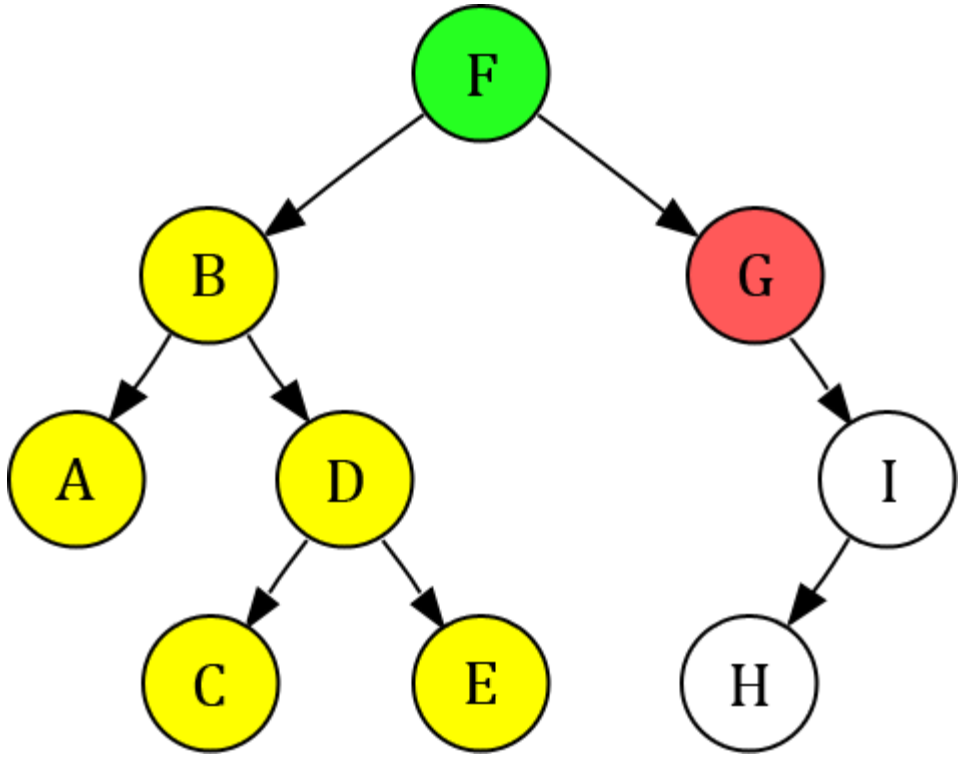
Current Node = G

Stack:

F	G								

Visited:

F	B	A	D	C	E	G			



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 01)

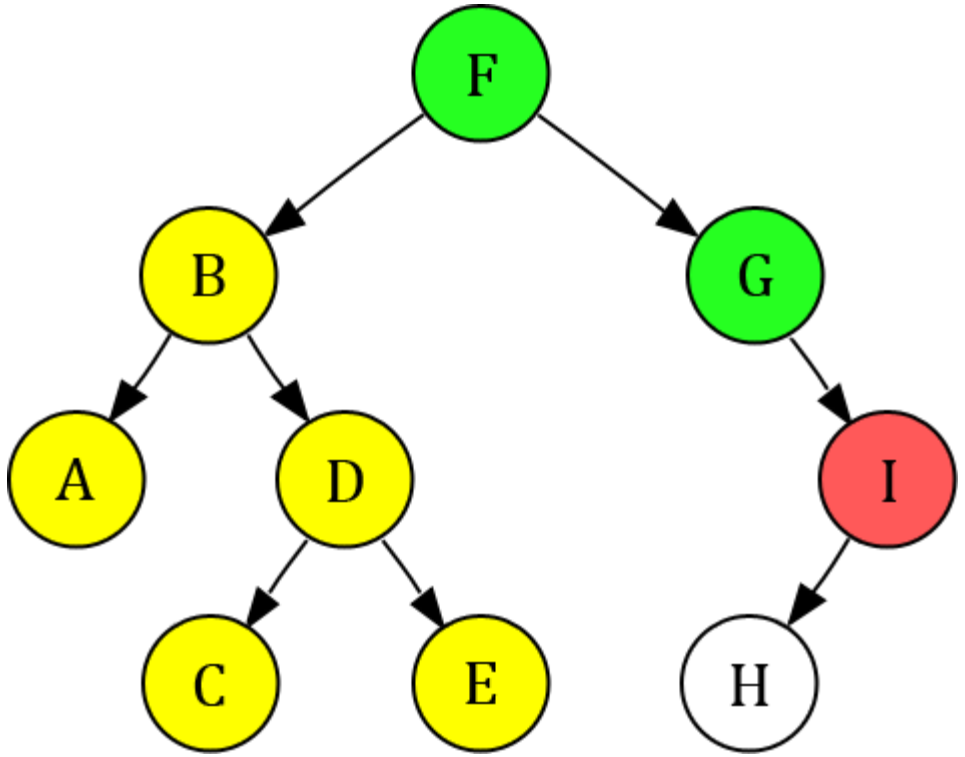
Current Node = I

Stack:

F	G	I							

Visited:

F	B	A	D	C	E	G	I		



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 01)

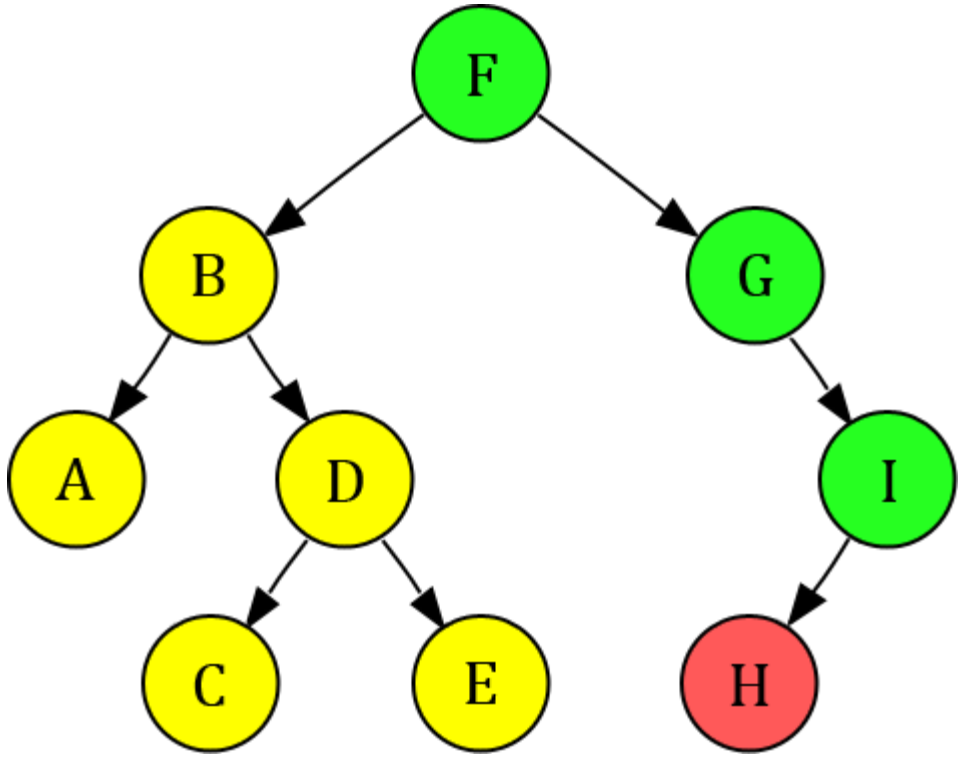
Current Node = H

Stack:

F	G	I	H						

Visited:

F	B	A	D	C	E	G	I	H	



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 01)

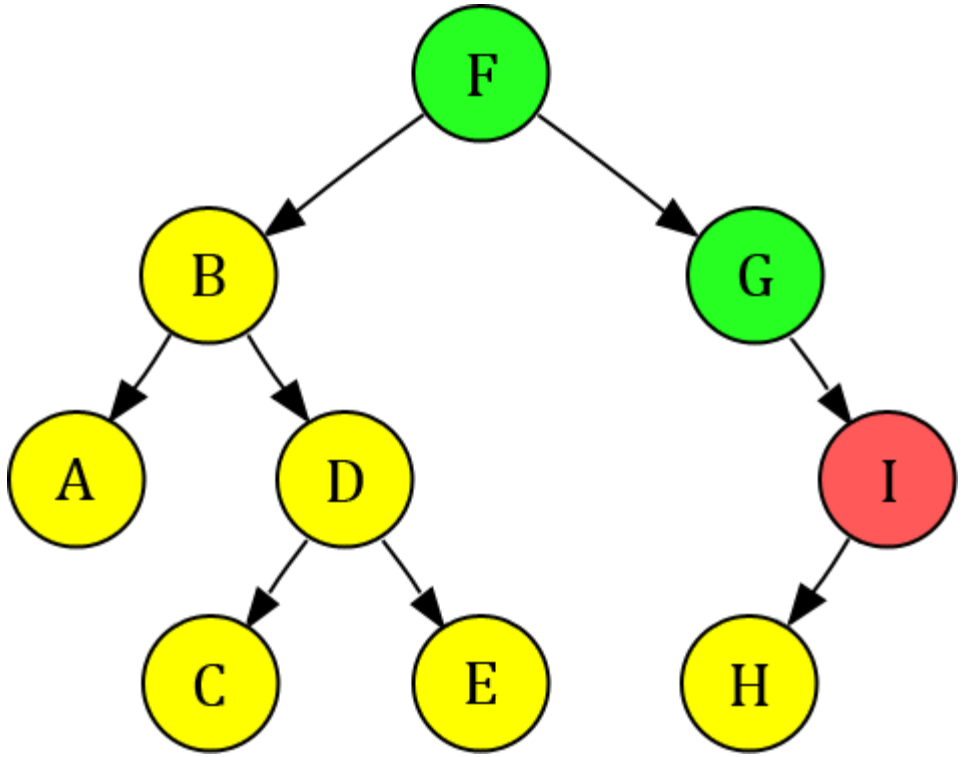
Current Node = I

Stack:

F	G	I							

Visited:

F	B	A	D	C	E	G	I	H	



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 01)

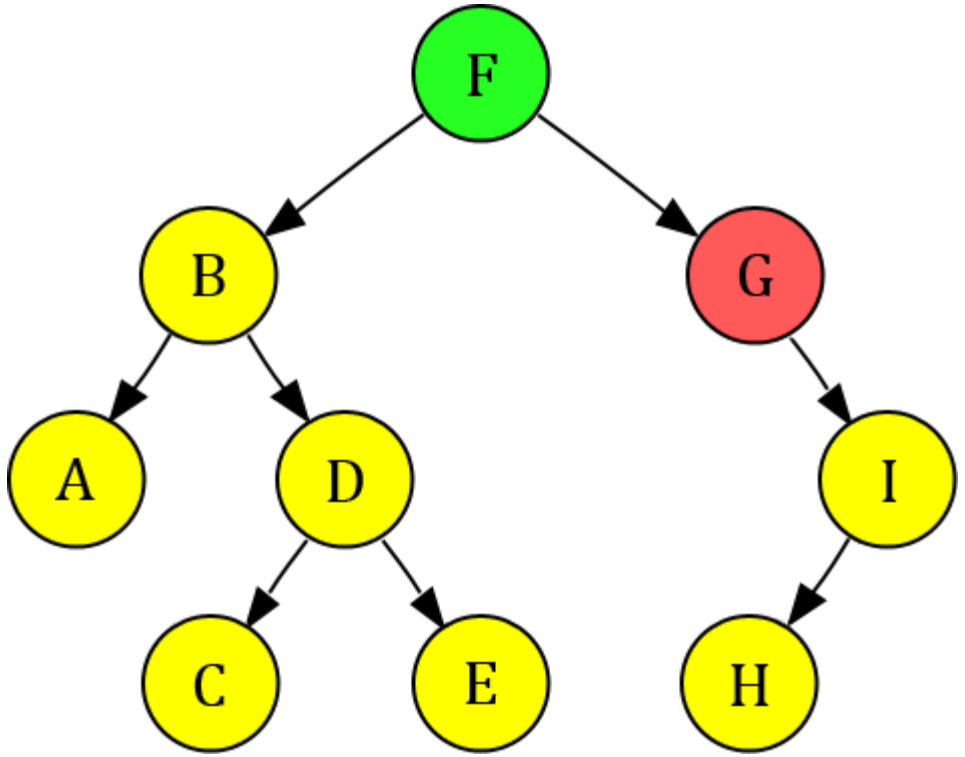
Current Node = G

Stack:

F	G								

Visited:

F	B	A	D	C	E	G	I	H	



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 01)

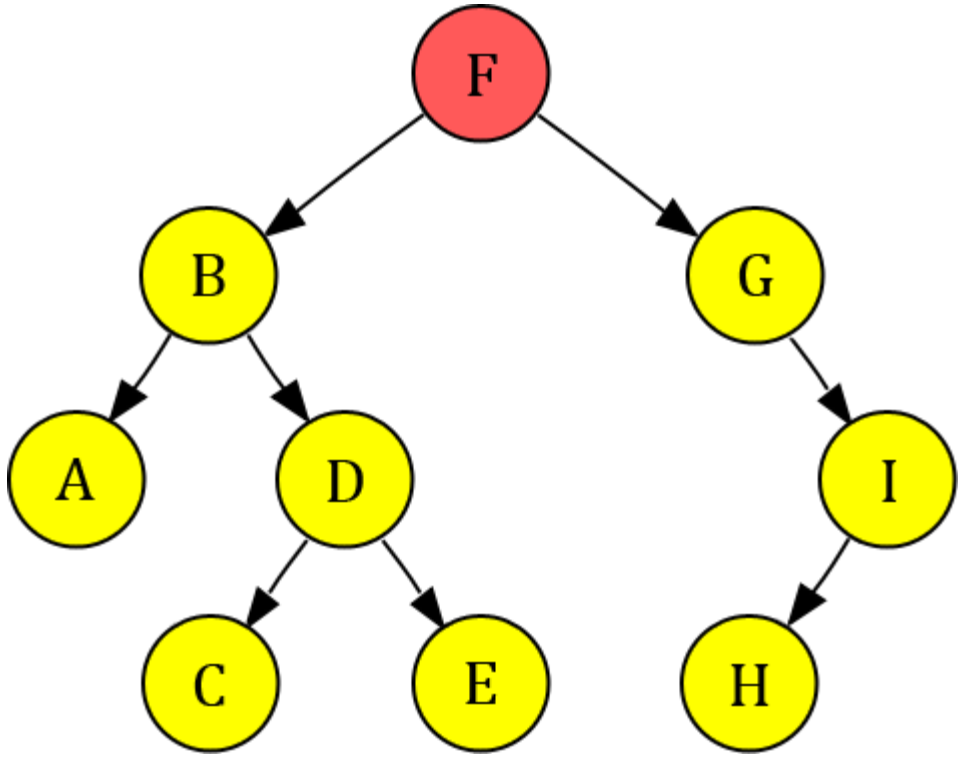
Current Node = F

Stack:

F									

Visited:

F	B	A	D	C	E	G	I	H	



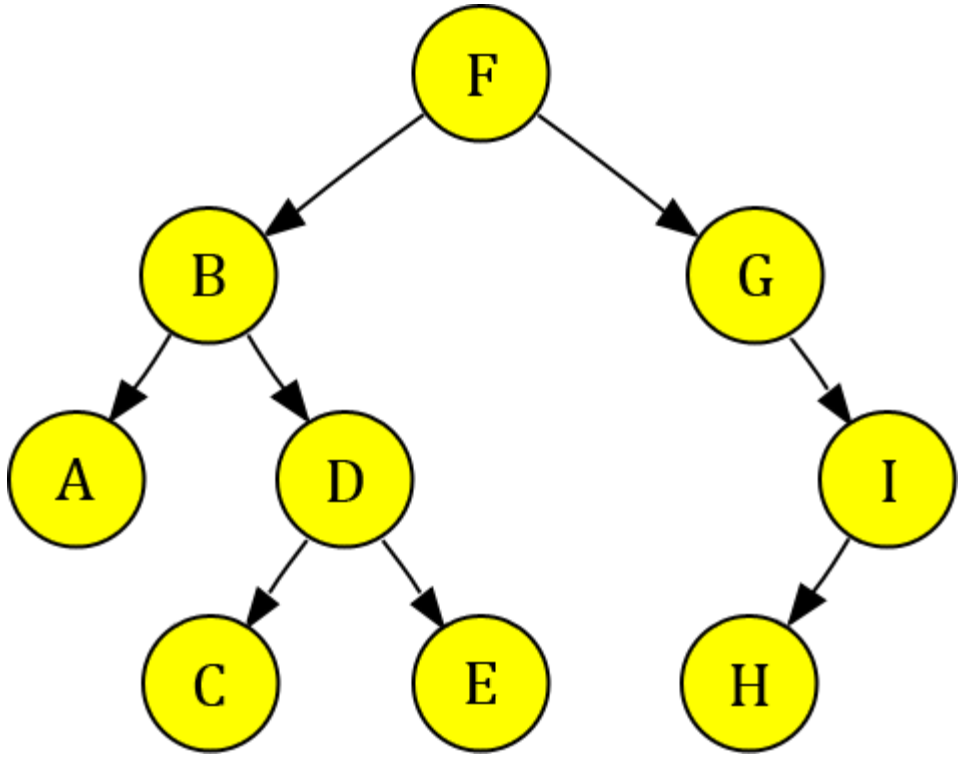
Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 01)

Stack:

Visited:

F	**B**	**A**	**D**	**C**	**E**	**G**	**I**	**H**	



DFS Traversal: **F B A D C E G I H**

Depth-First Search (Problem 02)

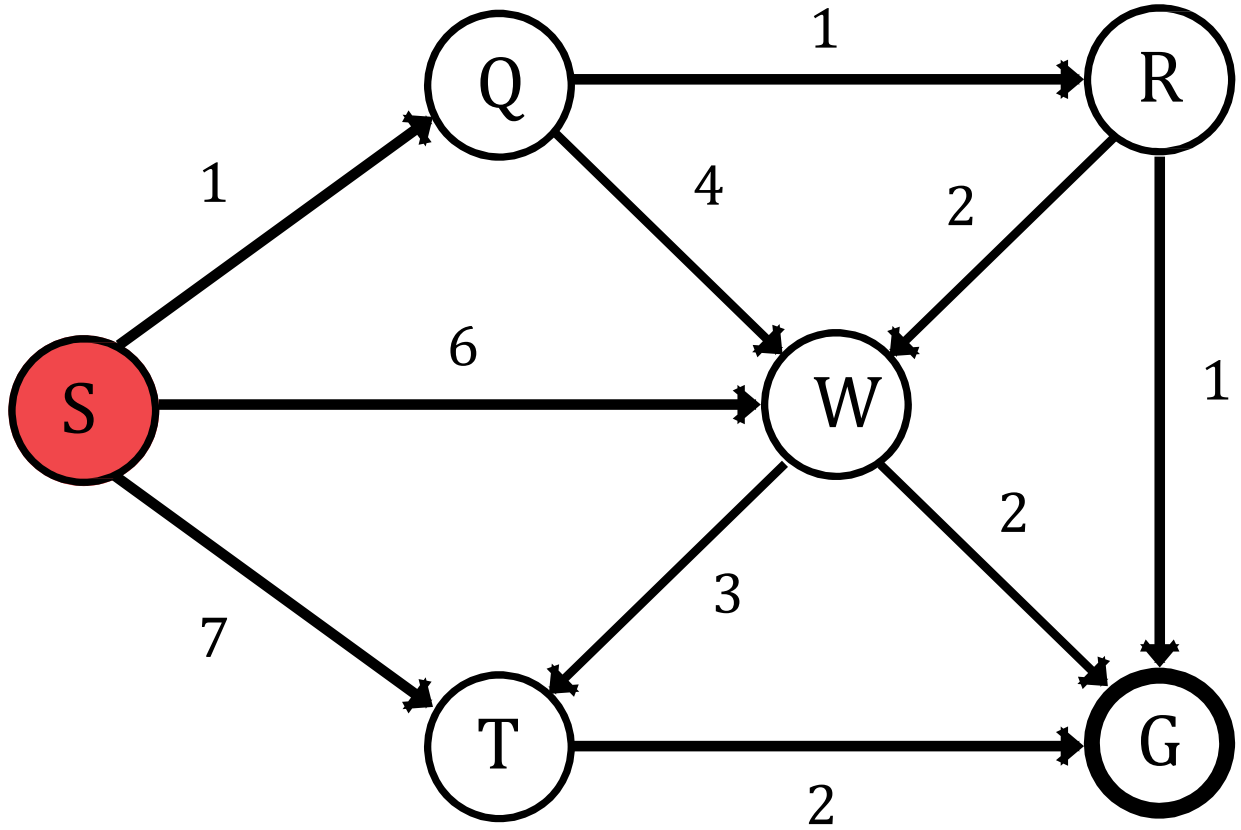
Current Node = S

Stack:

S					

Visited:

S					



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 02)

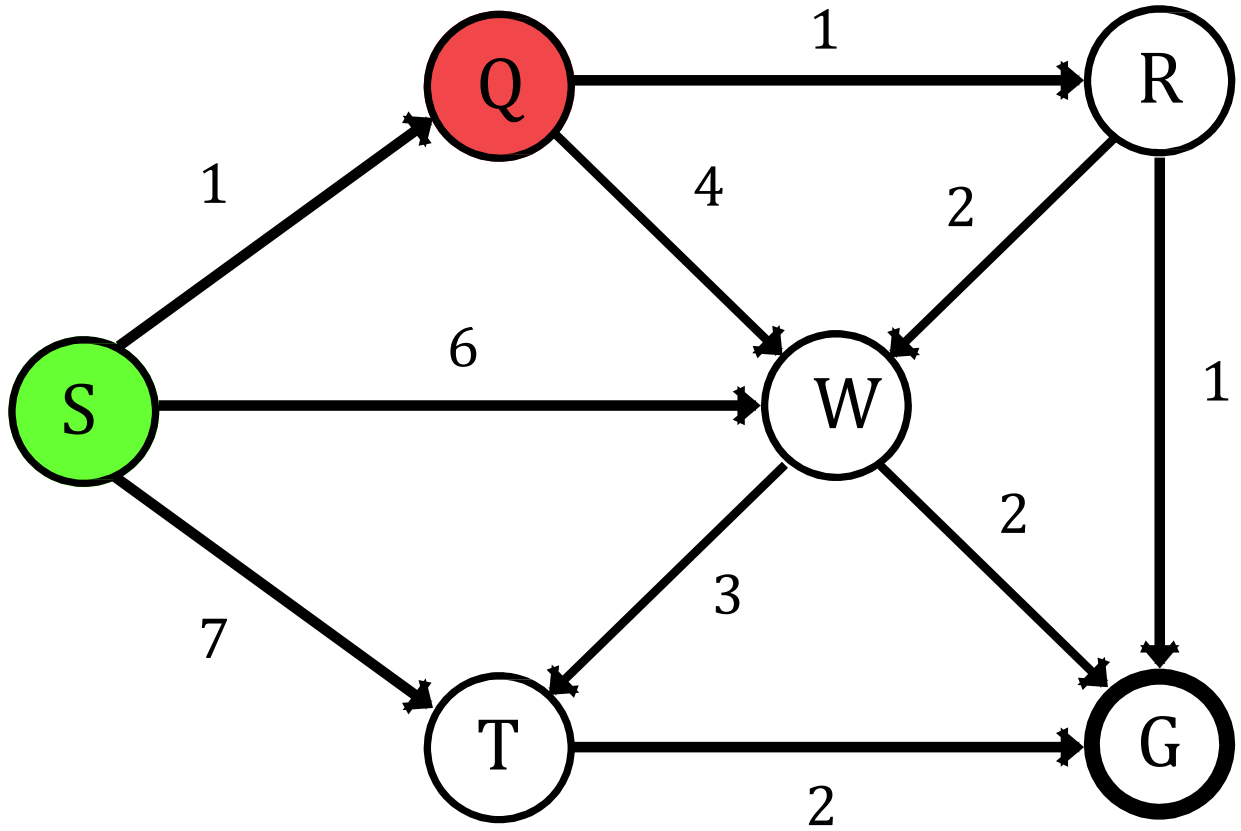
Current Node = Q

Stack:

S	**Q**				

Visited:

S	**Q**				



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 02)

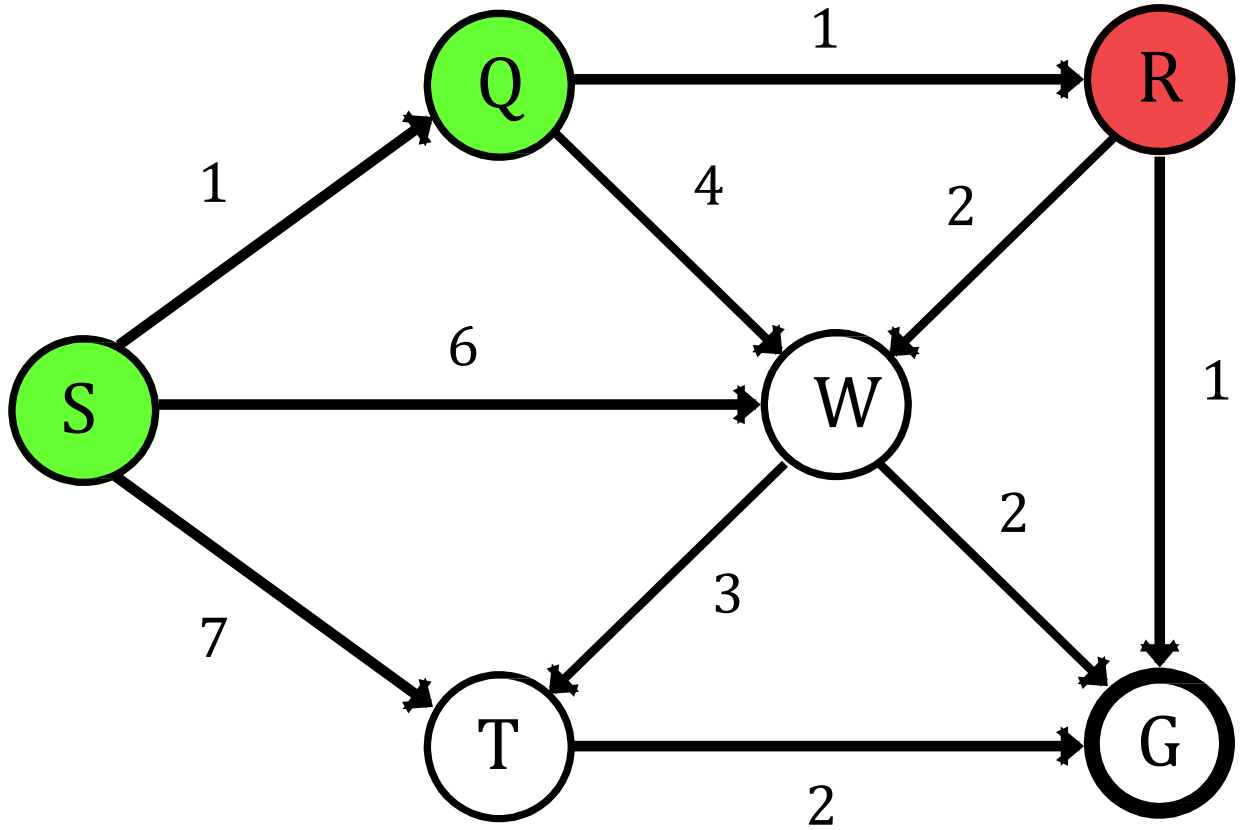
Current Node = R

Stack:

S	**Q**	**R**			

Visited:

S	**Q**	**R**			



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 02)

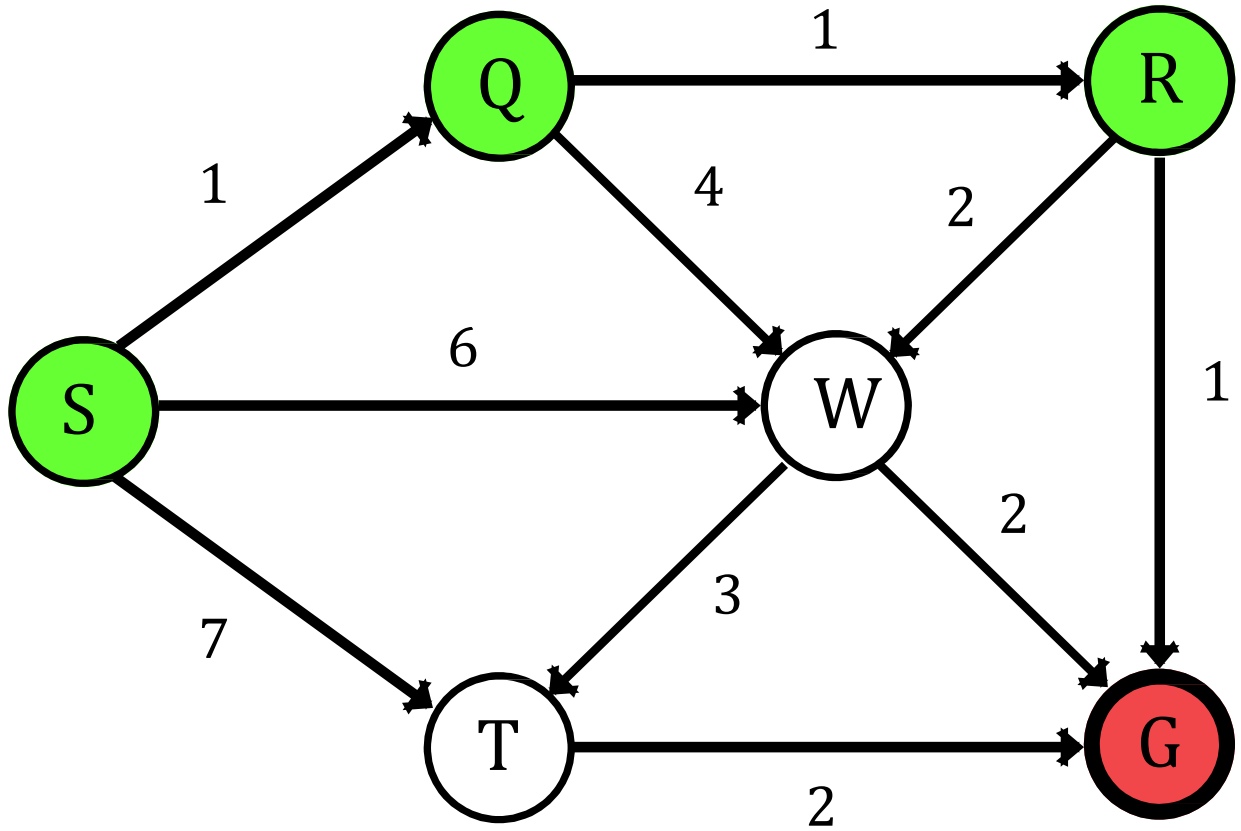
Current Node = G

Stack:

S	Q	R	G		

Visited:

S	Q	R	G		



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 02)

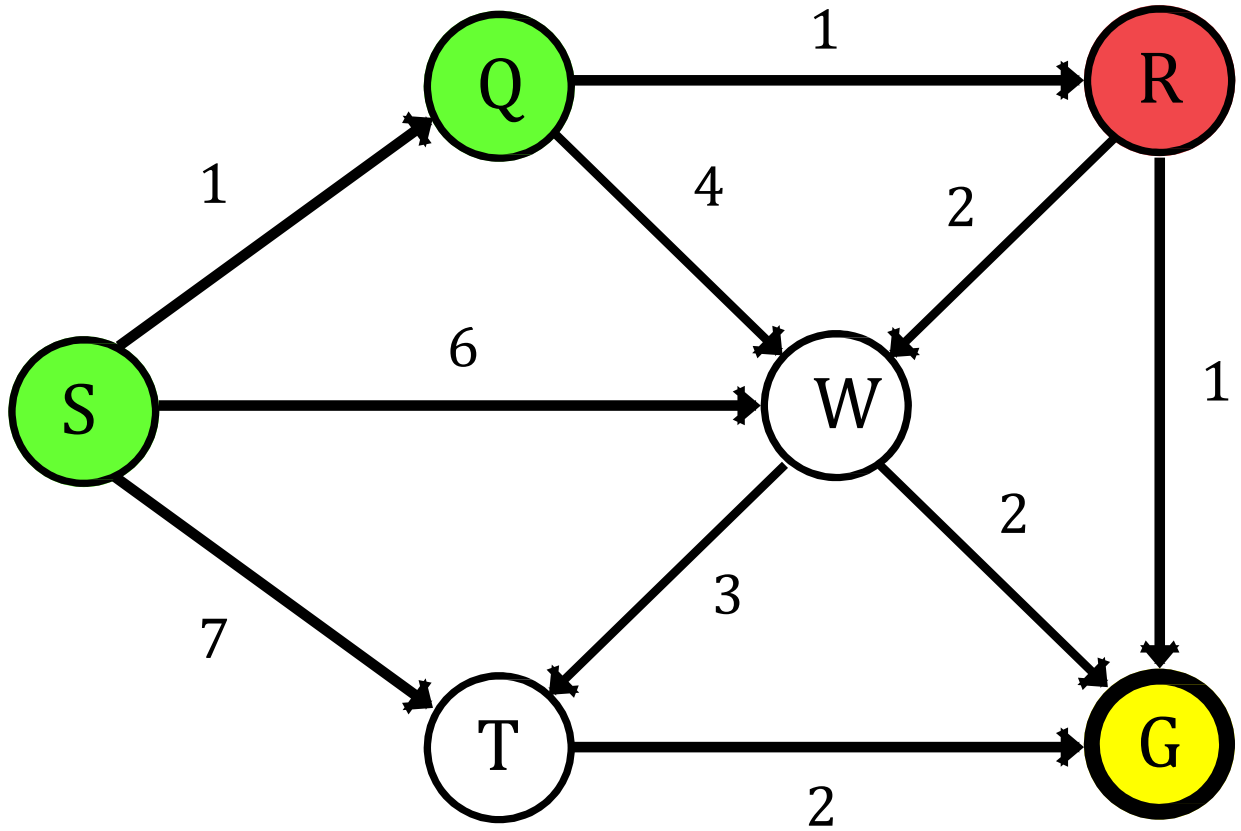
Current Node = R

Stack:

S	**Q**	**R**			

Visited:

S	**Q**	**R**	**G**		



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 02)

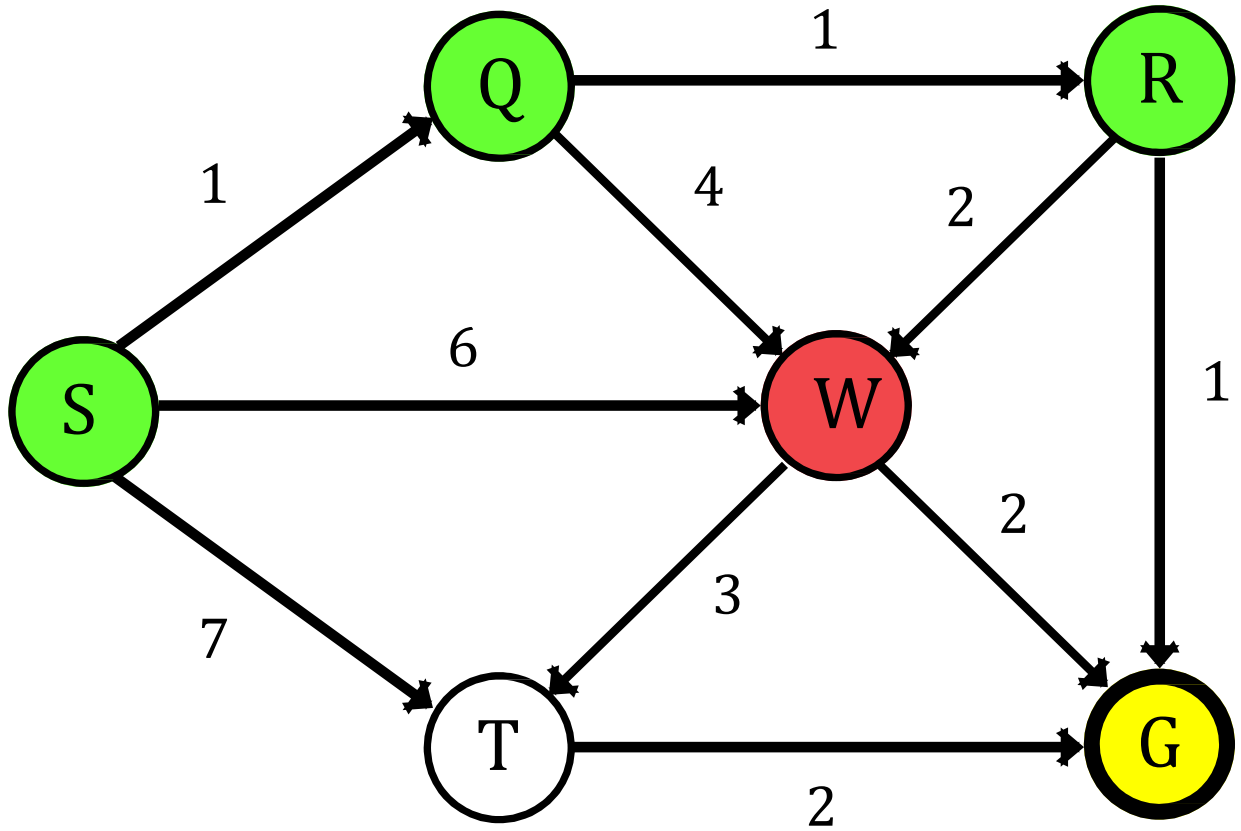
Current Node = W

Stack:

S	Q	R	W		

Visited:

S	Q	R	G	W	



Current Node Opened Node Closed Node

Depth-First Search (Problem 02)

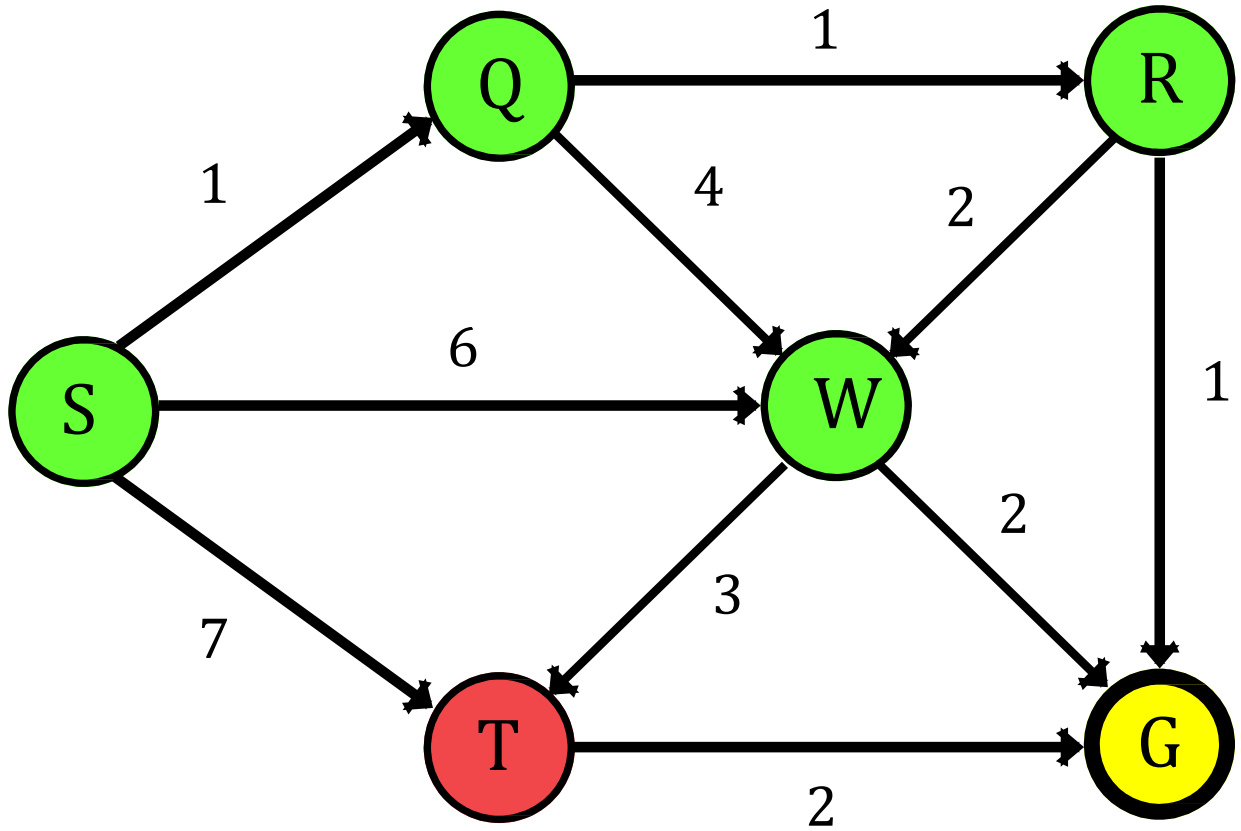
Current Node = T

Stack:

S	Q	R	W	T	

Visited:

S	Q	R	G	W	T



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 02)

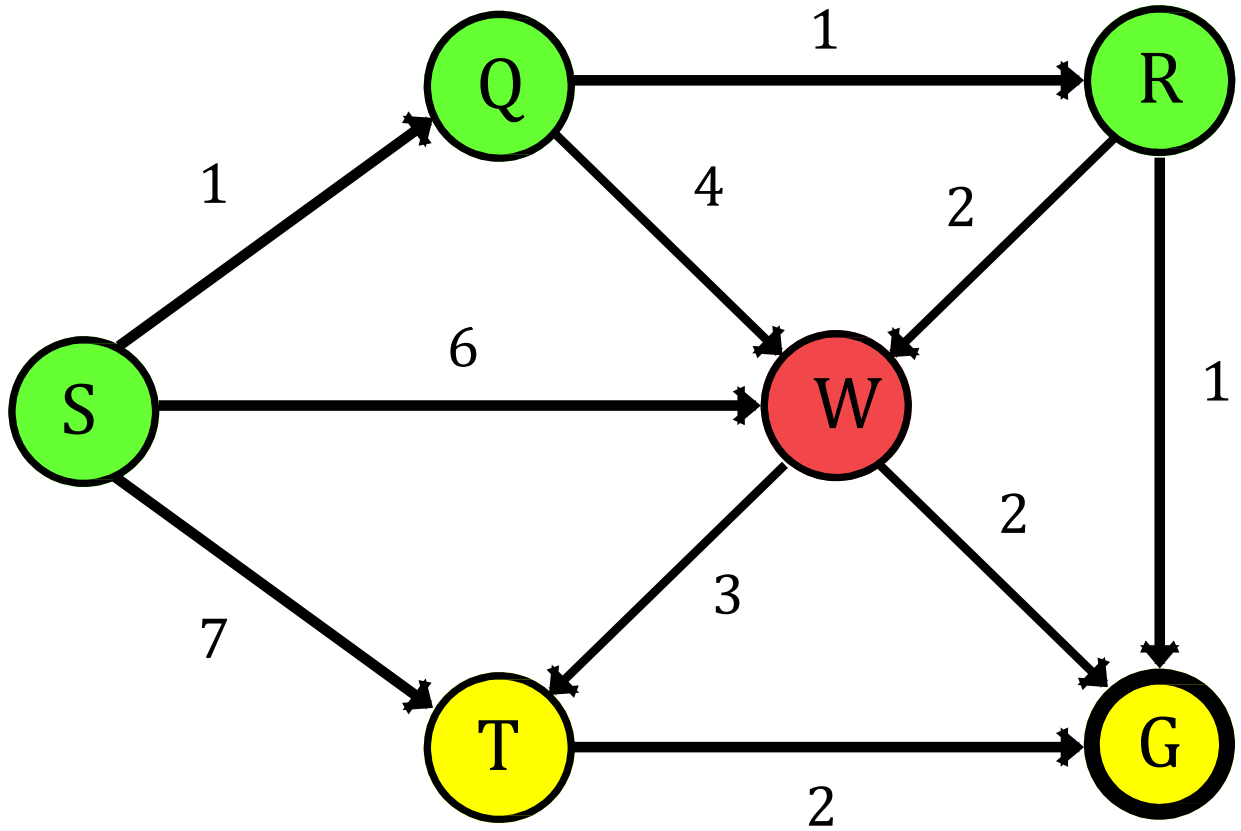
Current Node = W

Stack:

S	Q	R	W		

Visited:

S	Q	R	G	W	T



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 02)

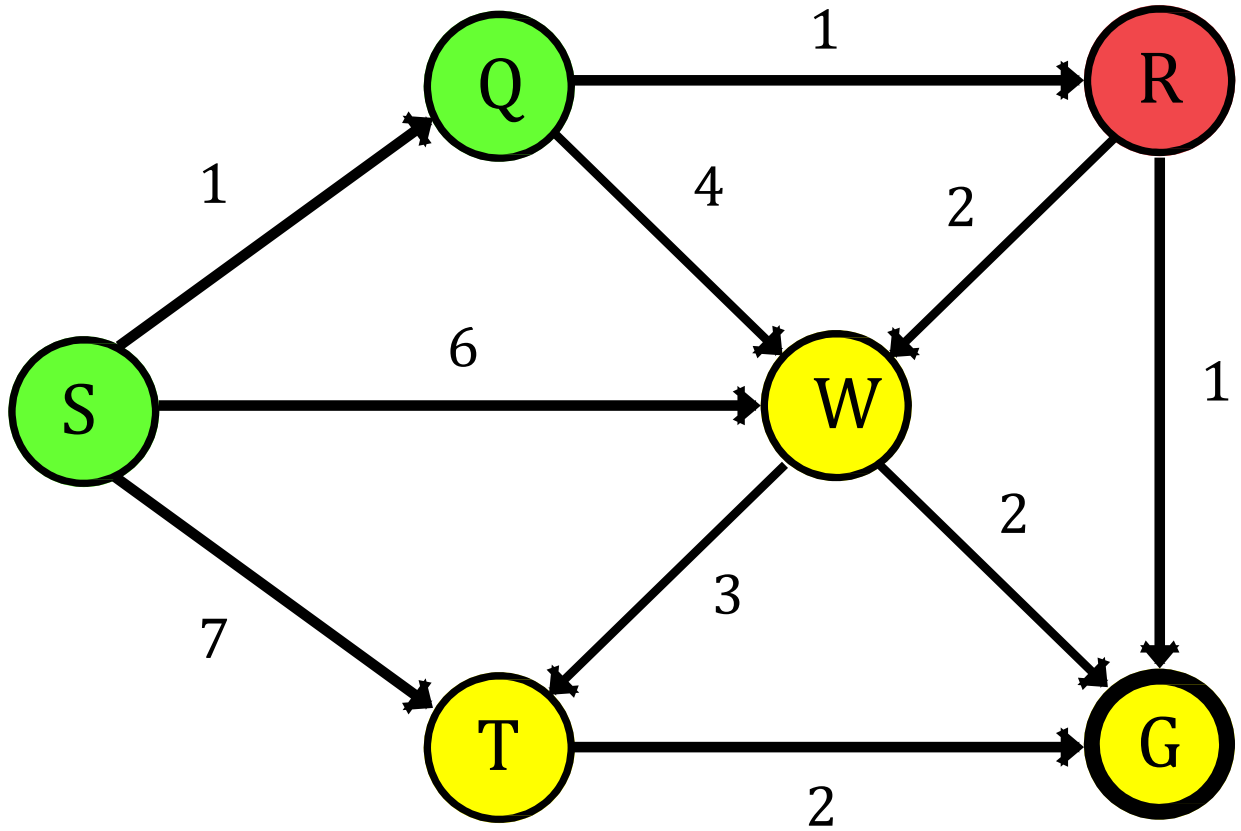
Current Node = R

Stack:

S	**Q**	**R**			

Visited:

S	**Q**	**R**	**G**	**W**	**T**



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 02)

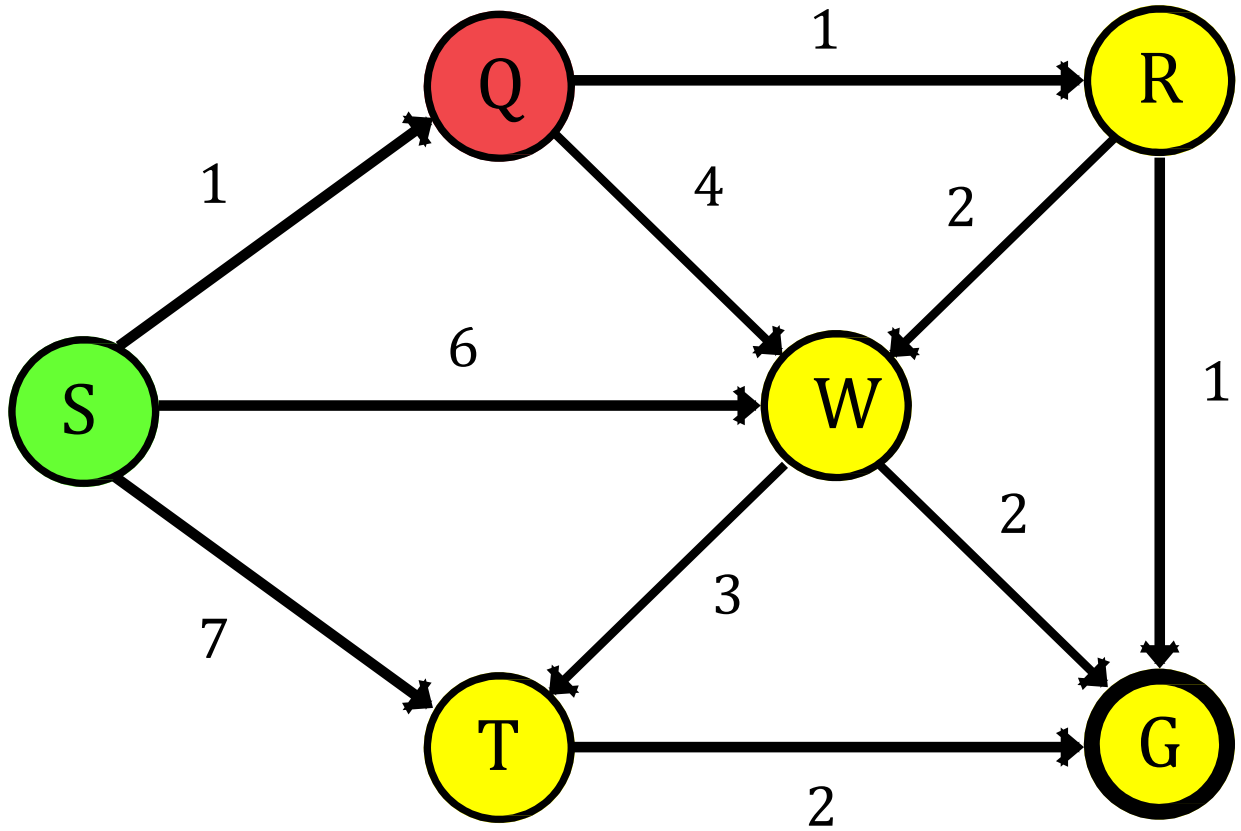
Current Node = Q

Stack:

S	**Q**				

Visited:

S	**Q**	**R**	**G**	**W**	**T**



Current Node **Opened Node** **Closed Node**

Depth-First Search (Problem 02)

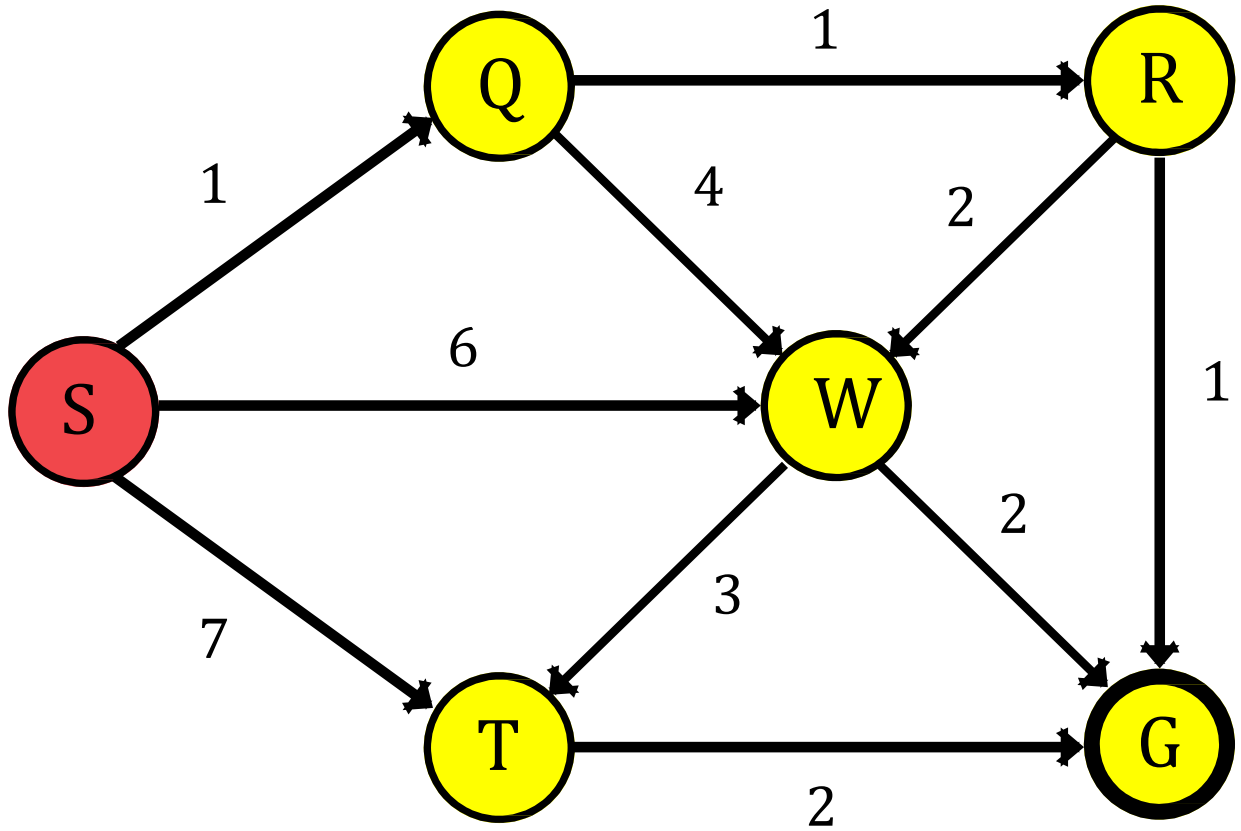
Current Node = S

Stack:

S					

Visited:

S	**Q**	**R**	**G**	**W**	**T**



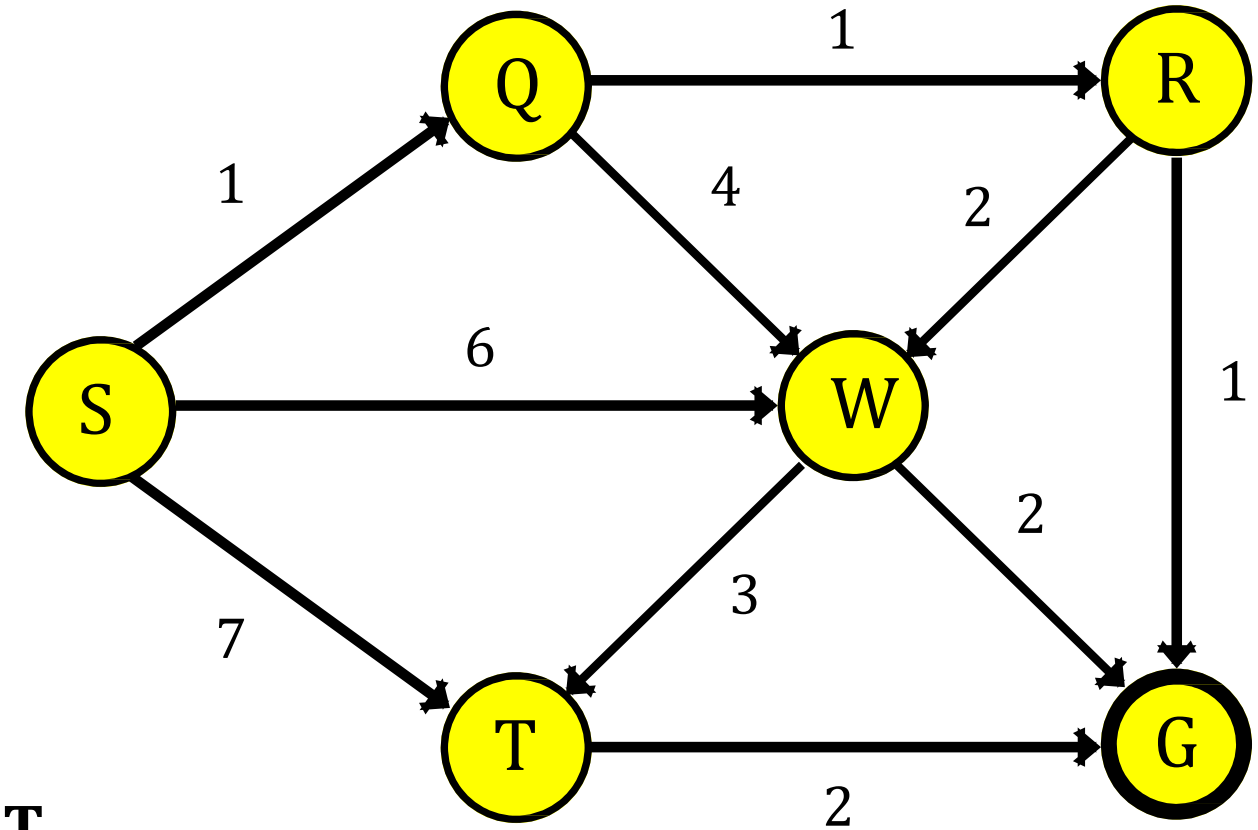
Current Node Opened Node Closed Node

Depth-First Search (Problem 02)

Stack:

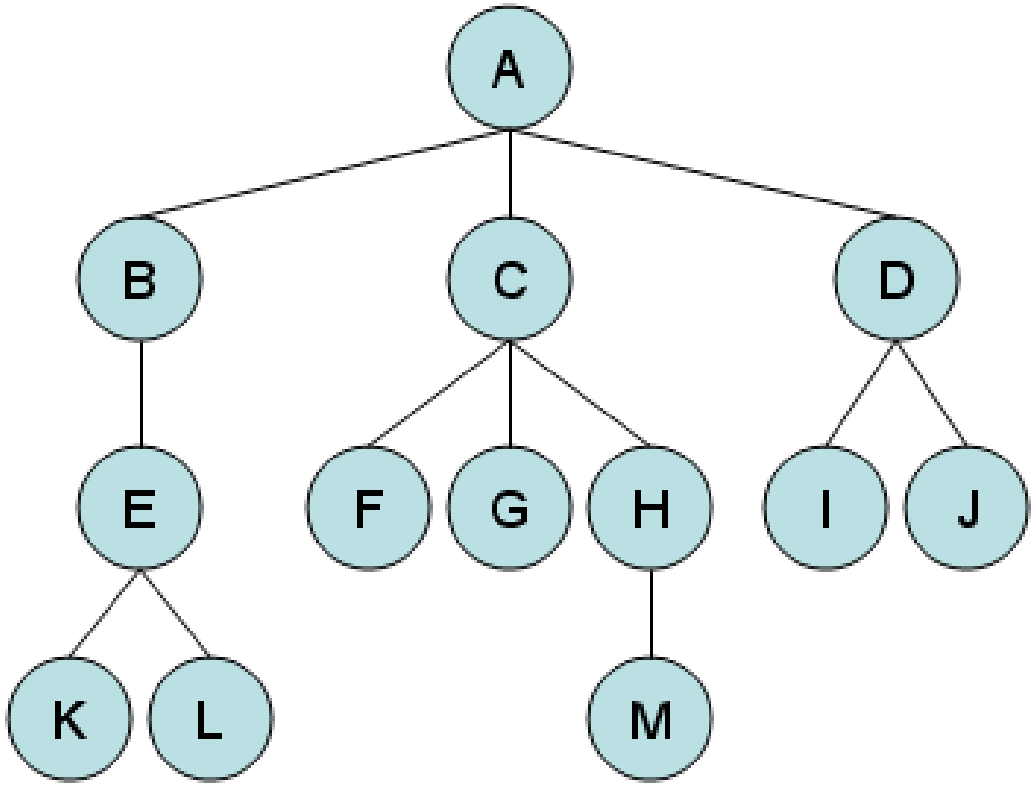
Visited:

S	**Q**	**R**	**G**	**W**	**T**



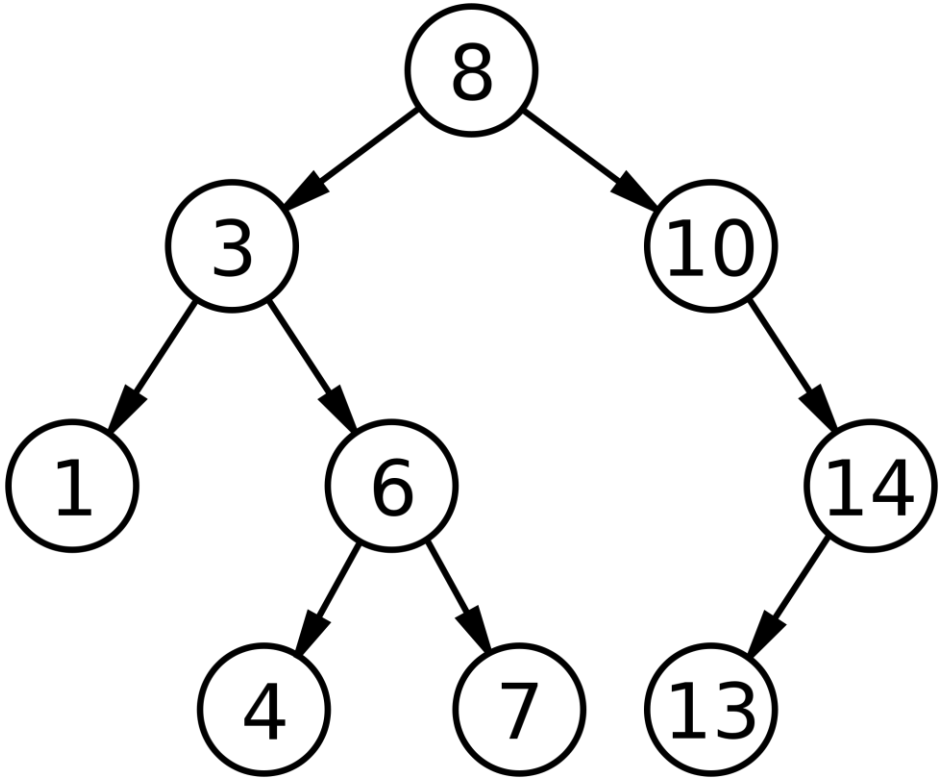
DFS Traversal: **S Q R G W T**

Depth-First Search (Problem 03)



DFS Traversal: **A B E K L C F G H M D I J**

Depth-First Search (Problem 04)



DFS Traversal: **8 3 1 6 4 7 10 14 13**

Depth-First Search

- Since we don't expand all nodes at a level, space complexity is modest. For branching factor b and depth d, we require *bd* number of nodes to be stored in memory. This is much better than b^d.
- In some cases, DFS can be faster than BFS. However, the worse case is still $O(b^d)$.
- If you have deep search trees (or infinite – which is quite possible), DFS may end up running off to infinity and may not be able to recover.
- Thus DFS is neither optimal nor complete.

Properties of Depth-First Search

Property	Description
Completeness	DFS is incomplete, as it may fail to find a solution if the search space contains cycles, causing it to get stuck in an infinite loop.
Optimality	DFS is suboptimal and does not guarantee finding the shortest path to a solution.
Time Complexity	The time complexity of DFS is $O(b^m)$, where b is the branching factor and m is the maximum depth of the search tree.
Space Complexity	DFS has a space complexity of $O(bm)$, where b is the branching factor and m is the maximum depth of the search tree.
Frontier	DFS maintains a stack or recursion as its frontier, exploring nodes deeply before backtracking.

Depth Limited Search

- Depth-first search will not find a goal if it searches down a path that has infinite length. So, in general, depth-first search is not guaranteed to find a solution, so it is not complete.
- This problem is eliminated by limiting the depth of the search to some value l. However, this introduces another way of preventing depth-first search from finding the goal: if the goal is deeper than l it will not be found.
- Regular depth-first search is a special case, for which $l=\infty$.

Depth Limited Search

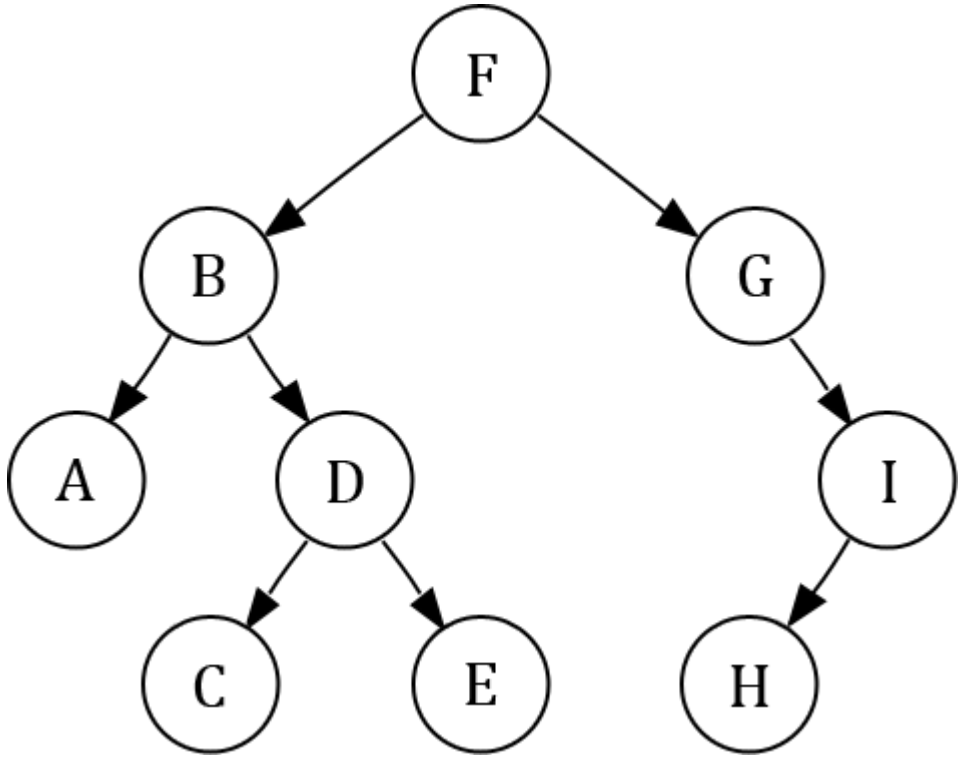
- Depth limited search is complete but not optimal.
- If we choose a depth limit that is too small, then depth limited search is not even complete.
- The time and space complexity of depth limited search is similar to depth first search.

Depth Limited Search (Problem 01)

Visited	Level		
	0	1	2
F	F		
F B	F	B	
F B A	F	B	A
F B A	F	B	
F B A D	F	B	D
F B A D	F	B	
F B A D	F		
F B A D G	F	G	
F B A D G I	F	G	I
F B A D G I	F	G	
F B A D G I	F		
F B A D G I			

Depth Level Limit:

L = 2



Depth Limited Search (Problem 01)

Depth Level Limit:

L = 2

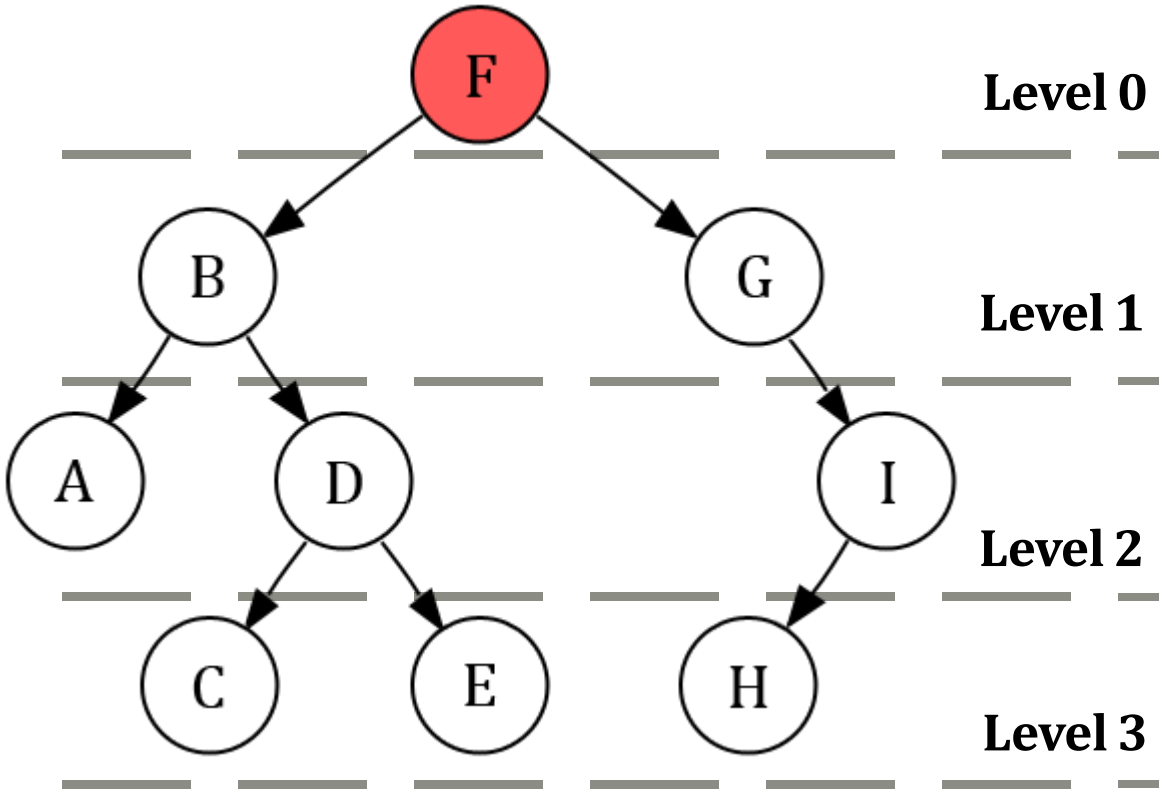
Stack:

F									

Visited:

F									

Current Node = F



Depth Limited Search (Problem 01)

Depth Level Limit:

L = 2

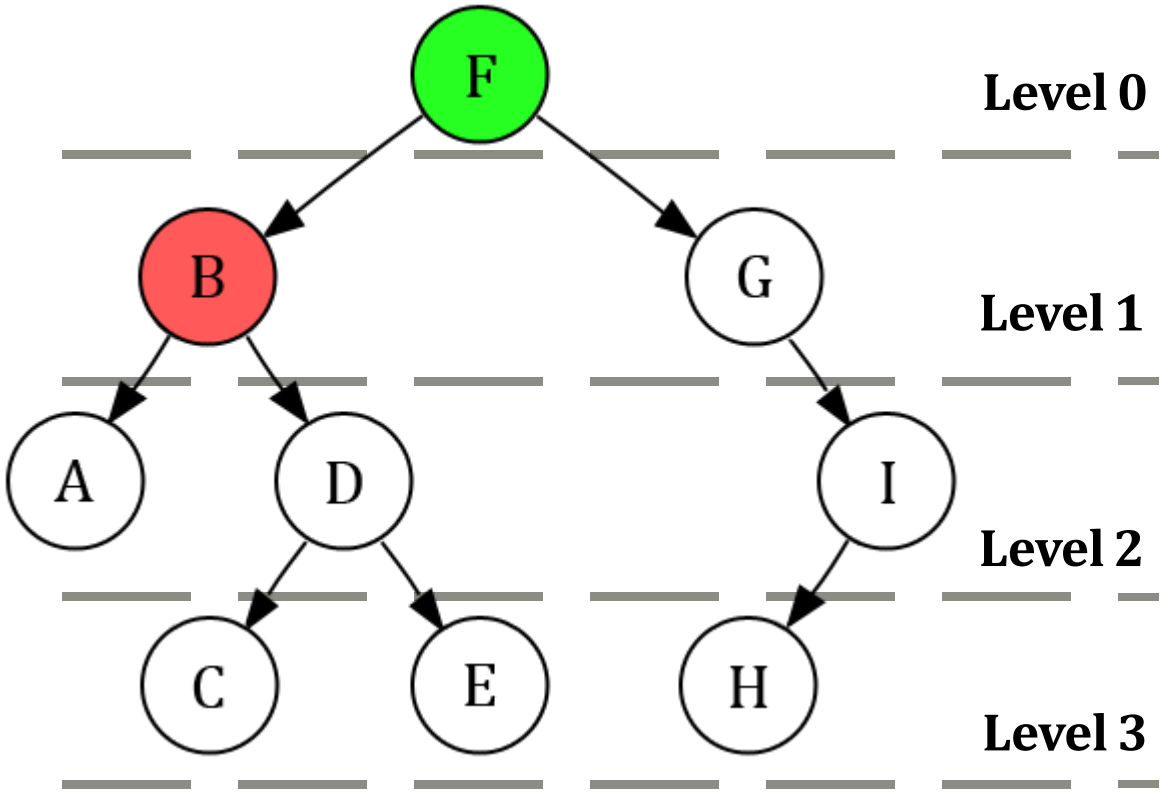
Stack:

F	B								

Visited:

F	B								

Current Node = B



Depth Limited Search (Problem 01)

Depth Level Limit:

L = 2

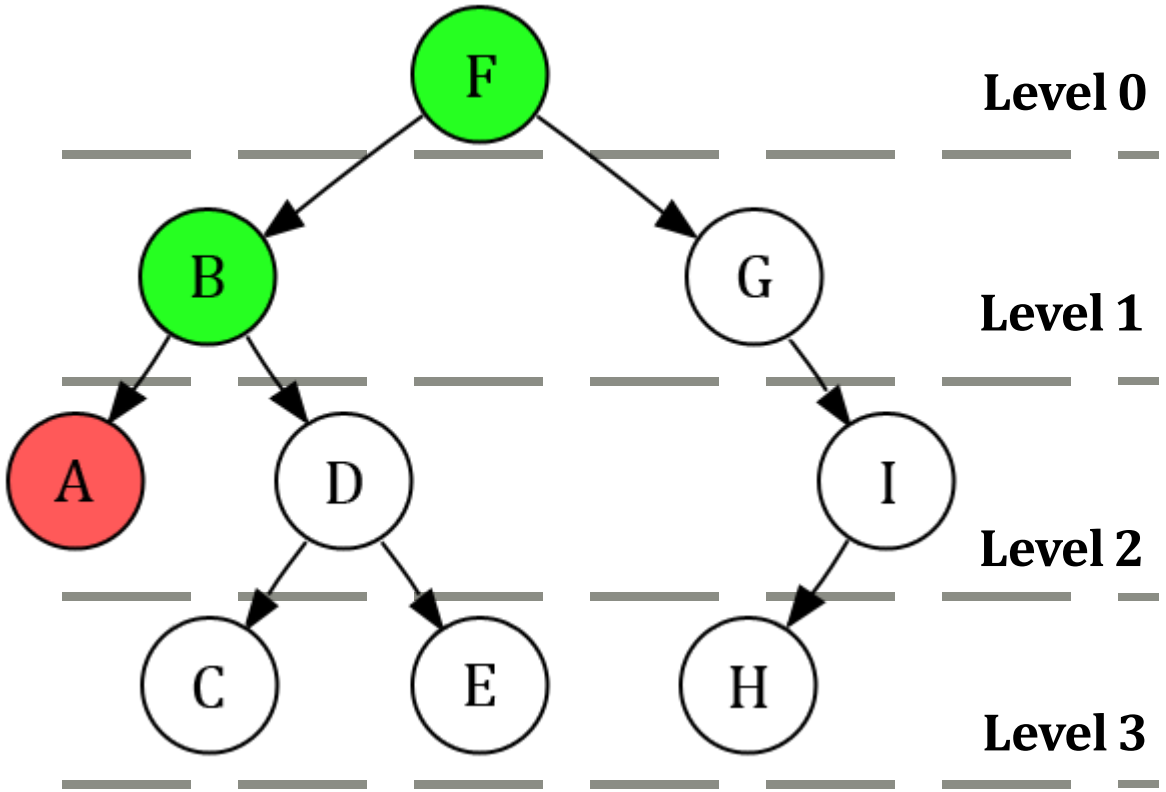
Stack:

F	B	A							

Visited:

F	B	A							

Current Node = A



Depth Limited Search (Problem 01)

Depth Level Limit:

L = 2

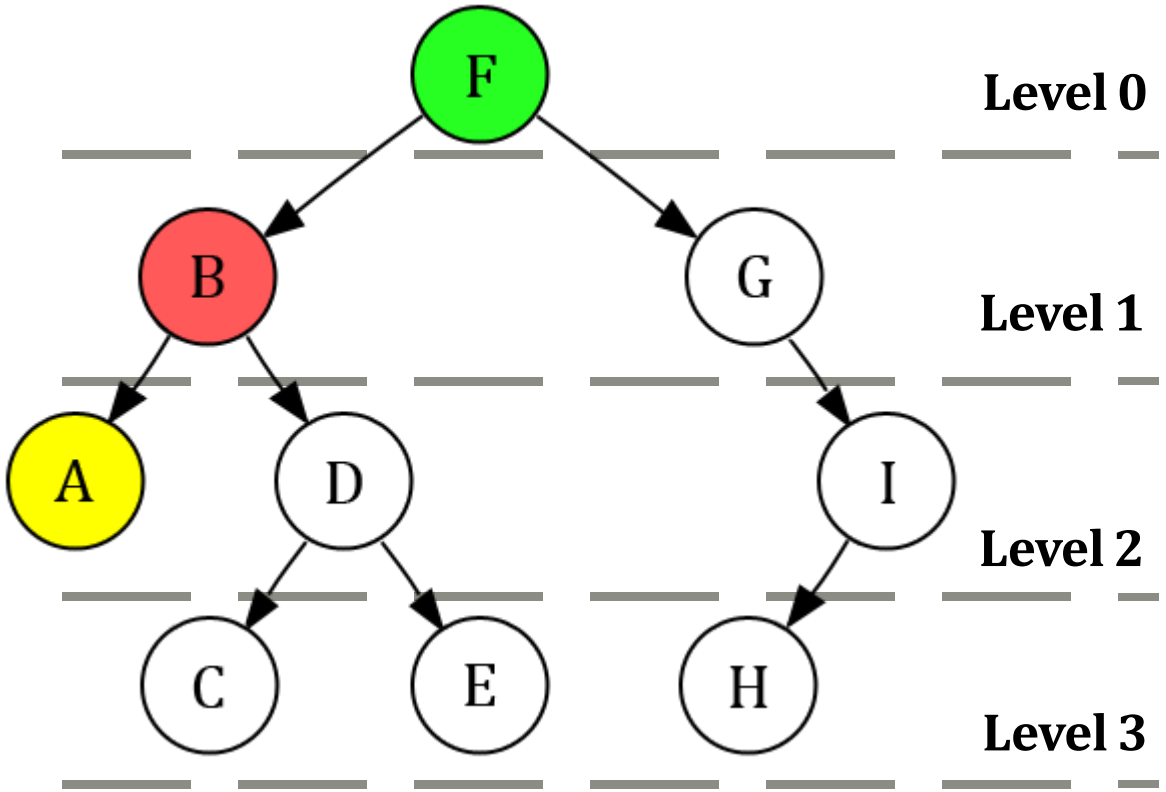
Stack:

F	B								

Visited:

F	B	A							

Current Node = B



Depth Limited Search (Problem 01)

Depth Level Limit:

L = 2

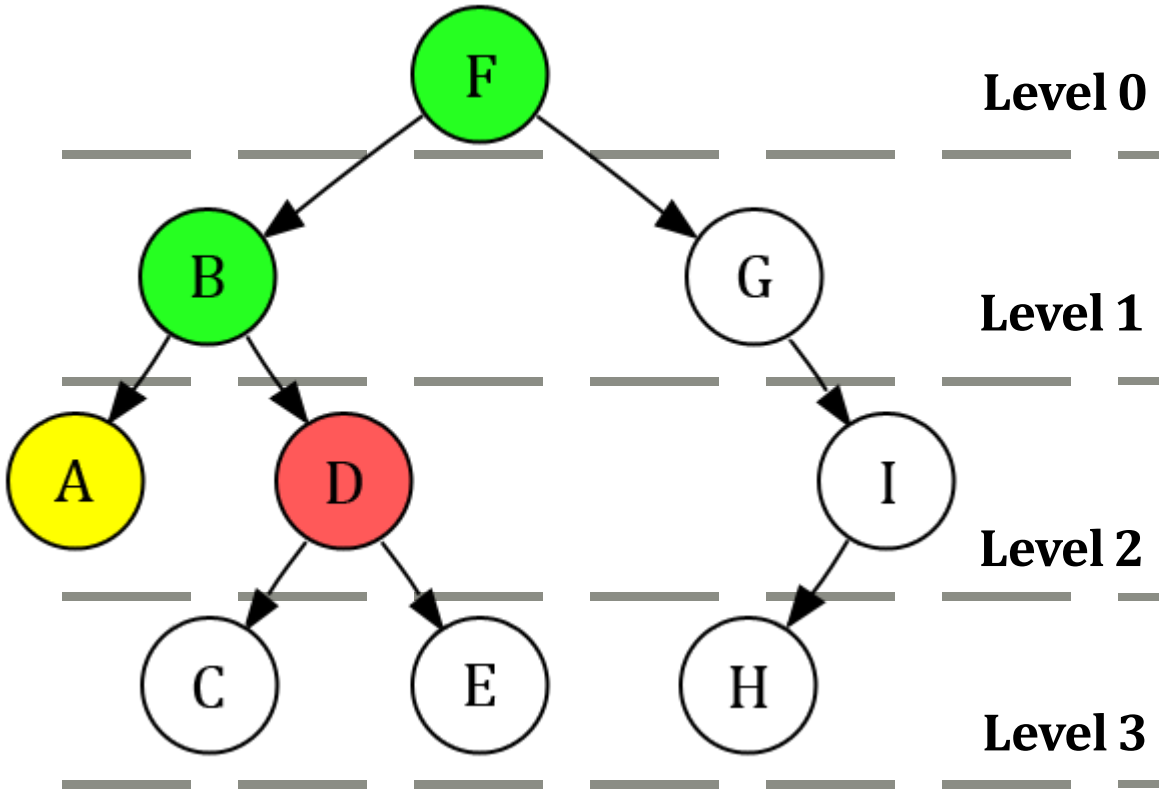
Stack:

F	B	D							

Visited:

F	B	A	D						

Current Node = D



Depth Limited Search (Problem 01)

Depth Level Limit:

L = 2

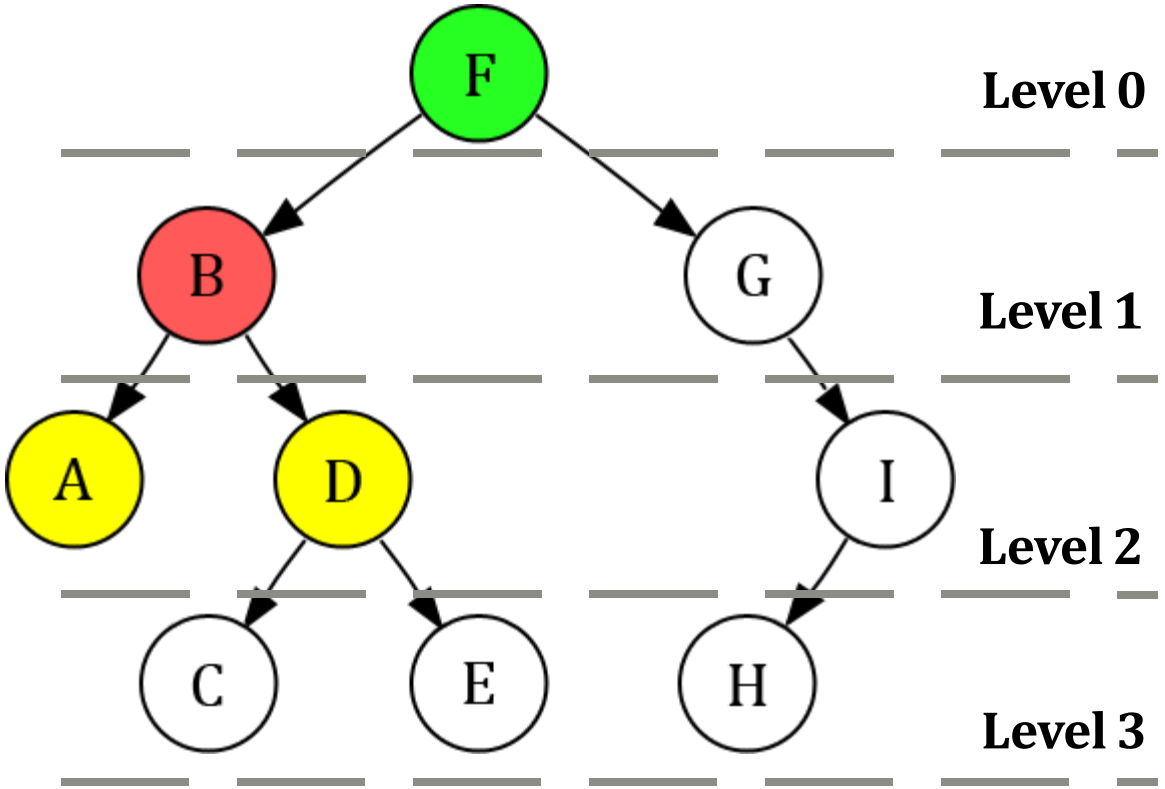
Stack:

F	B								

Visited:

F	B	A	D						

Current Node = B



Depth Limited Search (Problem 01)

Depth Level Limit:

L = 2

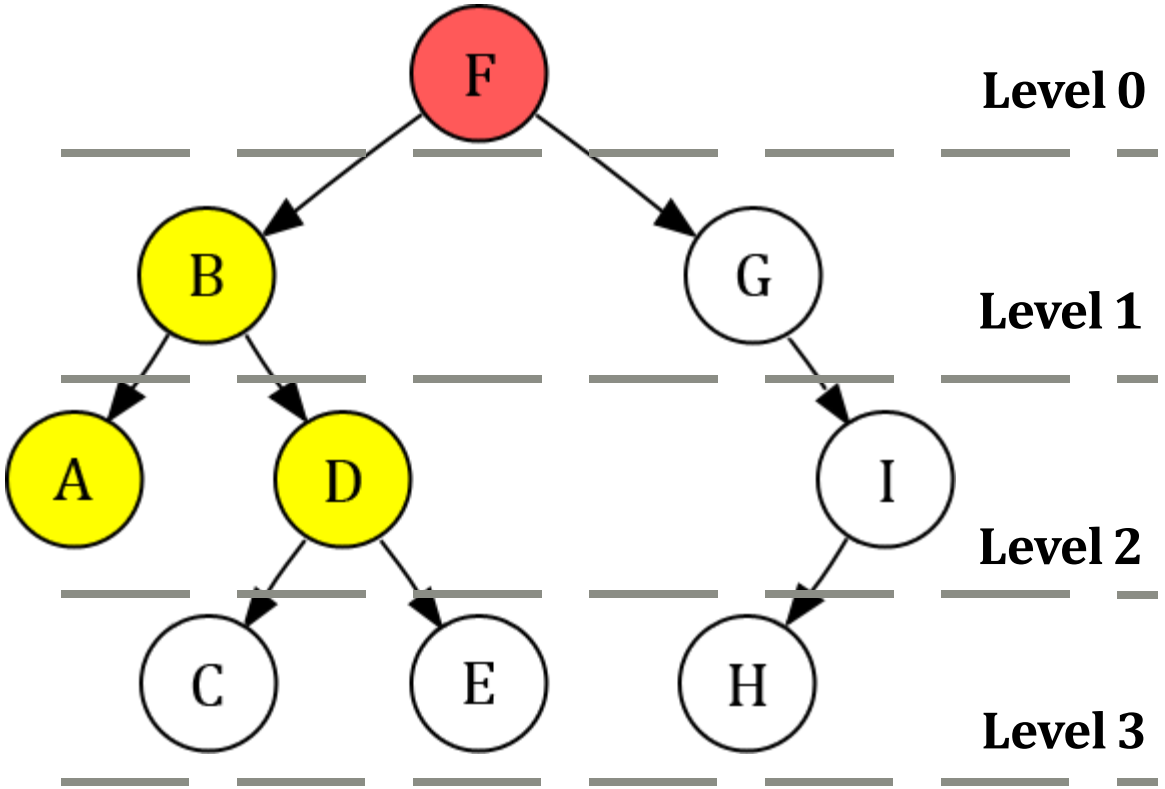
Stack:

F									

Visited:

F	B	A	D						

Current Node = F



Depth Limited Search (Problem 01)

Depth Level Limit:

L = 2

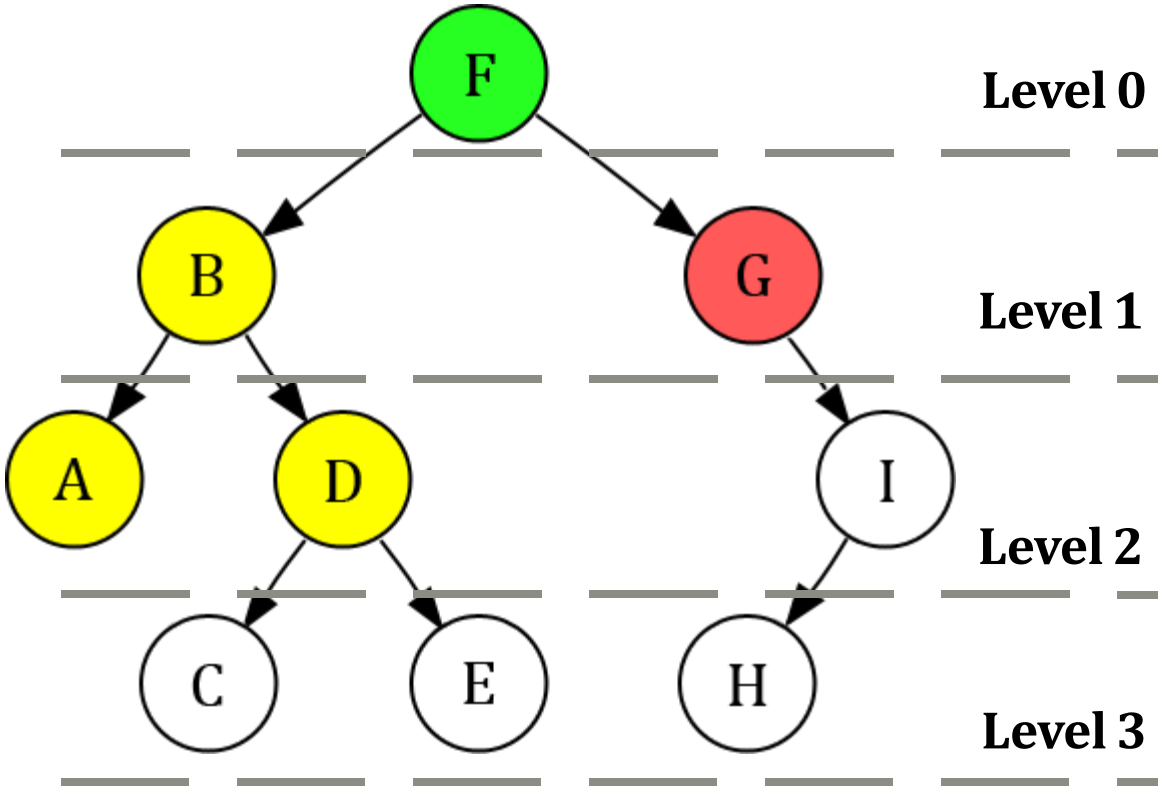
Stack:

F	G								

Visited:

F	B	A	D	G					

Current Node = G



Depth Limited Search (Problem 01)

Depth Level Limit:

L = 2

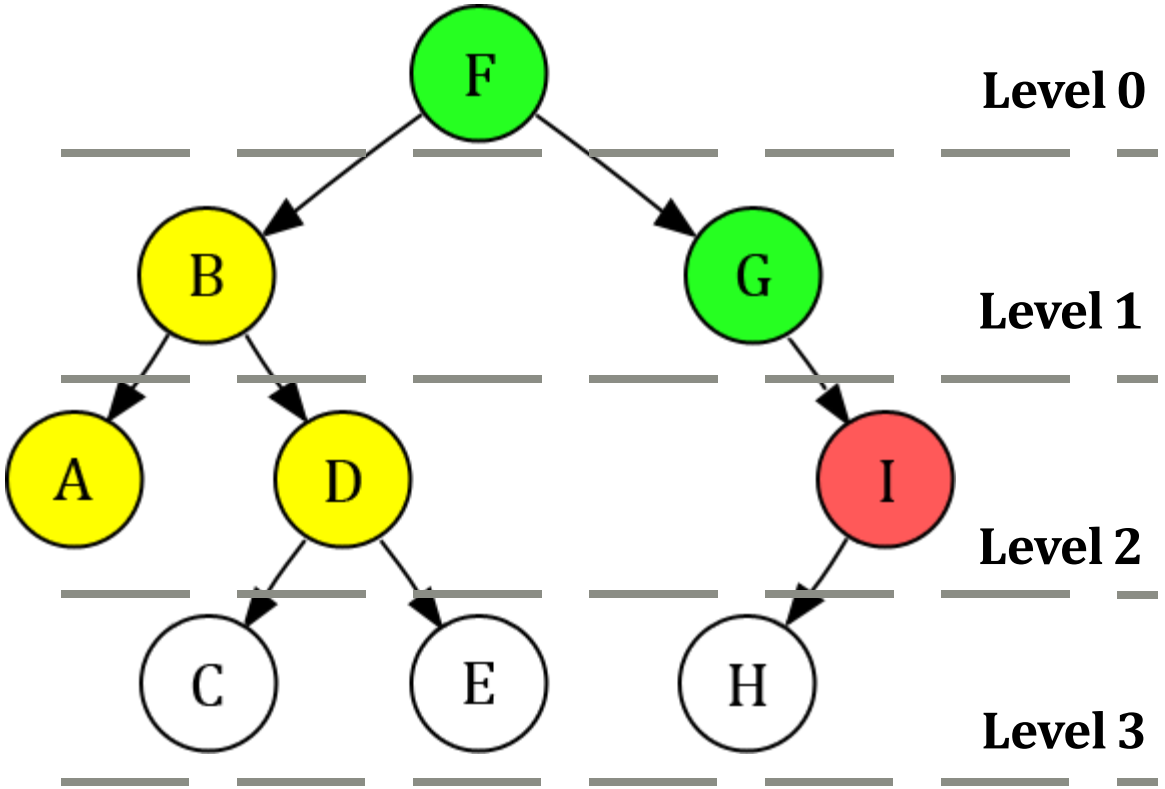
Stack:

F	G	I							

Visited:

F	B	A	D	G	I				

Current Node = I



Depth Limited Search (Problem 01)

Depth Level Limit:

L = 2

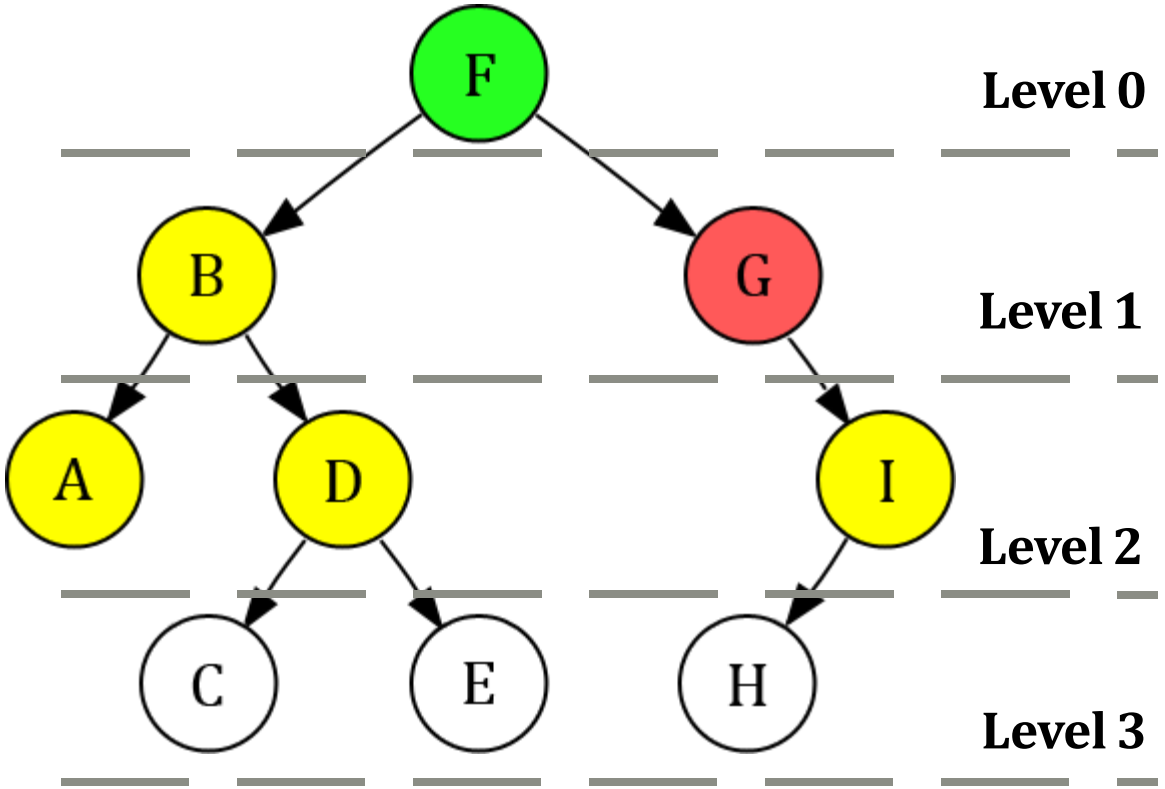
Stack:

F	G								

Visited:

F	B	A	D	G	I				

Current Node = G



Depth Limited Search (Problem 01)

Depth Level Limit:

L = 2

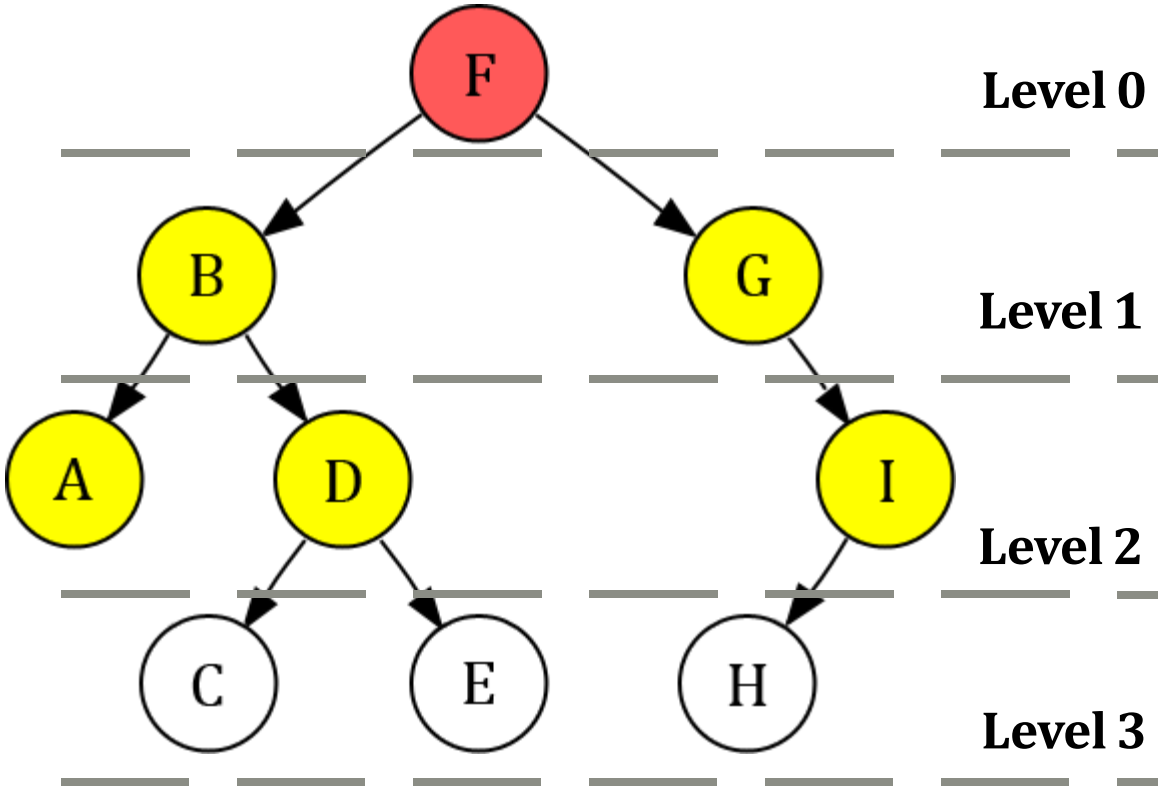
Stack:

F									

Visited:

F	B	A	D	G	I				

Current Node = F



Depth Limited Search (Problem 01)

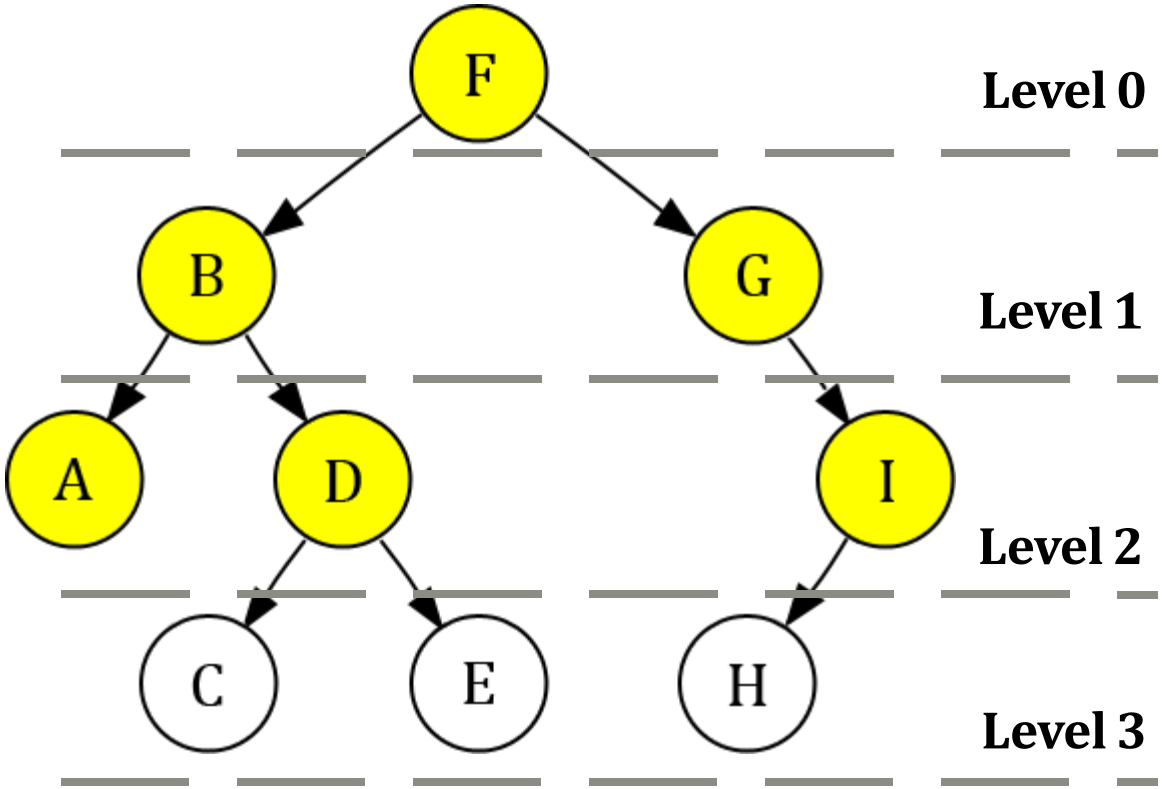
Depth Level Limit:

L = 2

Stack:

Visited:

F	B	A	D	G	I				



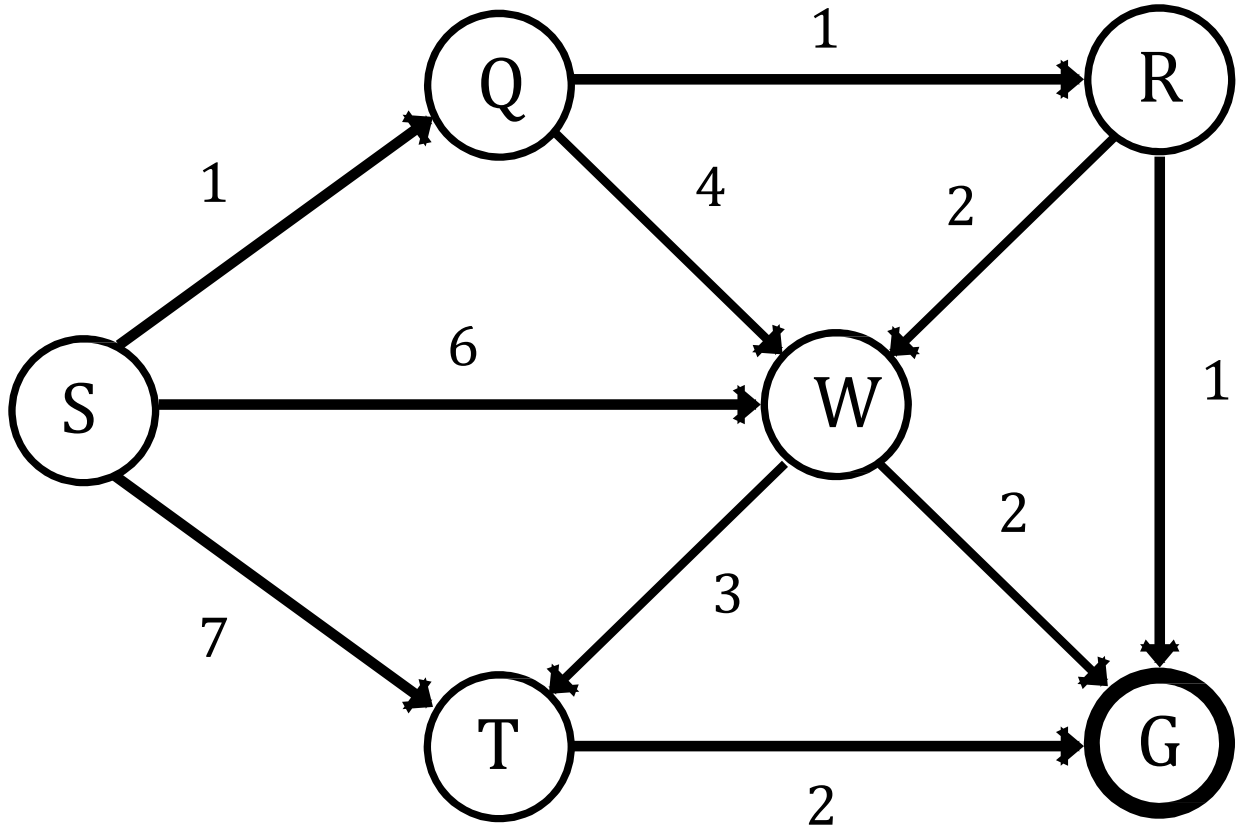
DFS Limited Traversal: **F B A D G I**

Depth Limited Search (Problem 02)

Visited	Level		
	0	1	2
S	S		
S Q	S	Q	
S Q R	S	Q	R
S Q R	S	Q	
S Q R W	S	Q	W
S Q R W	S	Q	
S Q R W	S		
S Q R W T	S	T	
S Q R W T G	S	T	G
S Q R W T G	S	T	
S Q R W T G	S		
S Q R W T G			

Depth Level Limit:

L = 2



Depth Limited Search (Problem 02)

Depth Level Limit:

L = 2

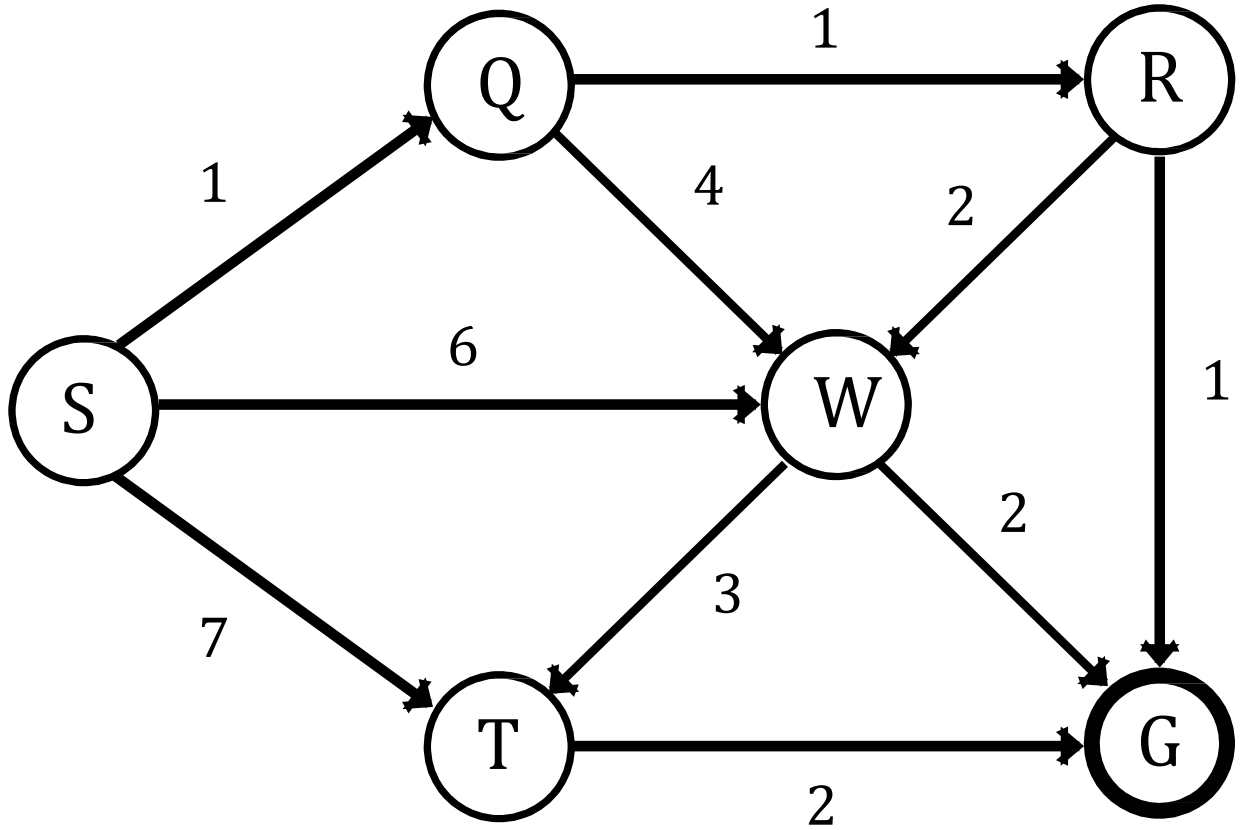
Stack:

S					

Visited:

S					

Current Node = S



Depth Limited Search (Problem 02)

Depth Level Limit:

L = 2

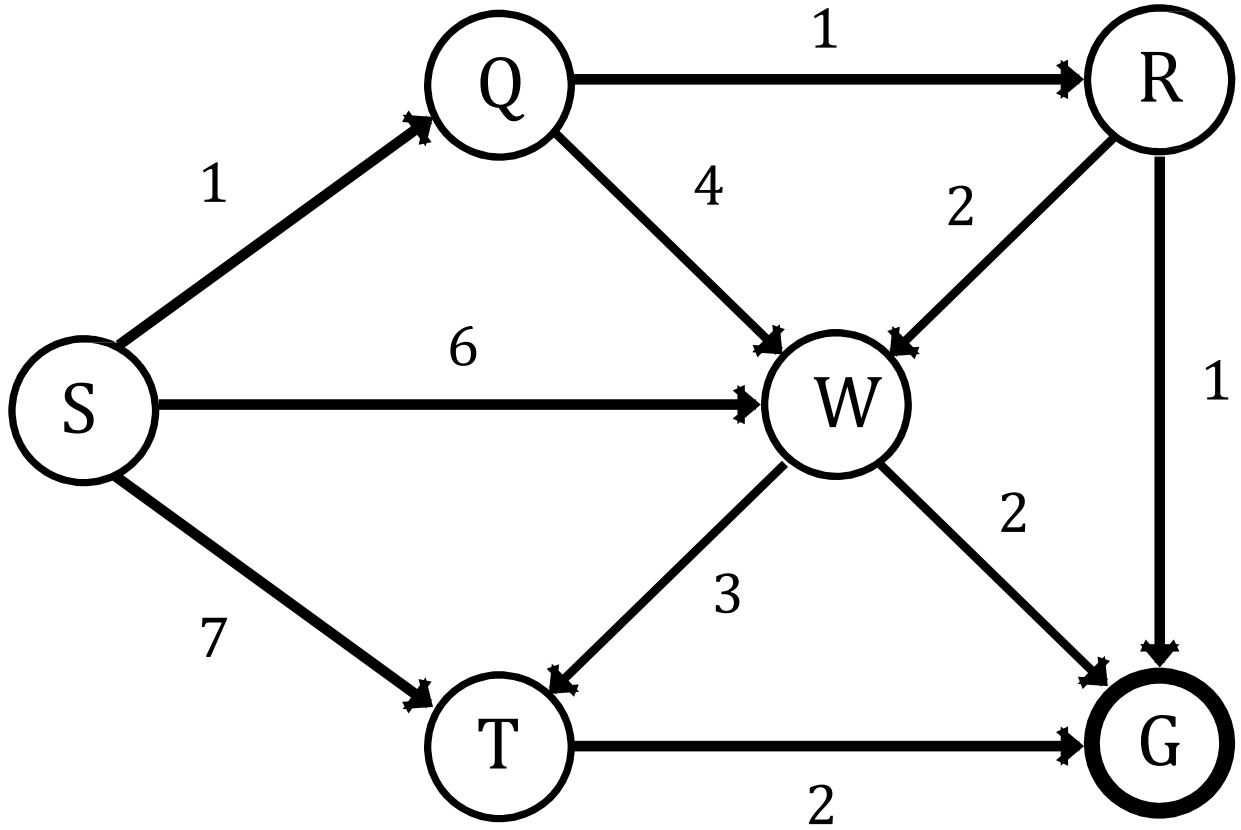
Stack:

S	**Q**				

Visited:

S	**Q**				

Current Node = Q



Depth Limited Search (Problem 02)

Depth Level Limit:

L = 2

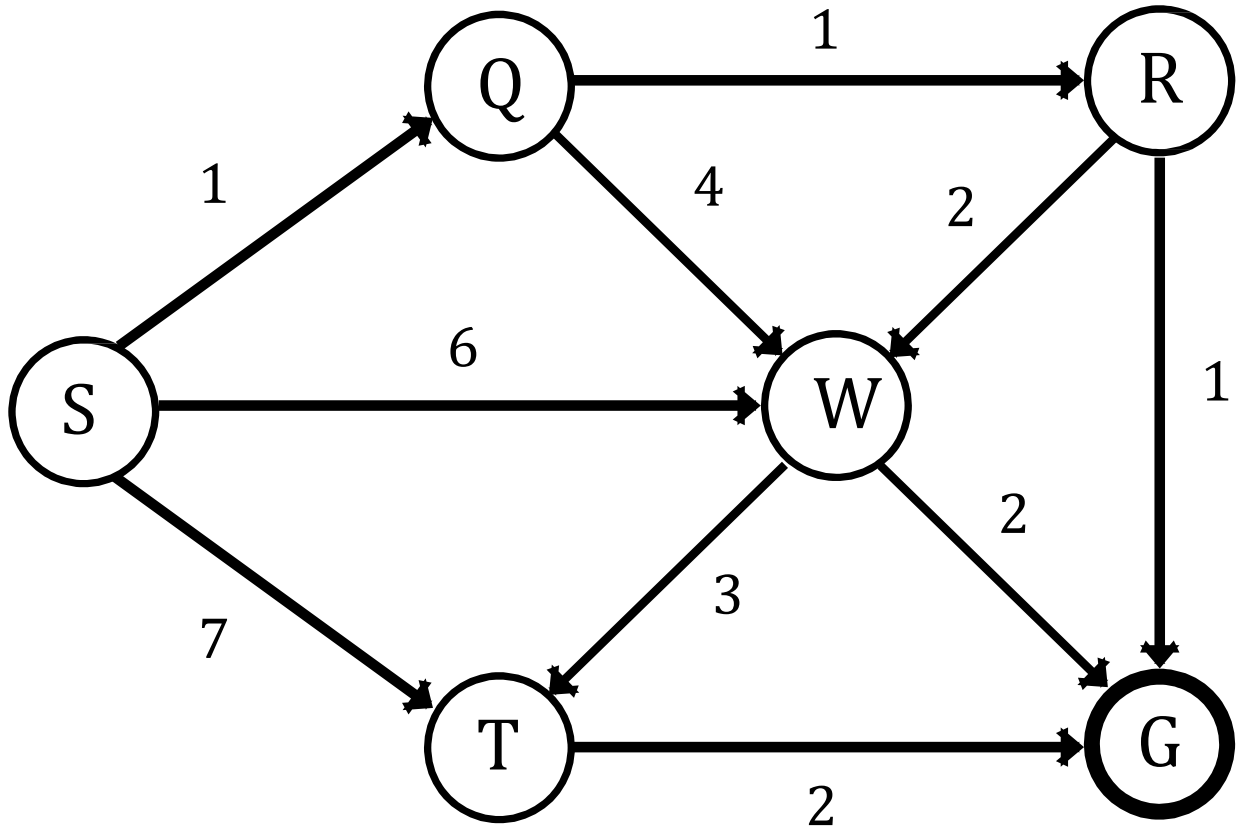
Stack:

S	**Q**	**R**			

Visited:

S	**Q**	**R**			

Current Node = R



Depth Limited Search (Problem 02)

Depth Level Limit:

L = 2

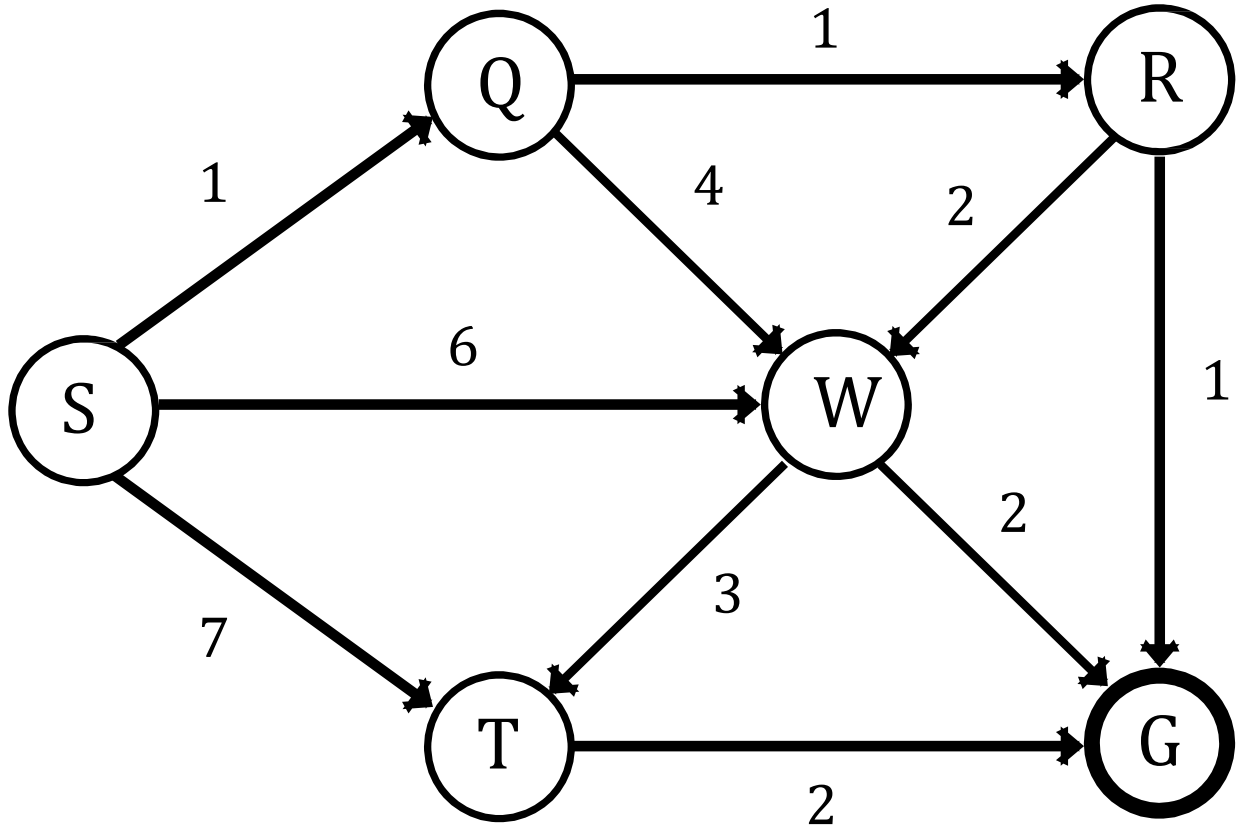
Stack:

S	**Q**				

Visited:

S	**Q**	**R**			

Current Node = Q



Depth Limited Search (Problem 02)

Depth Level Limit:

L = 2

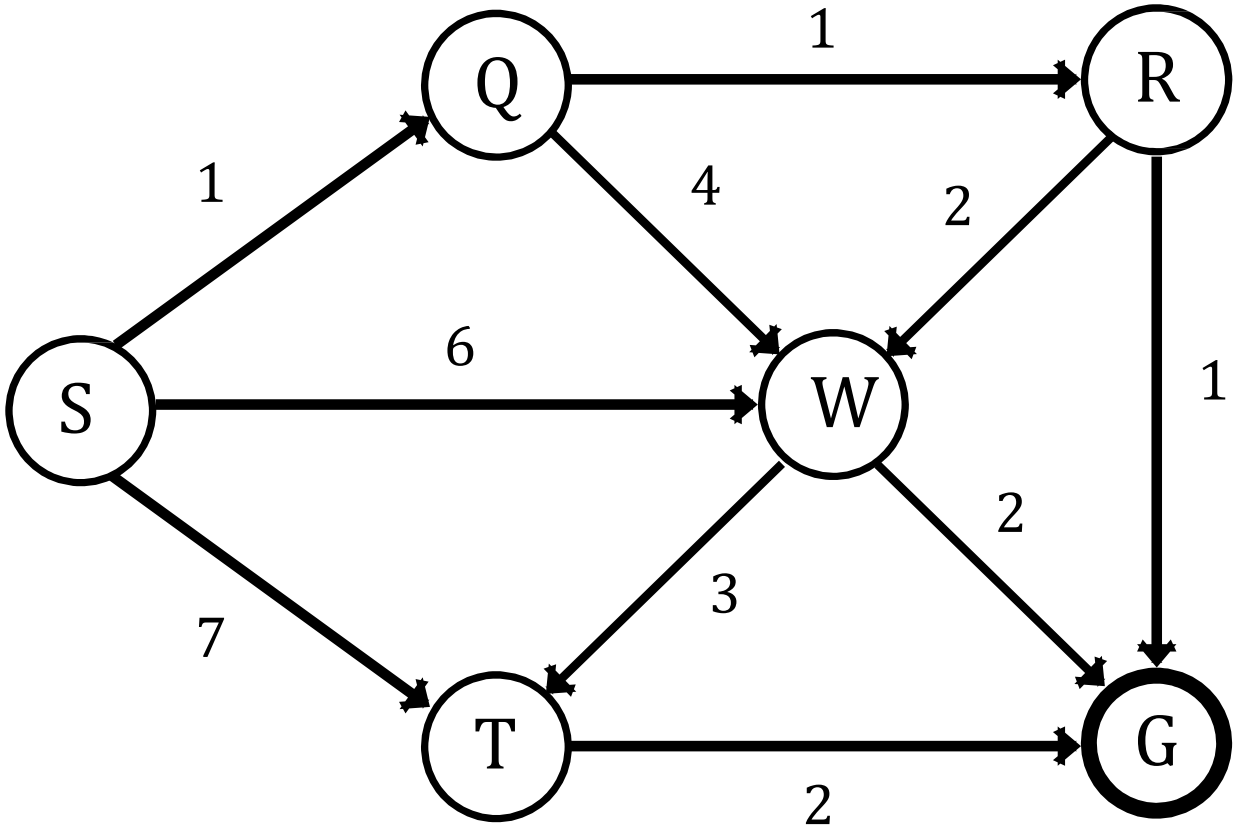
Stack:

S	**Q**	**W**			

Visited:

S	**Q**	**R**	**W**		

Current Node = W



Depth Limited Search (Problem 02)

Depth Level Limit:

L = 2

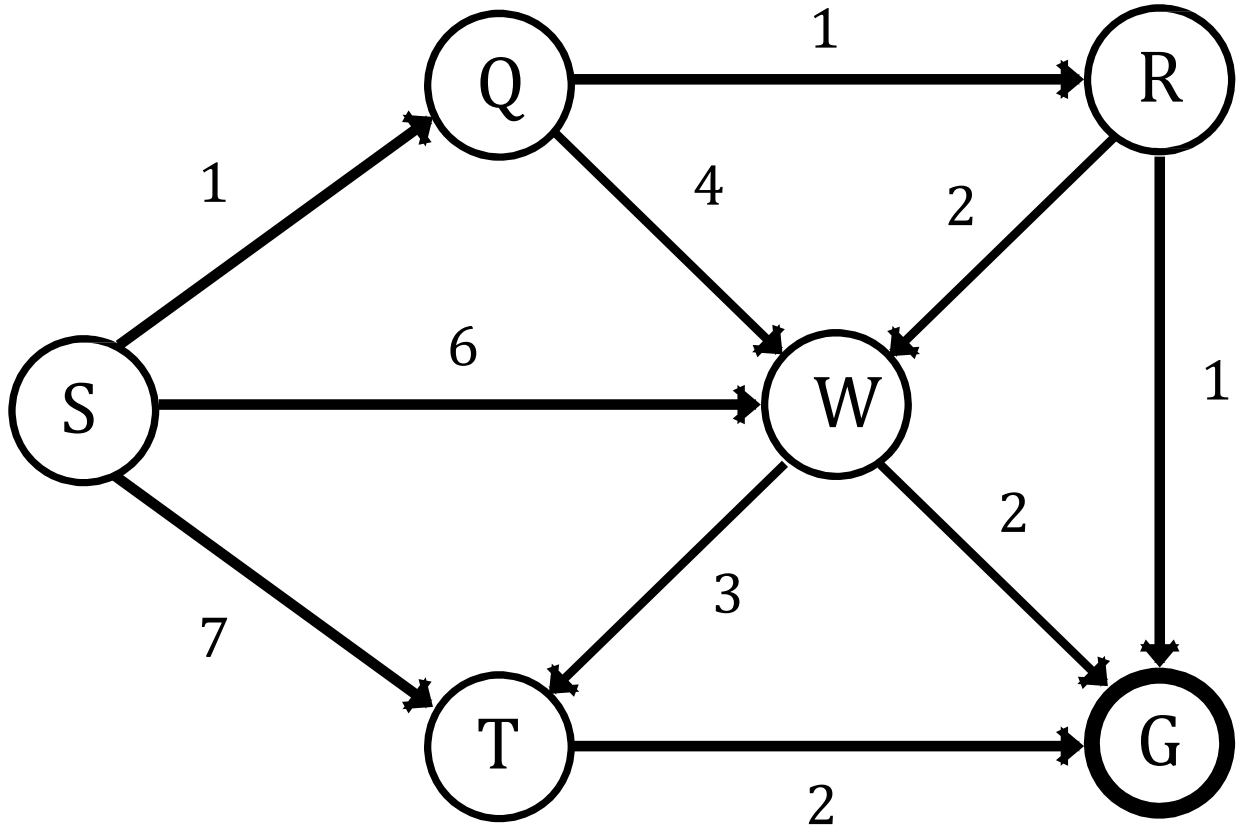
Stack:

S	**Q**				

Visited:

S	**Q**	**R**	**W**		

Current Node = Q



Depth Limited Search (Problem 02)

Depth Level Limit:

L = 2

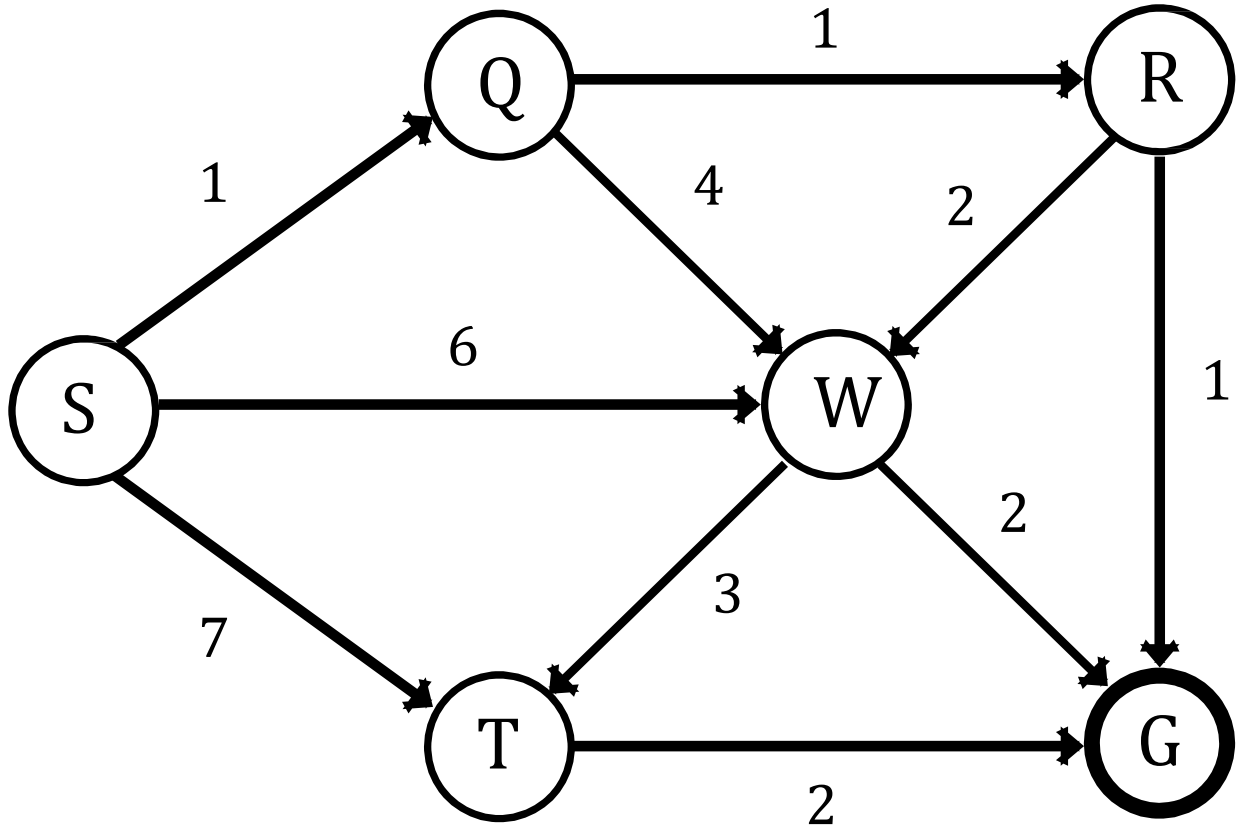
Stack:

S					

Visited:

S	**Q**	**R**	**W**		

Current Node = S



Depth Limited Search (Problem 02)

Depth Level Limit:

L = 2

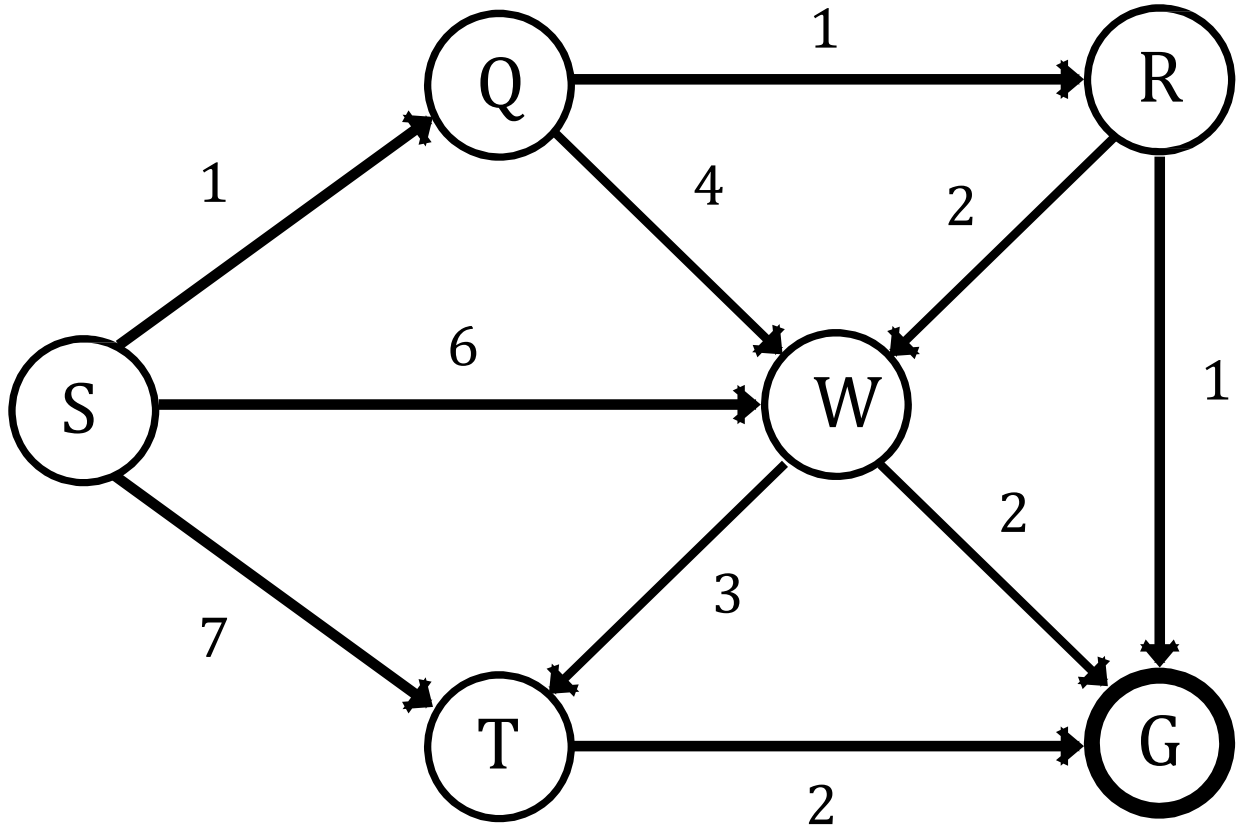
Stack:

S	**T**				

Visited:

S	**Q**	**R**	**W**	**T**	

Current Node = T



Depth Limited Search (Problem 02)

Depth Level Limit:

L = 2

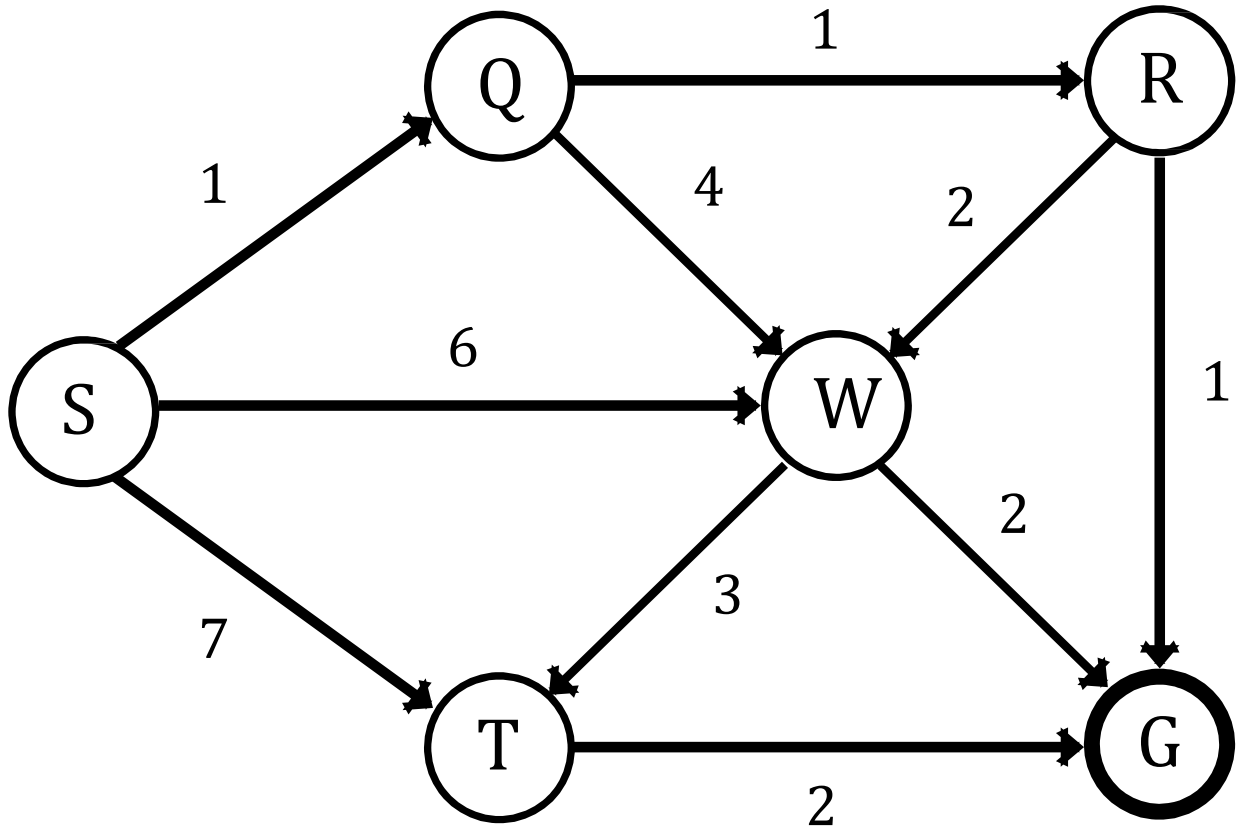
Stack:

S	**T**	**G**			

Visited:

S	**Q**	**R**	**W**	**T**	**G**

Current Node = G



Depth Limited Search (Problem 02)

Depth Level Limit:

L = 2

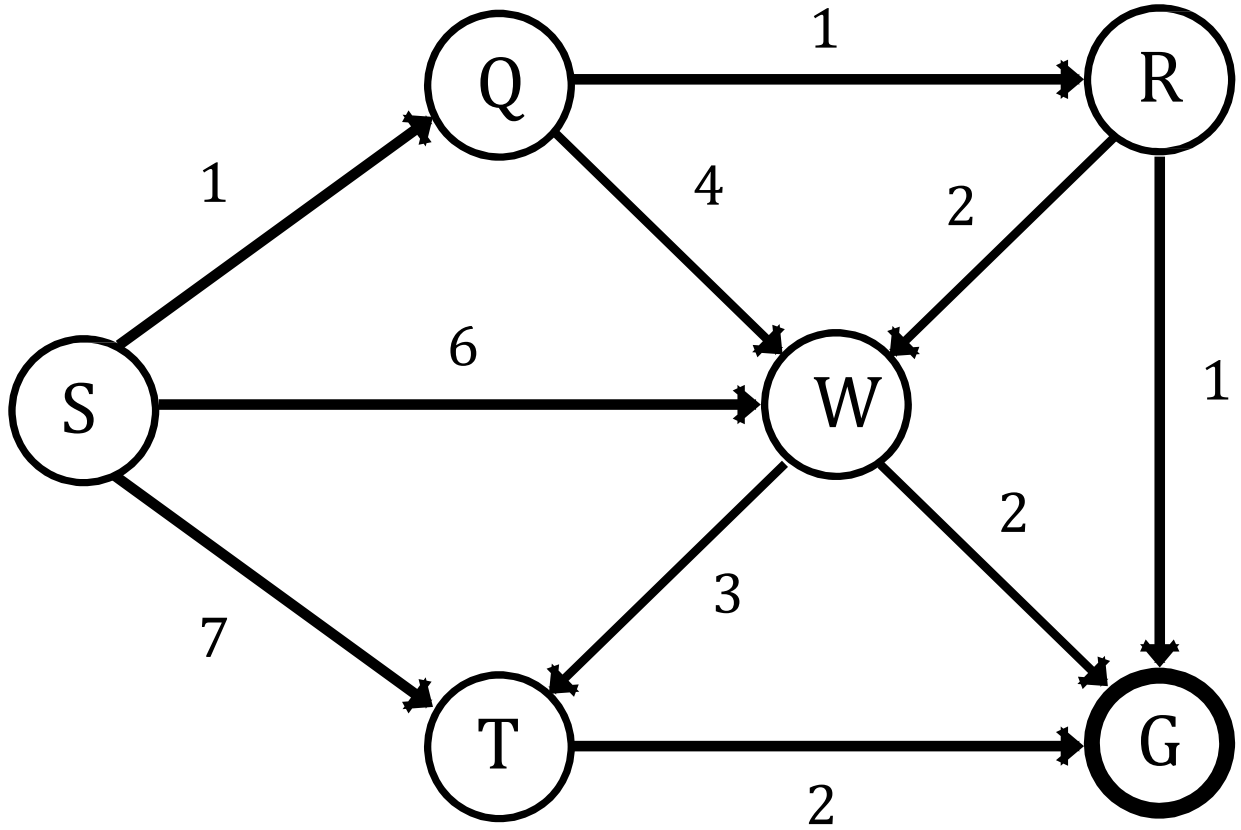
Stack:

S	T				

Visited:

S	Q	R	W	T	G

Current Node = T



Depth Limited Search (Problem 02)

Depth Level Limit:

L = 2

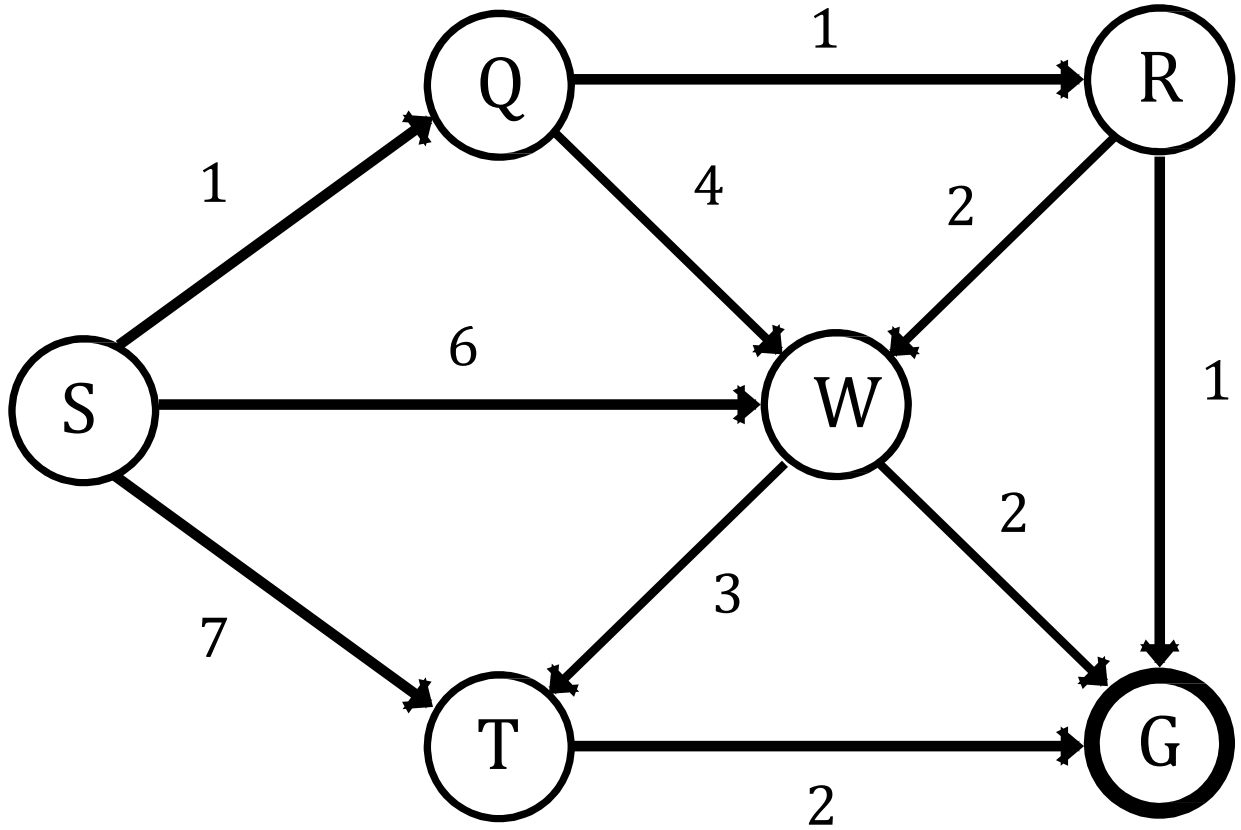
Stack:

S					

Visited:

S	**Q**	**R**	**W**	**T**	**G**

Current Node = S



Depth Limited Search (Problem 02)

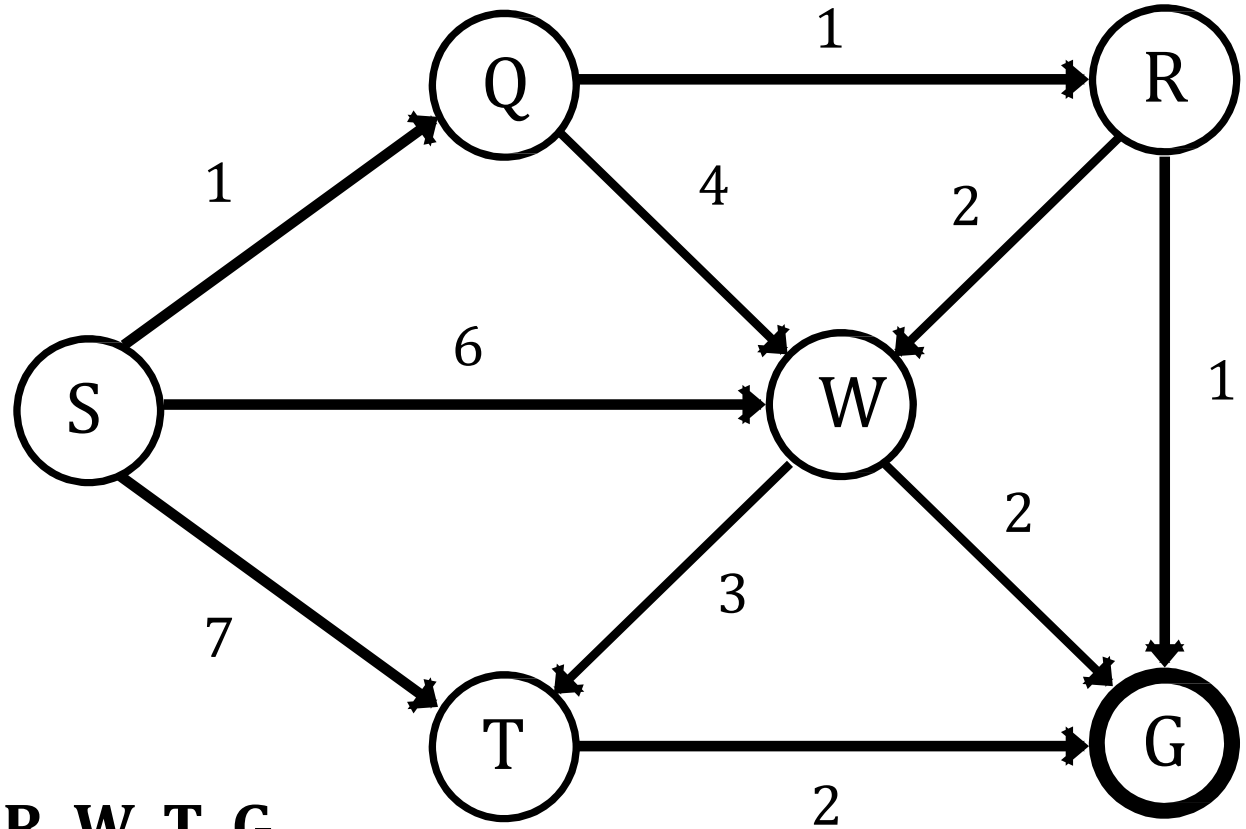
Depth Level Limit:

L = 2

Stack:

Visited:

S	**Q**	**R**	**W**	**T**	**G**



DFS Limited Traversal: **S Q R W T G**

Properties of Depth Limited Search

Property	Description
Completeness	Depth-Limited Search (DLS) is incomplete because it may fail to find a solution if the depth limit is exceeded.
Optimality	DLS is suboptimal, meaning it does not always find the optimal solution, especially if the depth limit is low.
Time Complexity	The time complexity of DLS is $O(b^l)$, where b is the branching factor and l is the depth limit.
Space Complexity	DLS has a space complexity of $O(bl)$, where b is the branching factor and l is the depth limit.
Frontier	DLS maintains a stack or recursion with a depth limit, ensuring that the search does not exceed the specified depth.

Depth-First Iterative Deeping Search (DFID)

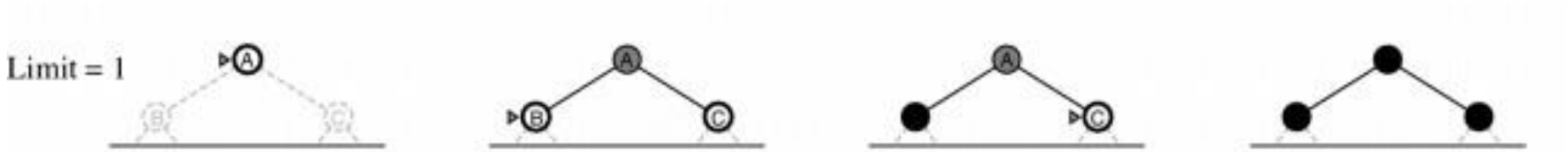
- Depth-first search will not find a goal if it searches down a path that has infinite length. So, in general, depth-first search is not guaranteed to find a solution, so it is not complete.
- This problem is eliminated by limiting the depth of the search to some value l. However, this introduces another way of preventing depth-first search from finding the goal: if the goal is deeper than l it will not be found.
- Regular depth-first search is a special case, for which $l=\infty$.

Depth-First Iterative Deeping Search (DFID)

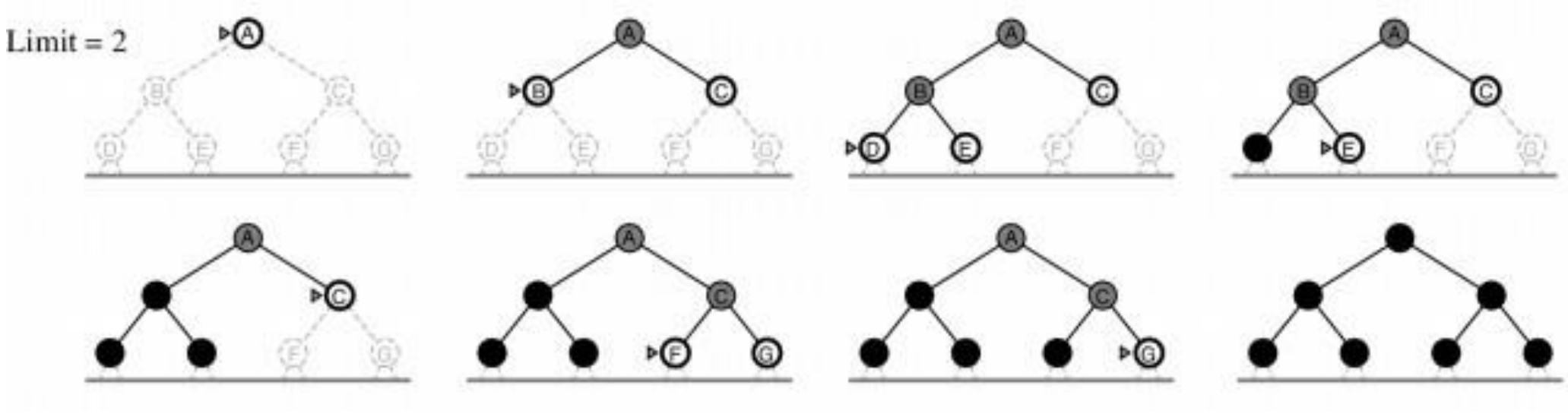
- If a depth-limited depth-first search limited to depth l does not find the goal, try it again with limit at l+1. Continue this until goal is found.
- Make depth-limited depth-first search complete by repeatedly applying it with greater values for the depth limit l.
- Feels like breadth-first search, in that a level is fully explored before extending the search to the next level. But, unlike breadth-first search, after one level is fully explored, all nodes already expanded are thrown away and the search starts with a clear memory.

Depth-First Iterative Deeping Search (DFID)
Limit = 0

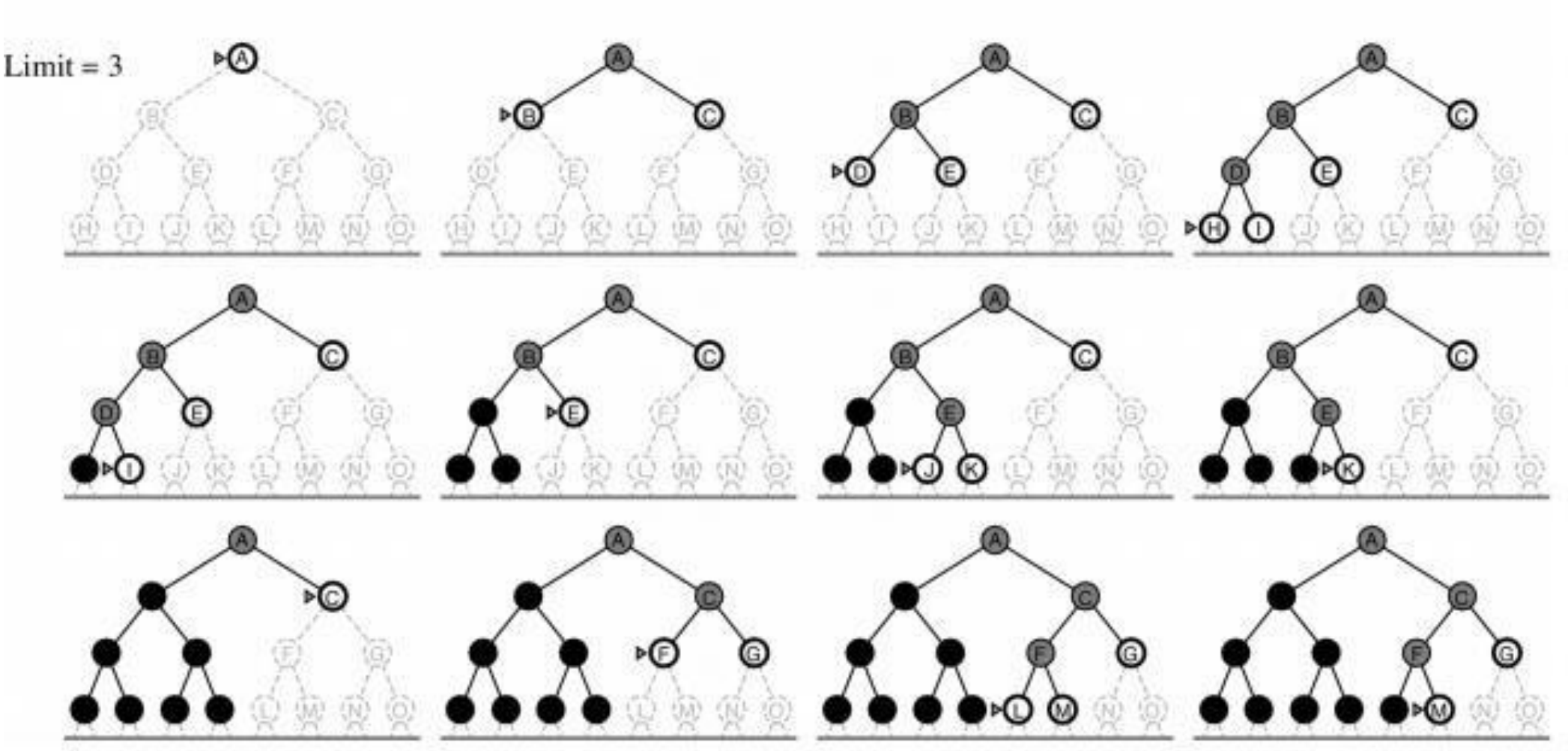
Depth-First Iterative Deeping Search (DFID)
Limit = 1



Depth-First Iterative Deeping Search (DFID)
Limit = 2



Depth-First Iterative Deeping Search (DFID)
Limit = 3



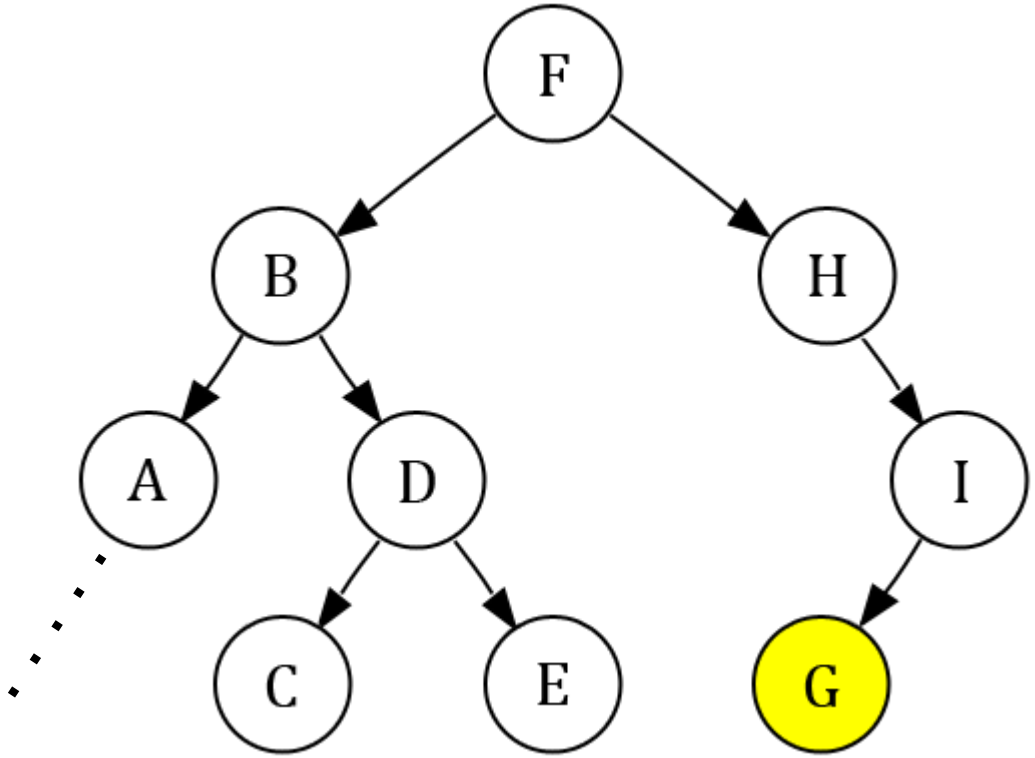
Properties of Depth-First Iterative Deeping Search (DFID)

Property	Description
Completeness	DFID is complete, as it is guaranteed to find a solution if one exists, provided that the branching factor is finite and the search space is finite.
Optimality	DFID is optimal for finding the shortest path in an unweighted graph, ensuring that the first solution found is the shortest path.
Time Complexity	The time complexity of DFID is $O(b^d)$, where b is the branching factor and d is the depth of the shallowest solution.
Space Complexity	DFID has a space complexity of $O(bd)$, where b is the branching factor and d is the depth of the shallowest solution.
Frontier	DFID maintains a stack or recursion with iterative depth limits, exploring nodes with increasing depth limits until a solution is found.

Evaluating Search Algorithms

Depth First Search

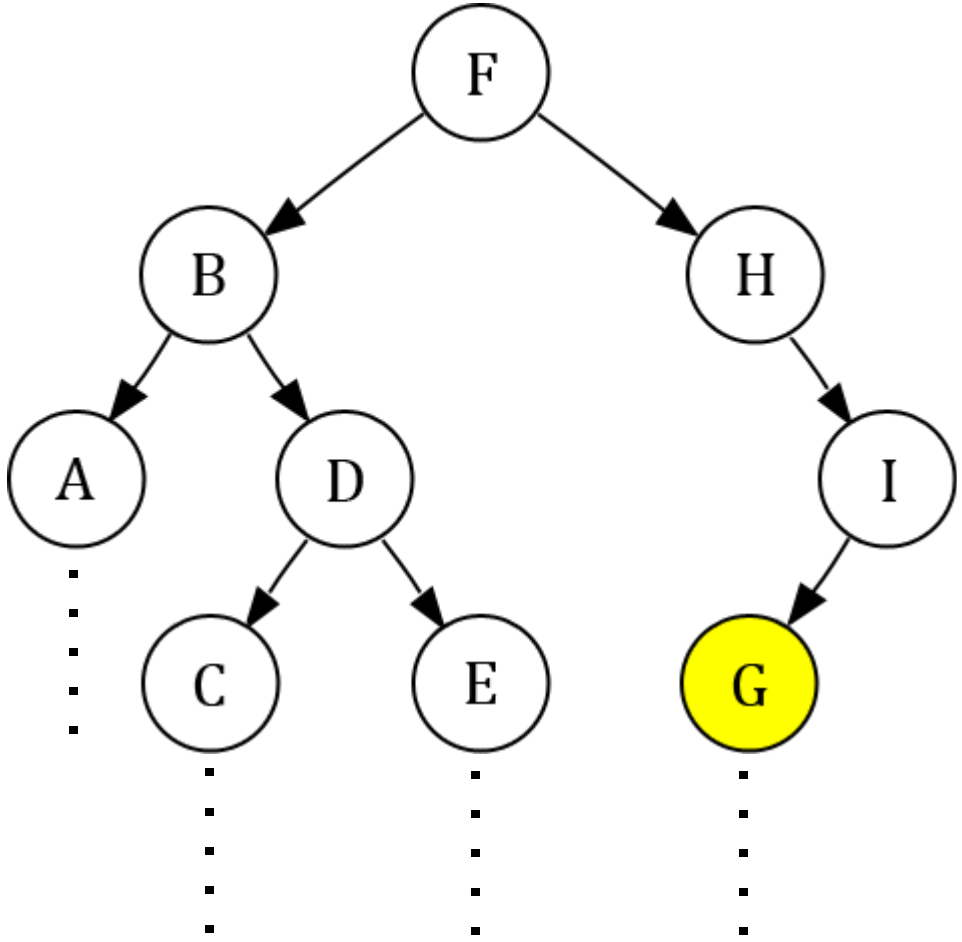
- If the first branch contains infinite nodes and the goal is G (at second branch).
- The DFS will not reach the goal because it will not be able to backtrack from first branch.
- **DFS is not complete** (It is not guaranteed to find the goal)



Evaluating Search Algorithms

Breadth First Search

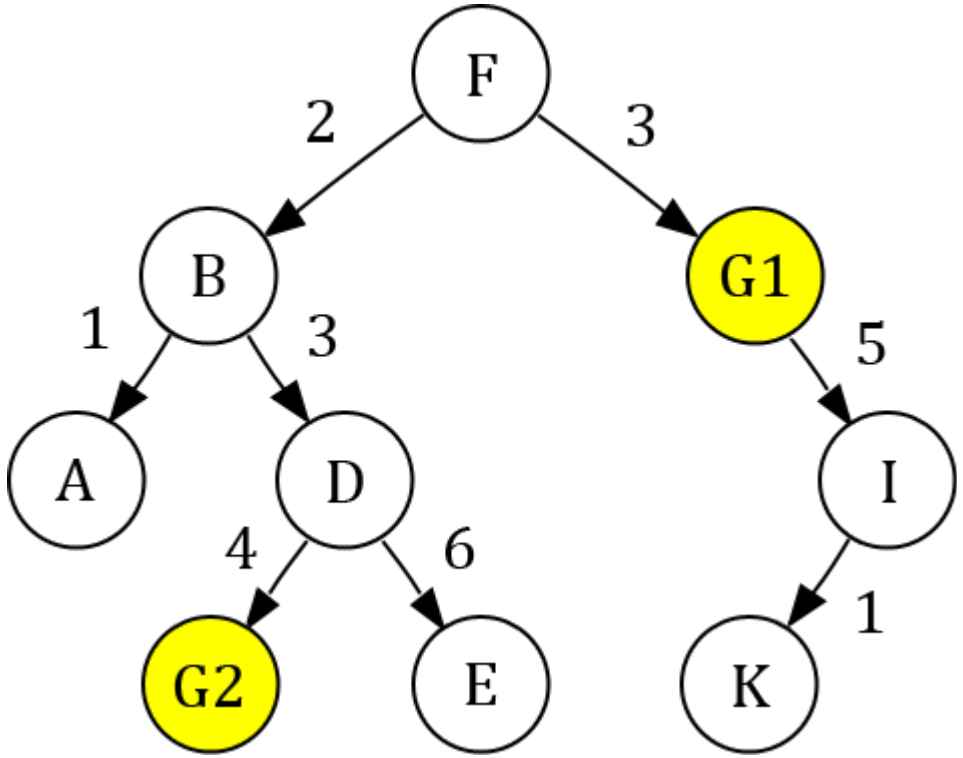
- If number of levels in the graph are infinite, but the goal is present in the solution.
- The BFS will always reach the goal.
- **BFS is complete** (It is guaranteed to find the goal, if it is present).



Evaluating Search Algorithms

Depth First Search

- If DFS is applied on this graph, it will find the goal G2.
- The path cost of G2 is 2+3+4=9.
- But there is another optimal goal present in the solution G1 having path cost 3.
- **DFS is not optimal** (It will not always return the optimal solution).



Uniform Cost Search

- Breadth-first search finds the shallowest goal state, but this may not always be the least-cost solution for a general path cost function.
- Uniform cost search modifies the breadth-first strategy by always expanding the lowest-cost node rather than the lowest-depth node.
- It maintain a priority Queue.

Uniform Cost Search

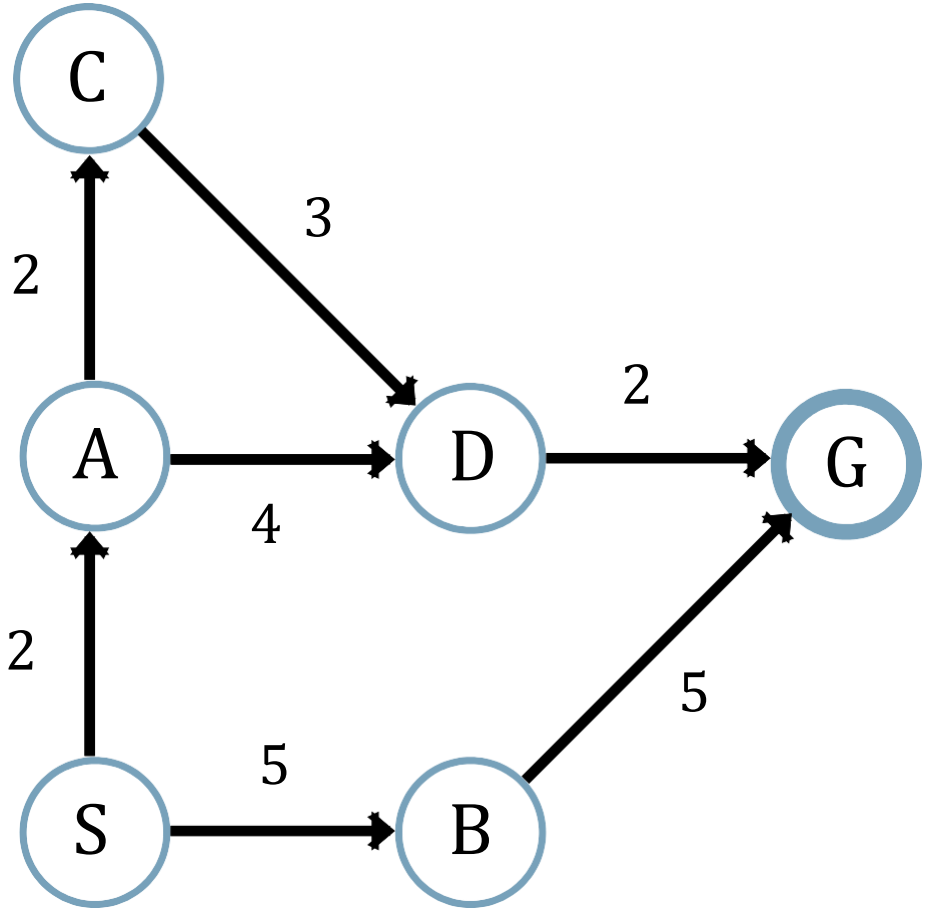
Algorithm:

- Step 01: Initialize Queue with Starting Node.
- Step 02: Pick minimum cost path from Queue.
- Step 03: If the minimum path is goal, stop algorithm you found the solution.
- Step 04: For each neighbor node v add expanded path < v ,P > to Queue.
- Step 05: Repeat Step 02 to Step 04 until Queue is empty

Uniform Cost Search (Problem 01)

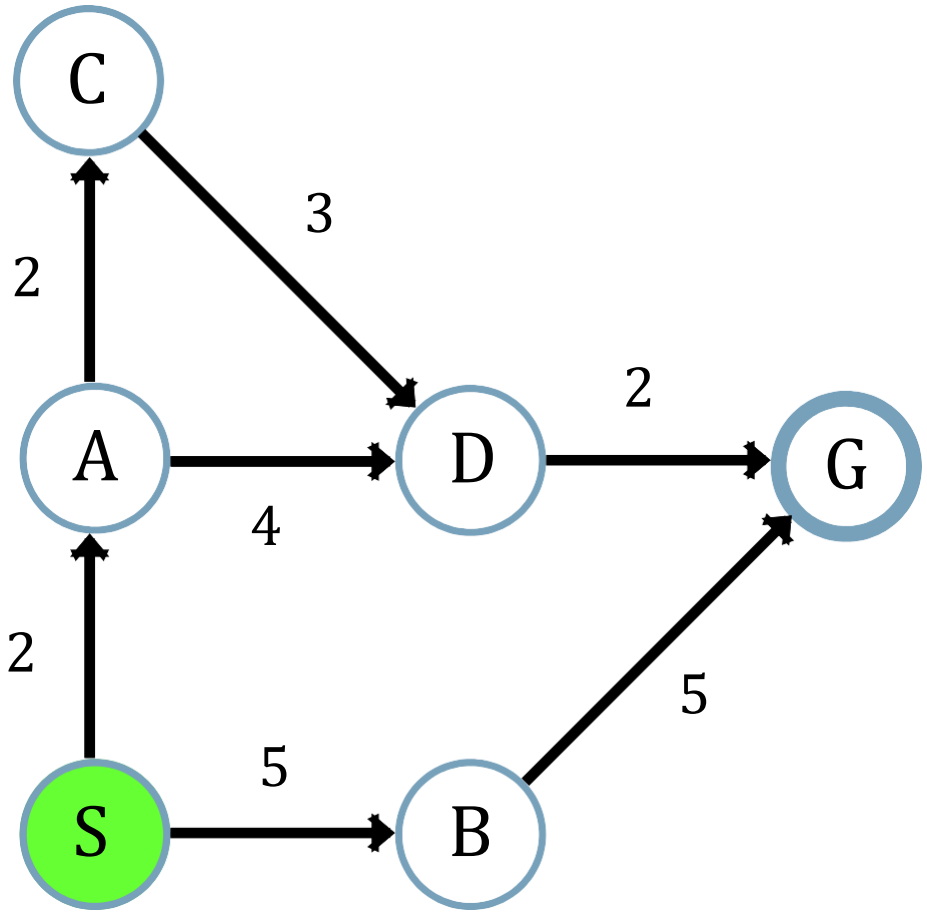
- S = Initial State
- G = Goal State

Queue	
Path	**Cost**



Uniform Cost Search (Problem 01)

Queue	
Path	**Cost**
<S>	0

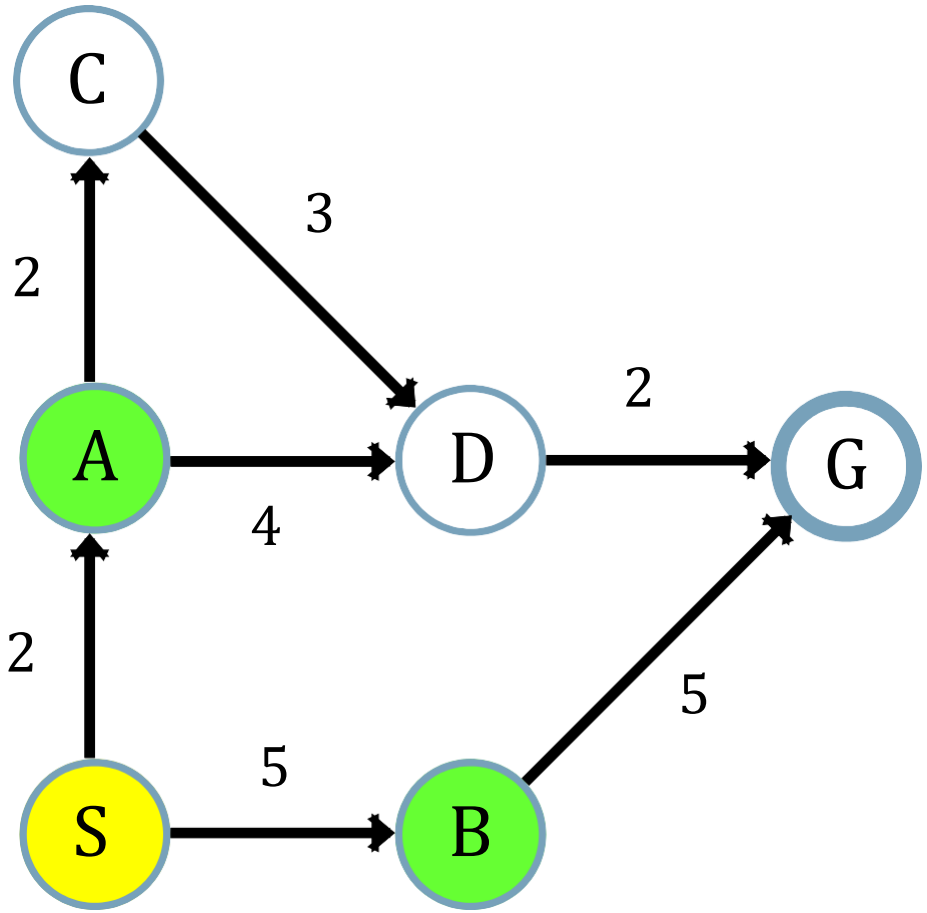


From the path we have:
<S> = 0

<A,S> = 2
<B,S> = 5

Uniform Cost Search (Problem 01)

Queue	
Path	**Cost**
<A,S>	2
<B,S>	5

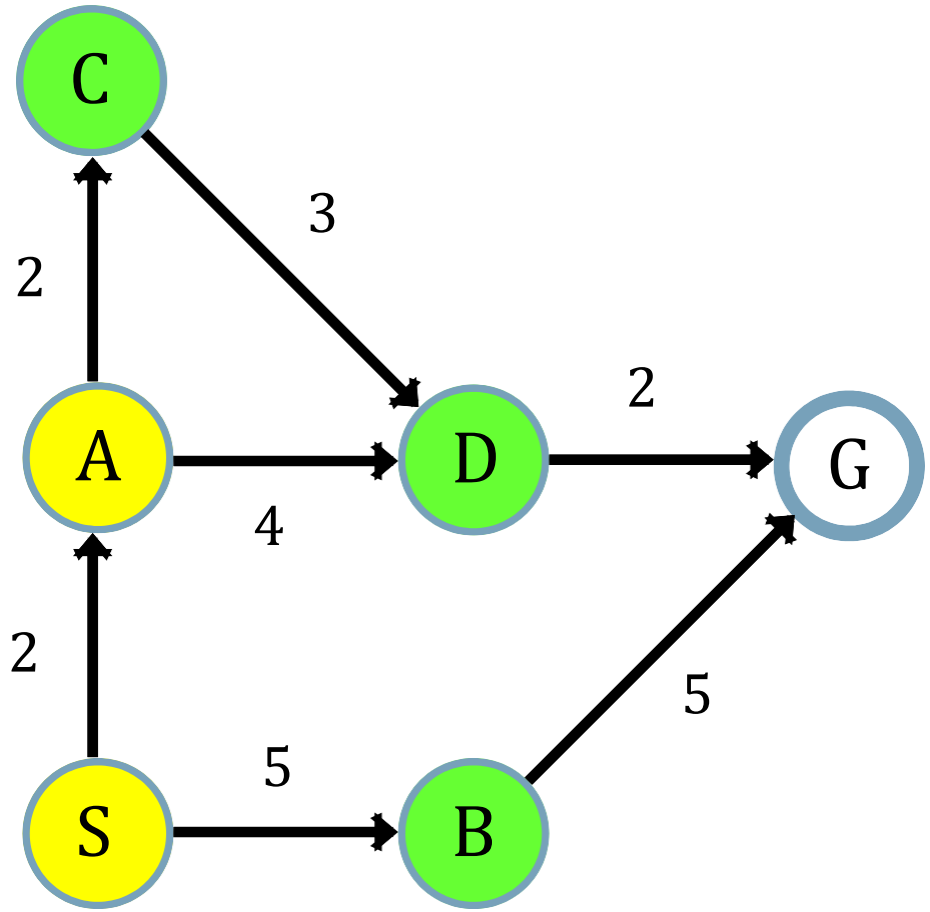


From the path we have:
<A,S> = 2

<C,A,S> = 4
<D,A,S> = 6

Uniform Cost Search (Problem 01)

Queue	
Path	**Cost**
<C,A,S>	4
<B,S>	5
<D,A,S>	6

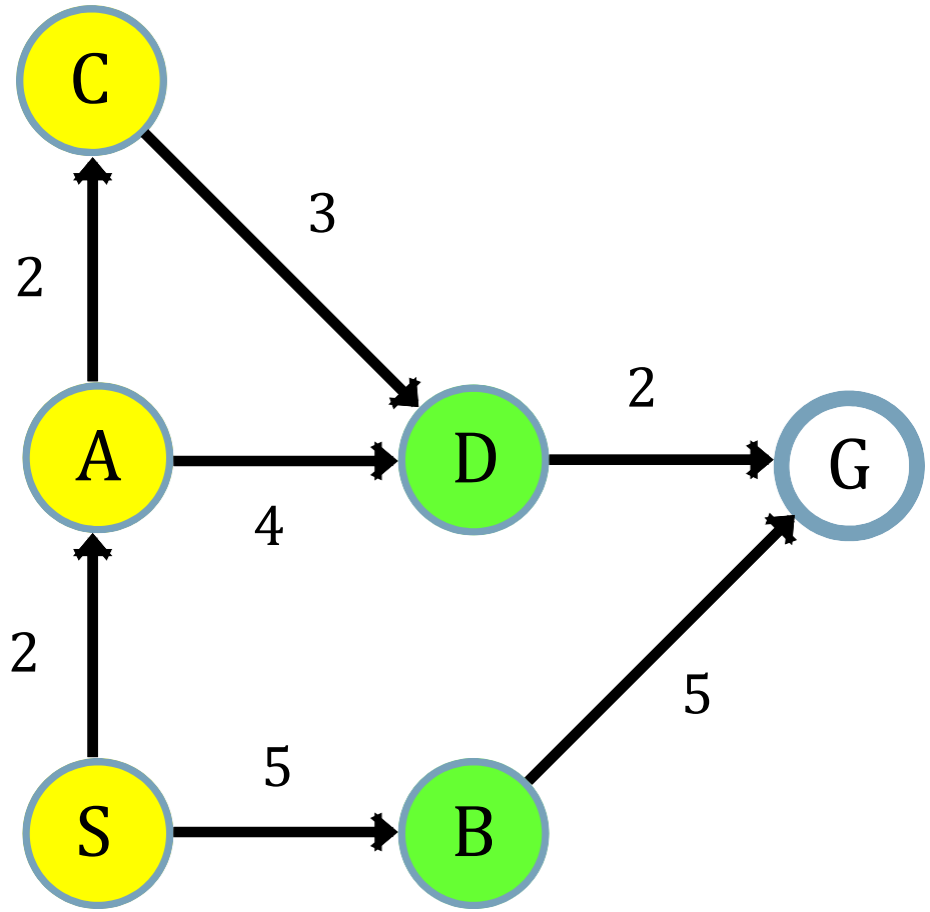


From the path we have:
<C,A,S> = 4

<D,C,A,S> = 7

Uniform Cost Search (Problem 01)

Queue	
Path	**Cost**
<B,S>	5
<D,A,S>	6
<D,C,A,S>	7

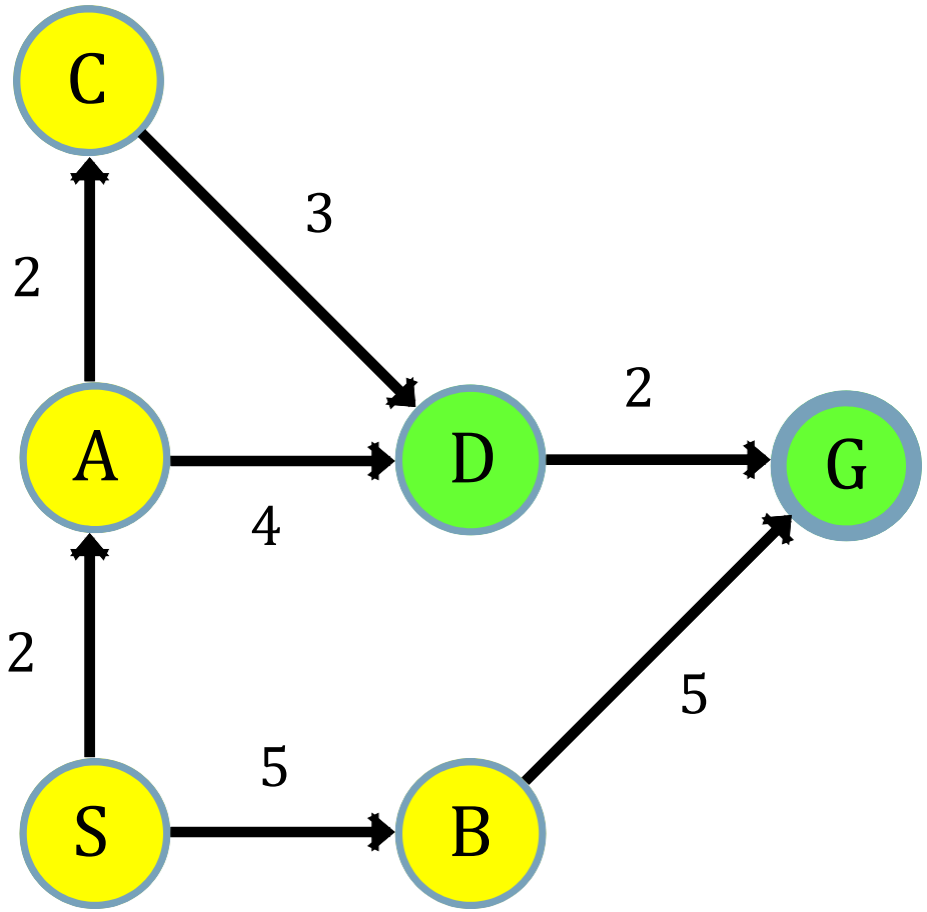


From the path we have:
<B,S> = 5

<G,B,S> = 10

Uniform Cost Search (Problem 01)

Queue	
Path	**Cost**
<D,A,S>	6
<D,C,A,S>	7
<G,B,S>	10

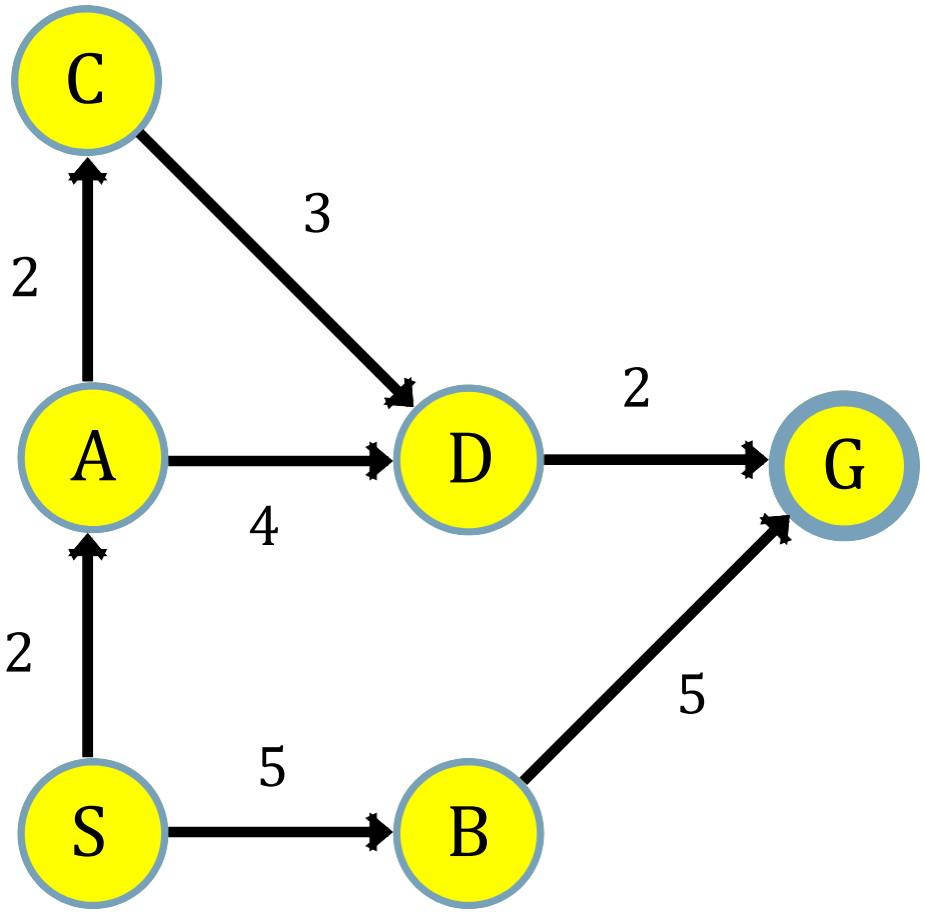


From the path we have:
<D,A,S> = 6

<G,D,A,S> = 8

Uniform Cost Search (Problem 01)

Queue	
Path	**Cost**
<D,C,A,S>	7
<G,D,A,S>	8
<G,B,S>	10



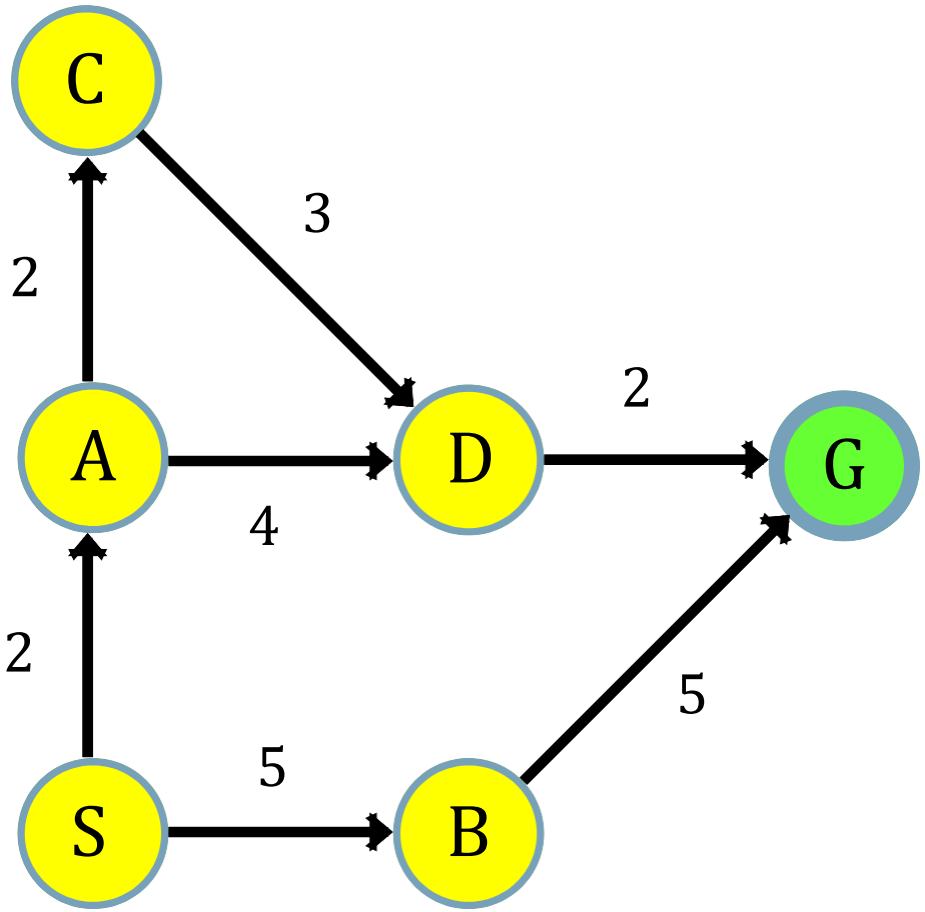
From the path we have:
<D,C,A,S> = 7

<G,D,C,A,S> = 9

Uniform Cost Search (Problem 01)

Queue	
Path	**Cost**
<G,D,A,S>	8
<G,D,C,A,S>	9
<G,B,S>	10

Shortest Path = Solution =
S → A → D → G

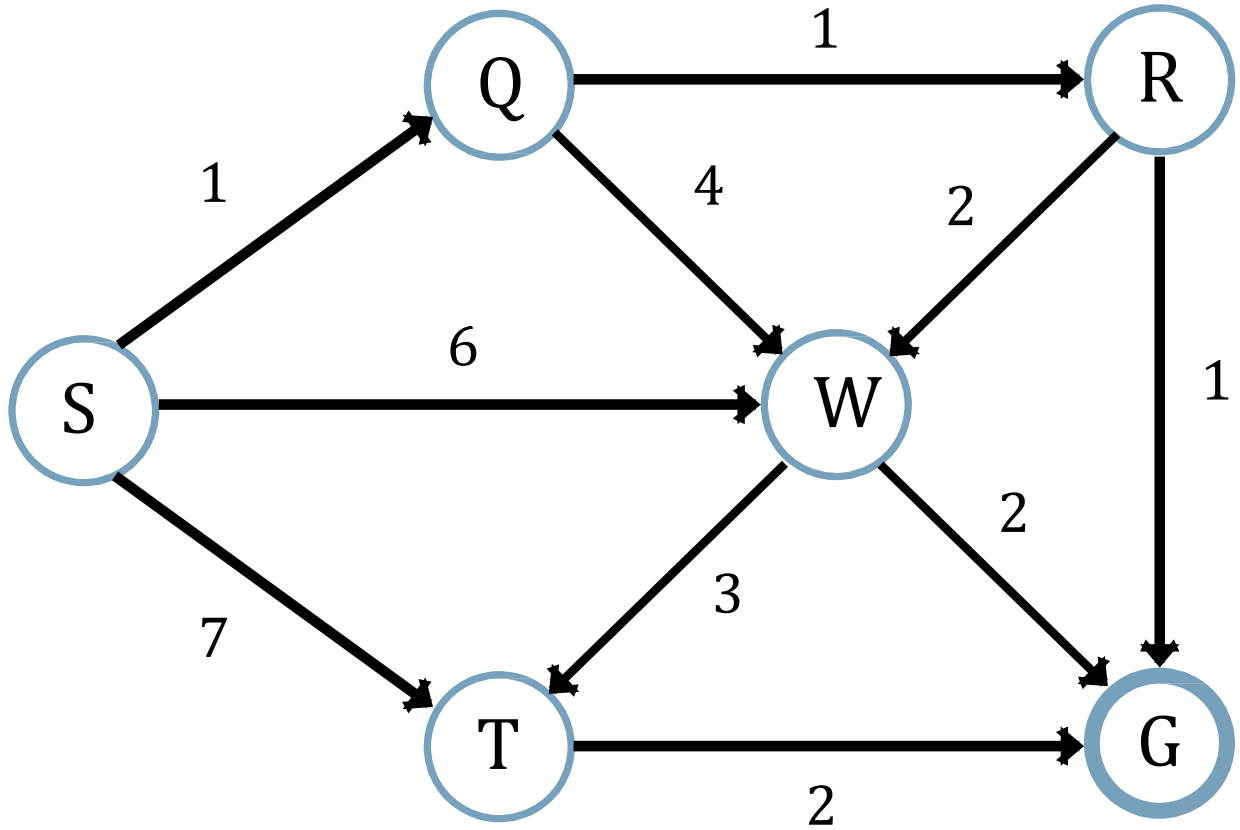


As minimum path is the goal path, so we have found a solution.

Uniform Cost Search (Problem 02)

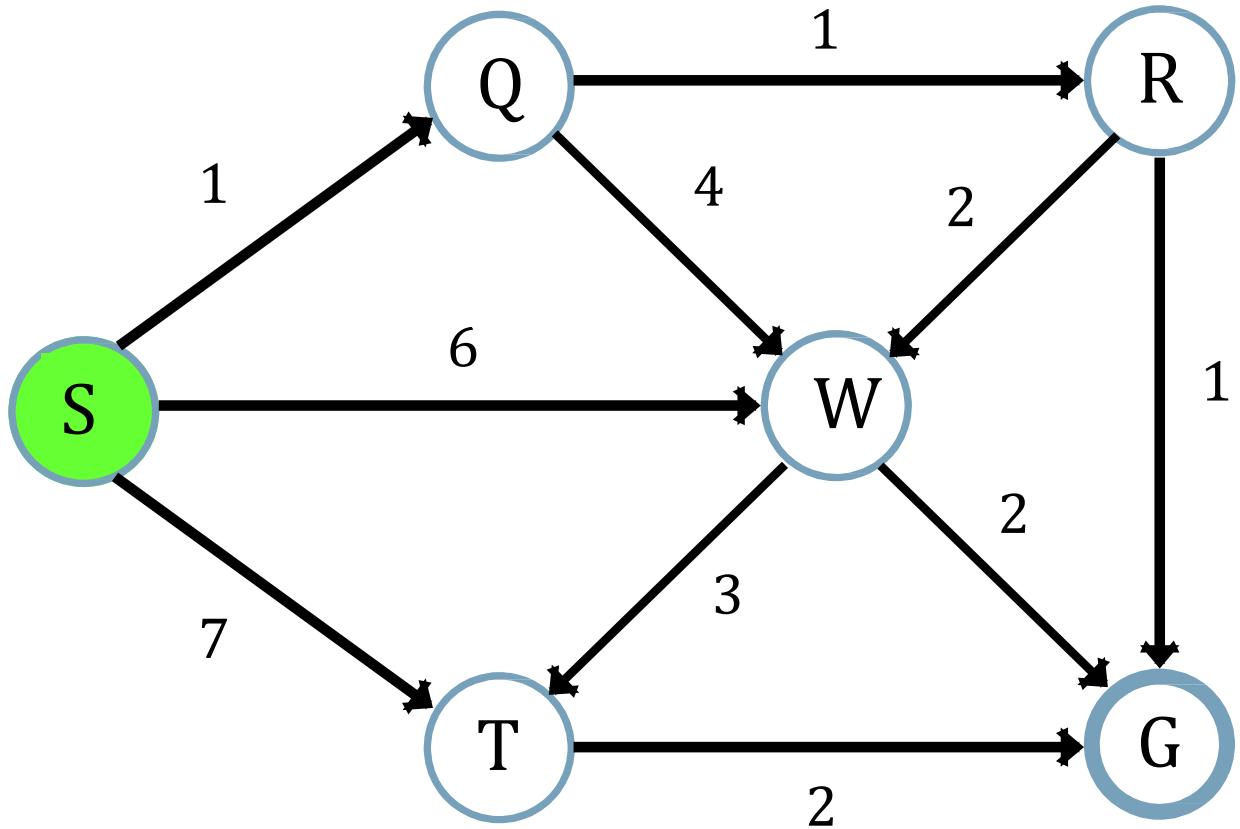
- S = Initial State
- G = Goal State

Queue	
Path	**Cost**



Uniform Cost Search (Problem 02)

Queue	
Path	**Cost**
<S>	0

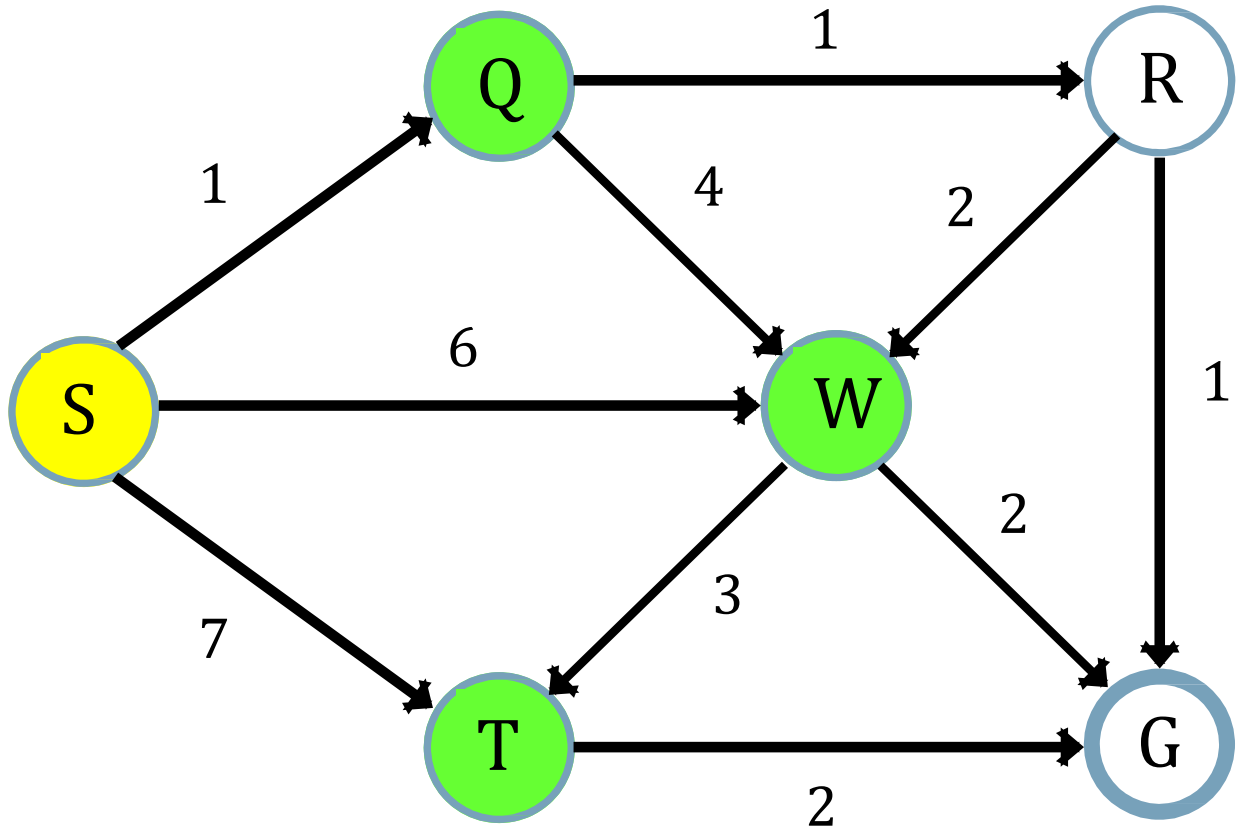


From the path we have:
<S> = 0

<Q,S> = 1
<W,S> = 6
<T,S> = 7

Uniform Cost Search (Problem 02)

Queue	
Path	**Cost**
<Q,S>	1
<W,S>	6
<T,S>	7

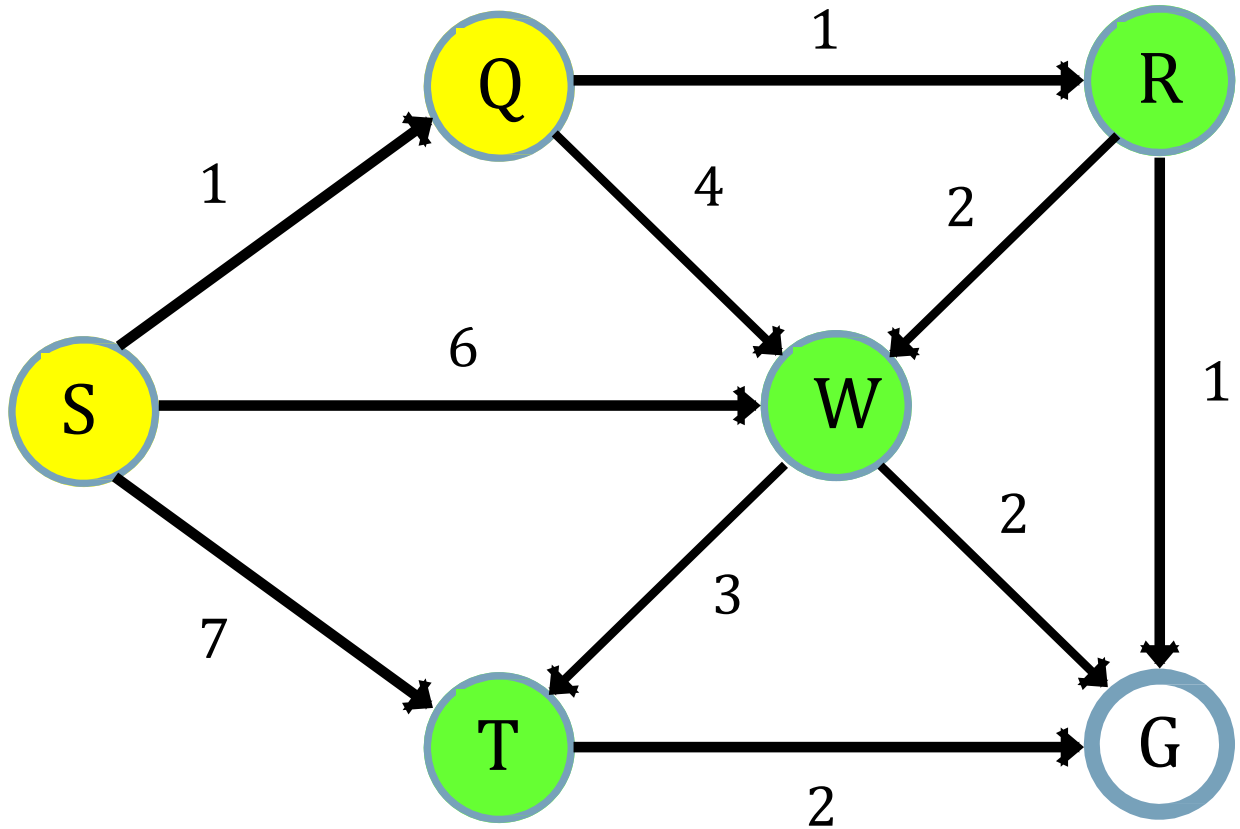


From the path we have:
<Q,S> = 1

<R,Q,S> = 2
<W,Q,S> = 5

Uniform Cost Search (Problem 02)

Queue	
Path	**Cost**
<R,Q,S>	2
<W,Q,S>	5
<W,S>	6
<T,S>	7

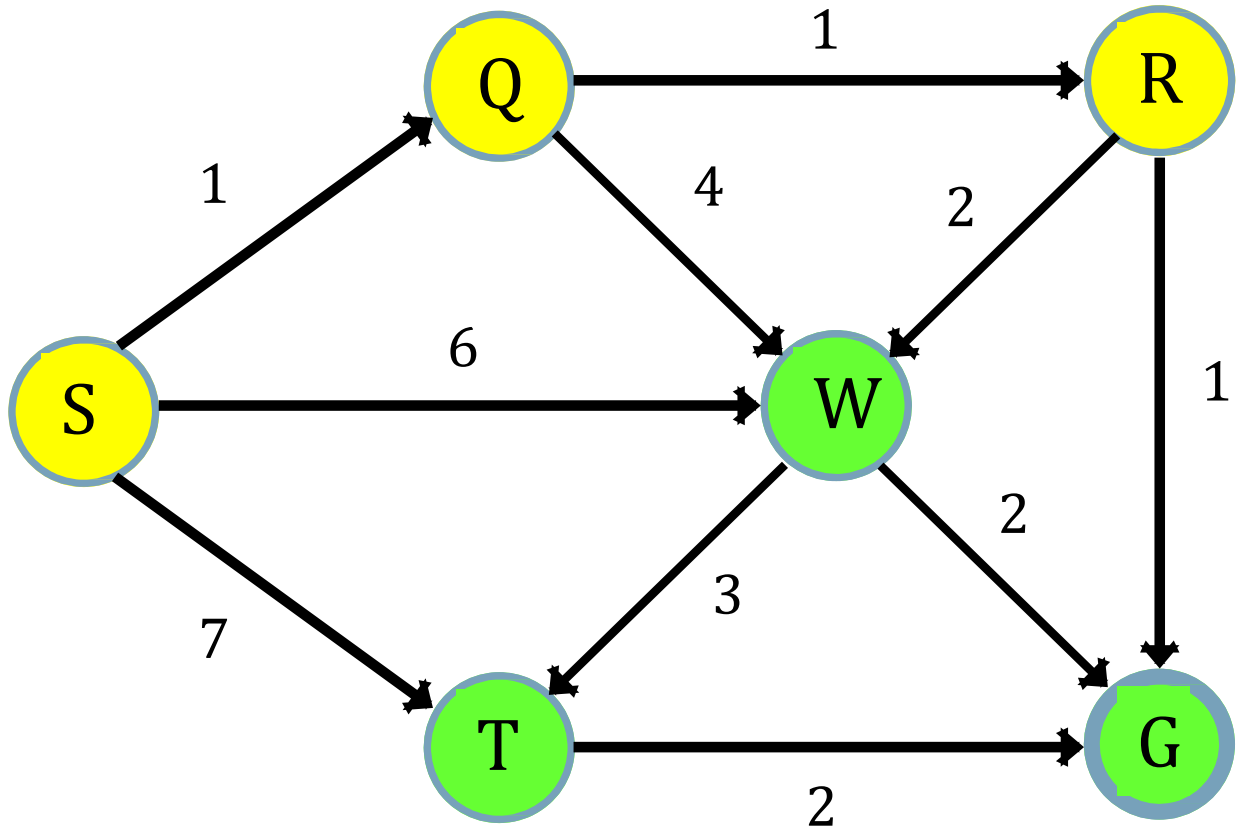


From the path we have:
<R,Q,S> = 2

<W,R,Q,S> = 4
<G,R,Q,S> = 3

Uniform Cost Search (Problem 02)

Queue	
Path	**Cost**
<G,R,Q,S>	3
<W,R,Q,S>	4
<W,Q,S>	5
<W,S>	6
<T,S>	7



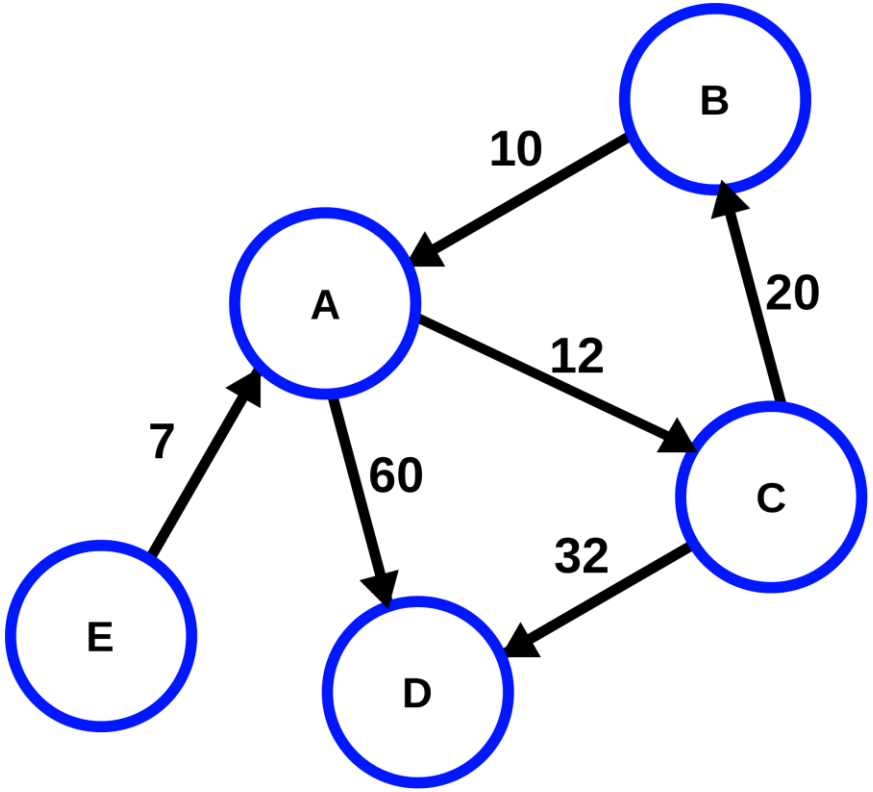
Shortest Path = Solution =
S → Q → R → G

As minimum path is the goal path, so we have found a solution.

Uniform Cost Search (Problem 03)

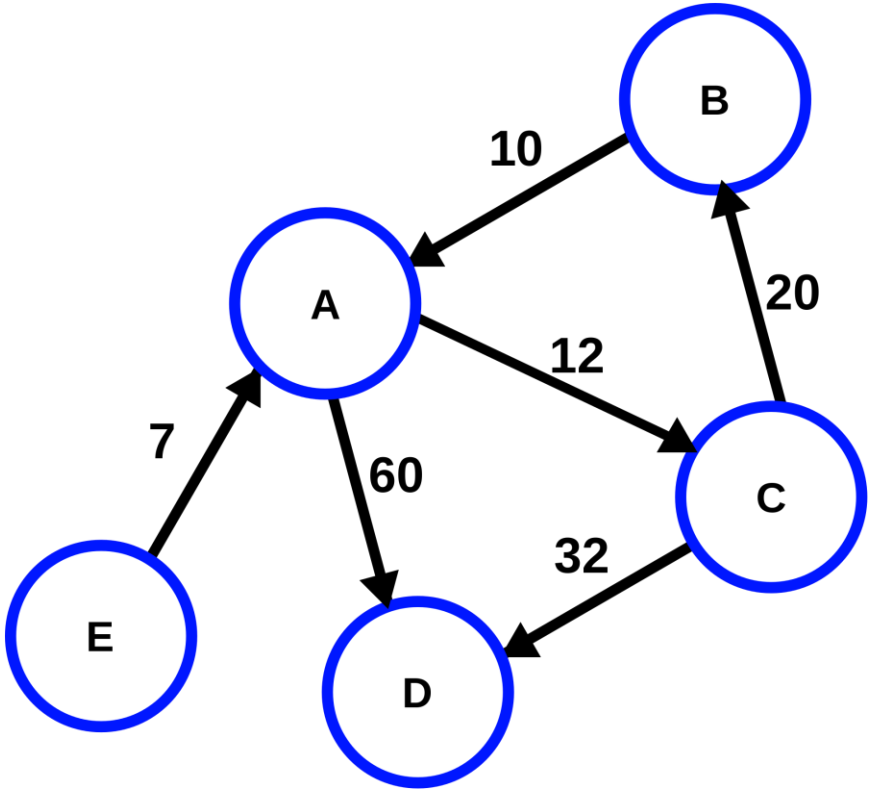
- B = Initial State
- D = Goal State

Queue	
Path	**Cost**



Uniform Cost Search (Problem 03)

Queue	
Path	**Cost**
<B>	0

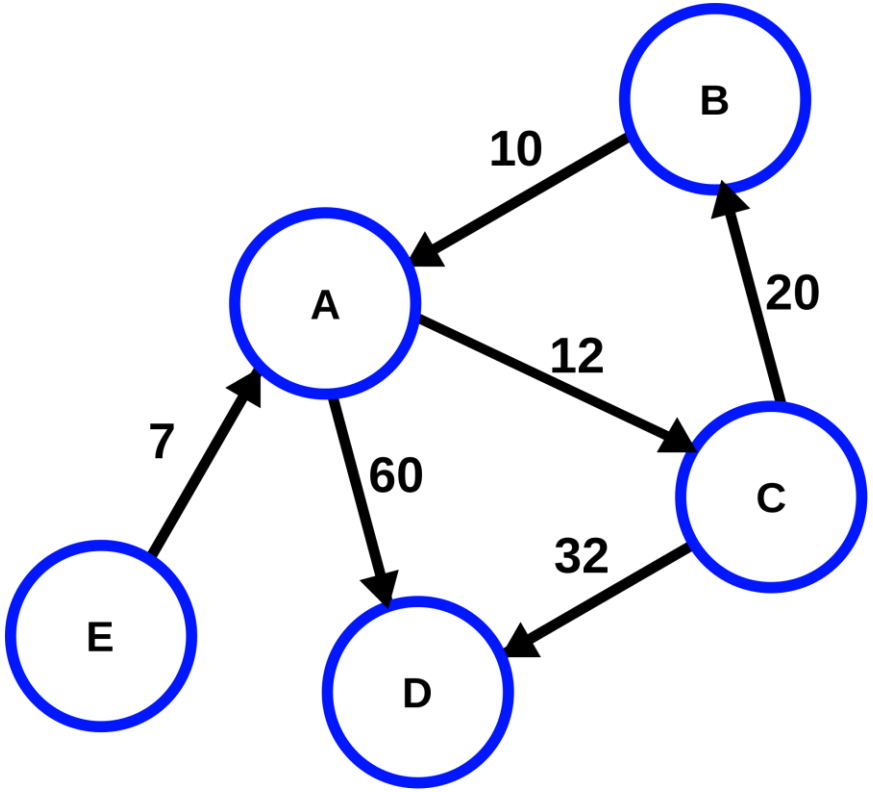


From the path we have:
<B> = 0

<A,B> = 10

Uniform Cost Search (Problem 03)

Queue	
Path	**Cost**
<A,B>	10

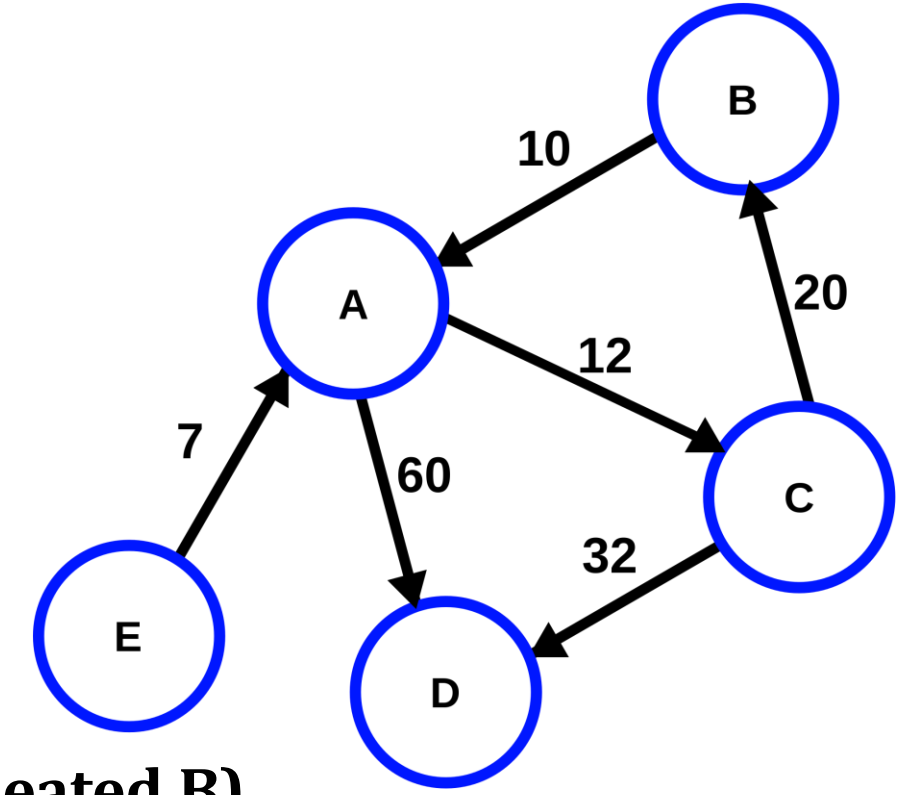


From the path we have:
<A,B> = 10

<C,A,B> = 22
<D,A,B> = 70

Uniform Cost Search (Problem 03)

Queue	
Path	**Cost**
<C,A,B>	22
<D,A,B>	70



From the path we have:

<C,A,B> = 22

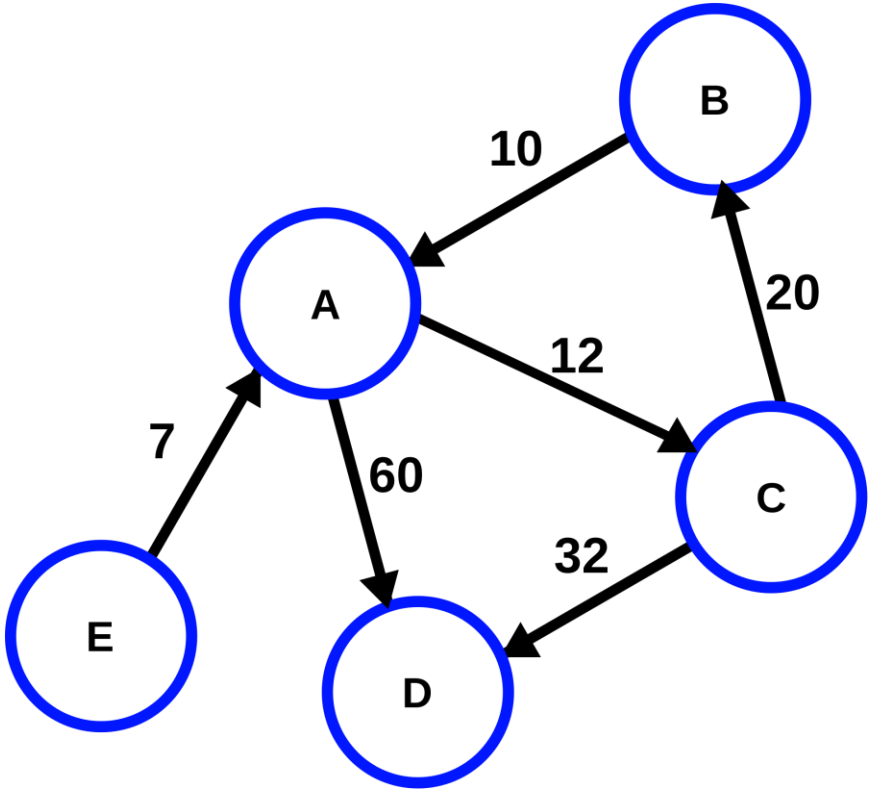
<B,C,A,B> = 42 (It is a cycle as repeated B)

<D,C,A,B> = 54

Avoid Cycles (Discard them)

Uniform Cost Search (Problem 03)

Queue	
Path	**Cost**
<D,C,A,B>	54
<D,A,B>	70

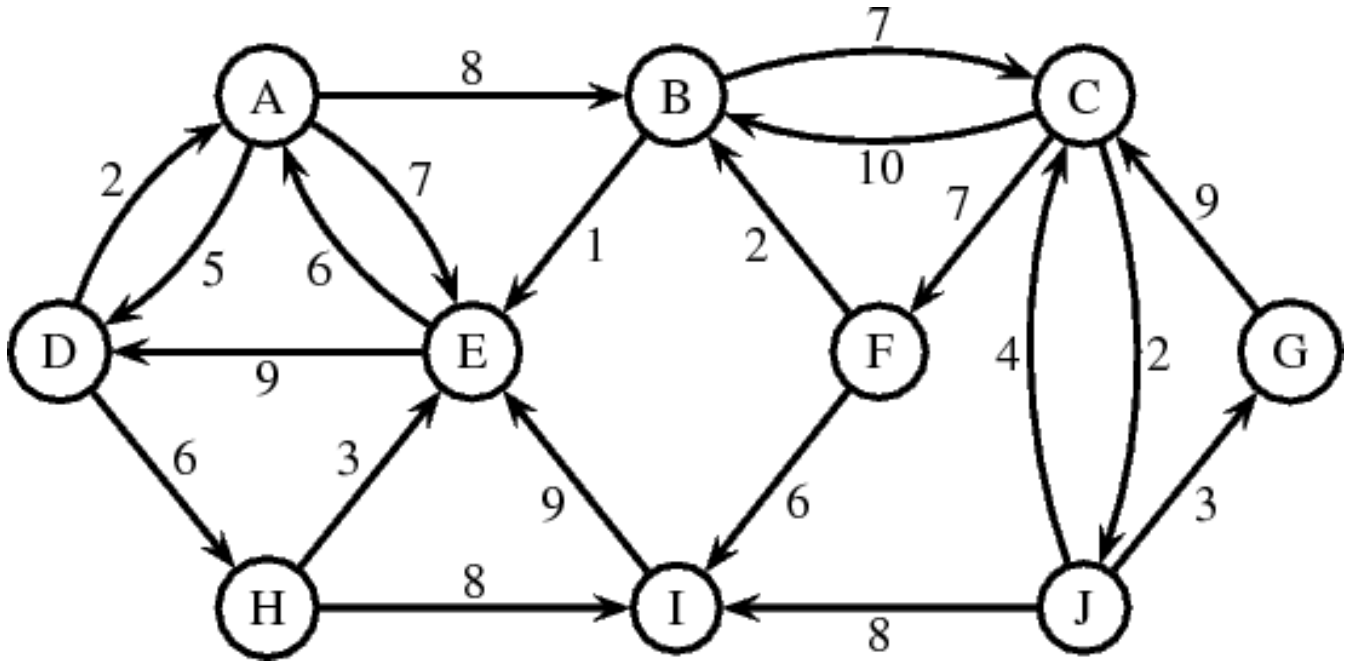


Shortest Path = Solution =
B → A → C → D

As minimum path is the goal path, so we have found a solution.

Uniform Cost Search (Problem 04)

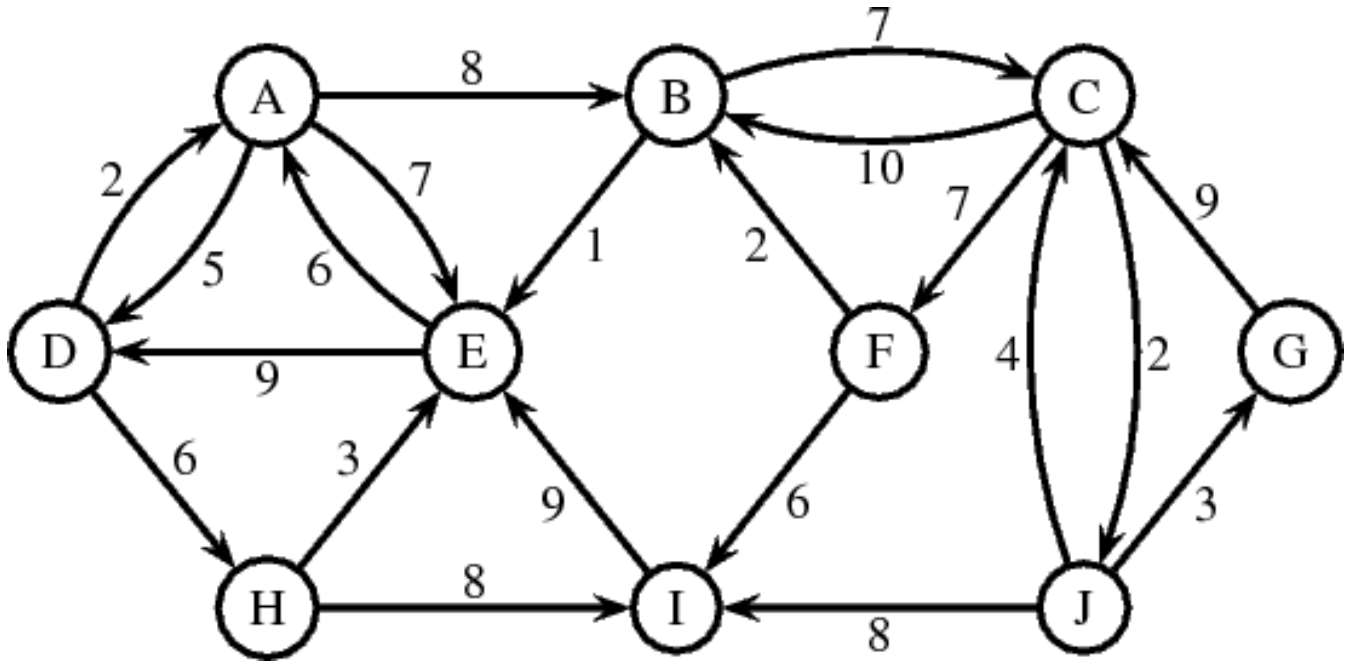
- D = Initial State
- G = Goal State



Queue	
Path	**Cost**

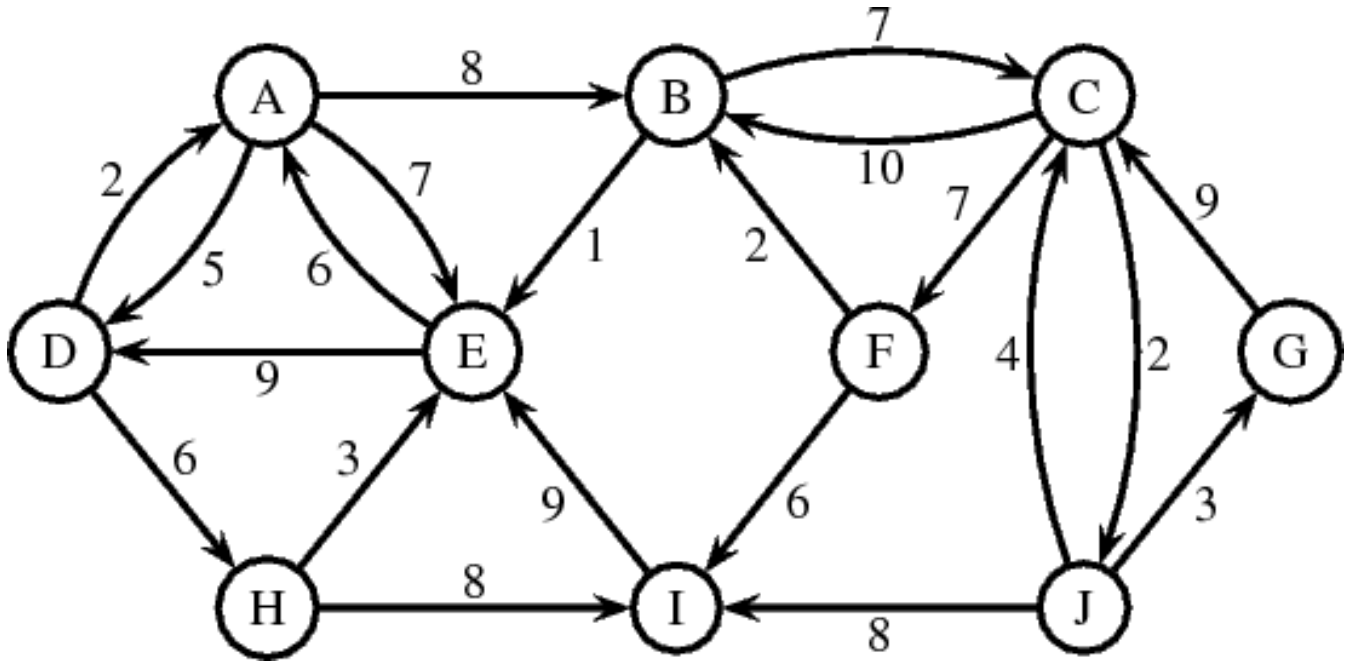
Uniform Cost Search (Problem 04)

Queue	
Path	**Cost**
<D>	0



Uniform Cost Search (Problem 04)

Queue	
Path	**Cost**
<A,D>	2
<H,D>	6



From the path we have:

<A,D> = 2

<B,A,D> = 10

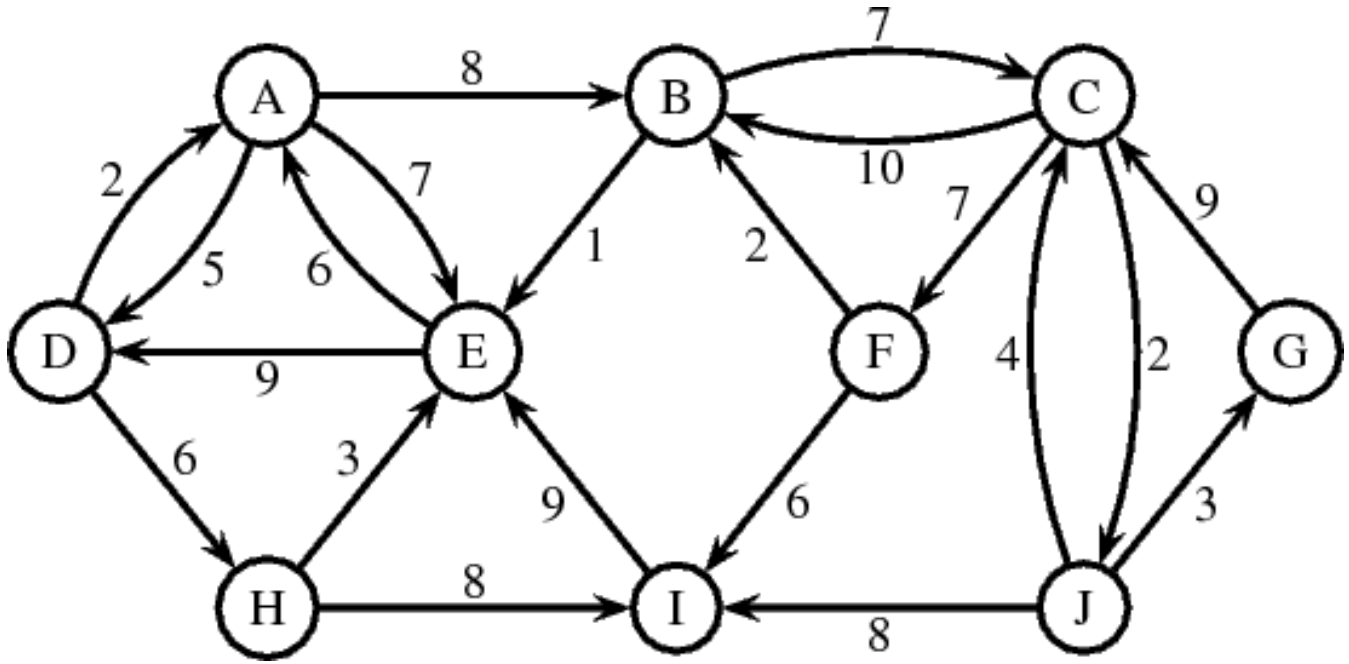
<D,A,D> = 7 (It is a cycle as repeated D)

<E,A,D> = 9

Avoid Cycles (Discard them)

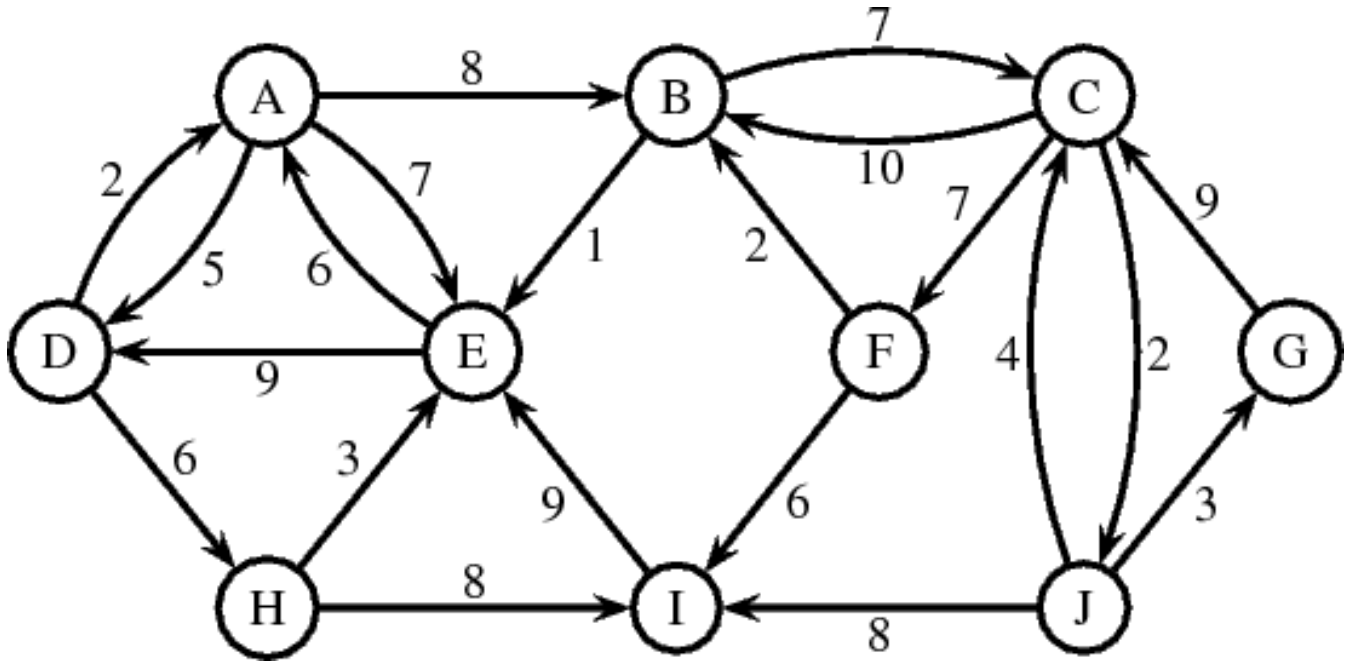
Uniform Cost Search (Problem 04)

Queue	
Path	**Cost**
<H,D>	6
<E,A,D>	9
<B,A,D>	10



Uniform Cost Search (Problem 04)

Queue	
Path	**Cost**
<E,H,D>	9
<E,A,D>	9
<B,A,D>	10
<I,H,D>	14



From the path we have:

<E,H,D> = 9

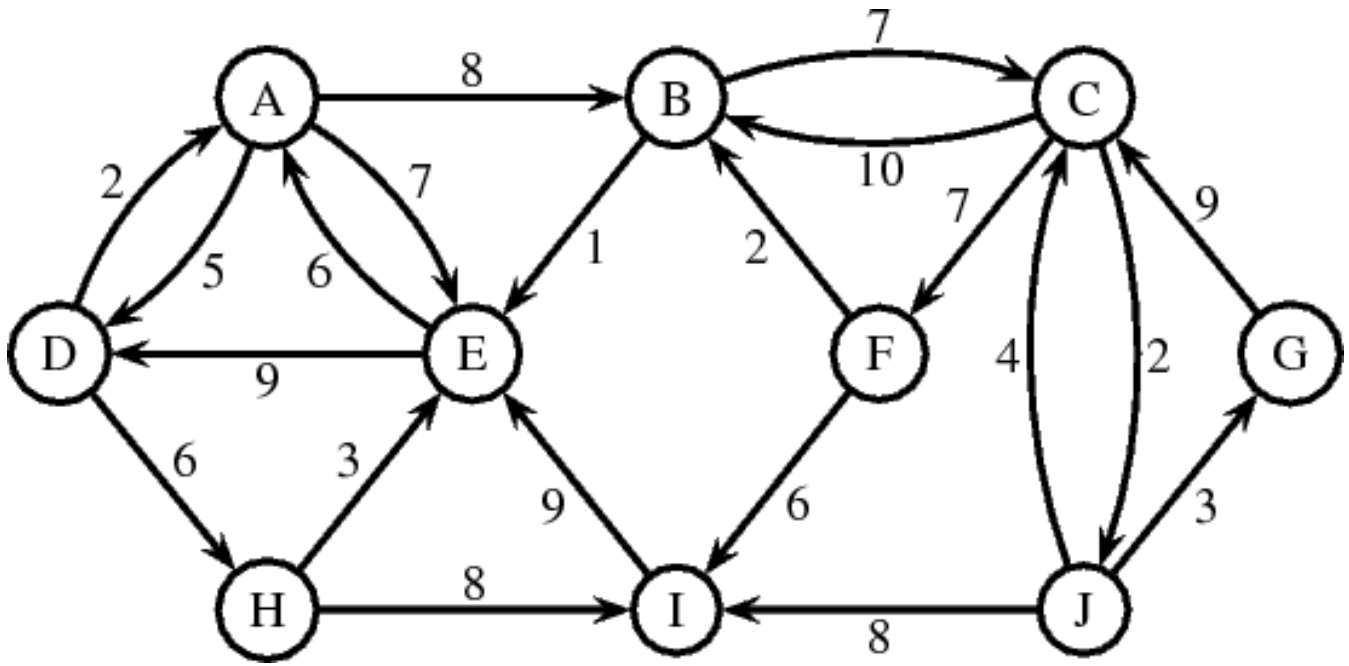
<A,E,H,D> = 15

<D,E,H,D> = 18 (It is a cycle as repeated D)

Avoid Cycles (Discard them)

Uniform Cost Search (Problem 04)

Queue	
Path	**Cost**
<E,A,D>	9
<B,A,D>	10
<I,H,D>	14
<A,E,H,D>	15

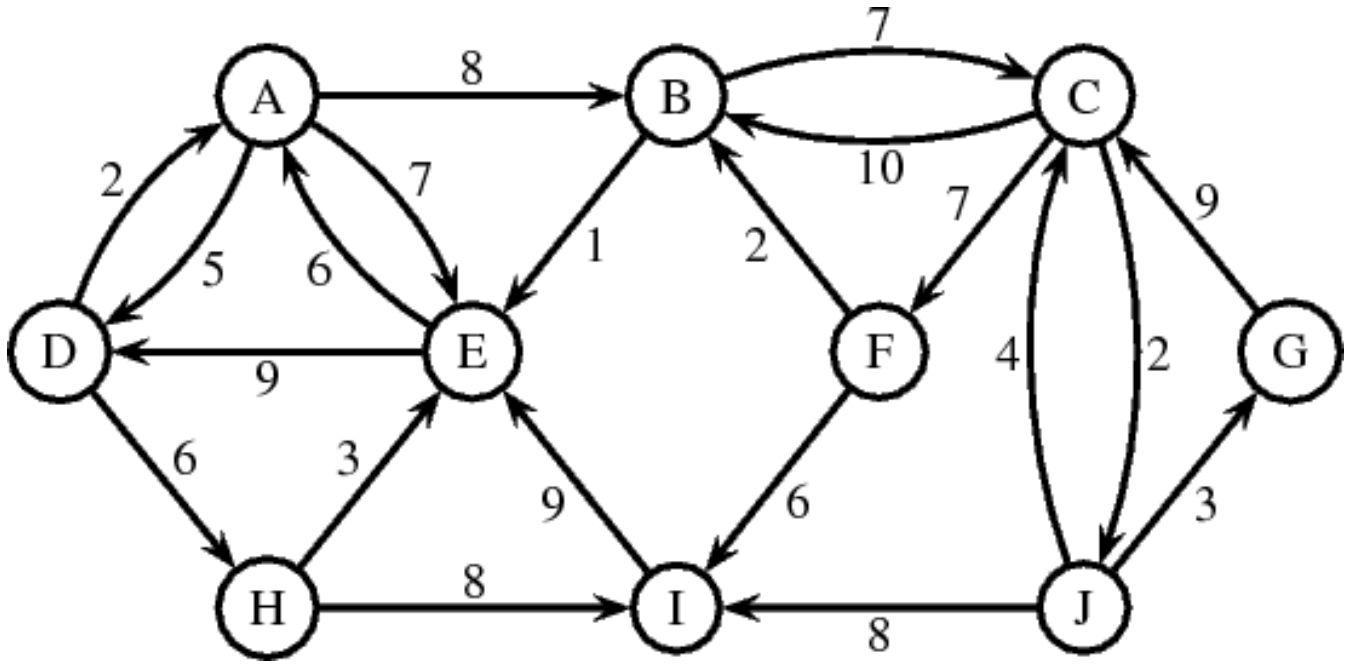


From the path we have:
<E,A,D> = 9

<A,E,A,D> = 15 (Cycle)
<D,E,A,D> = 18 (Cycle)
Avoid Cycles (Discard them)

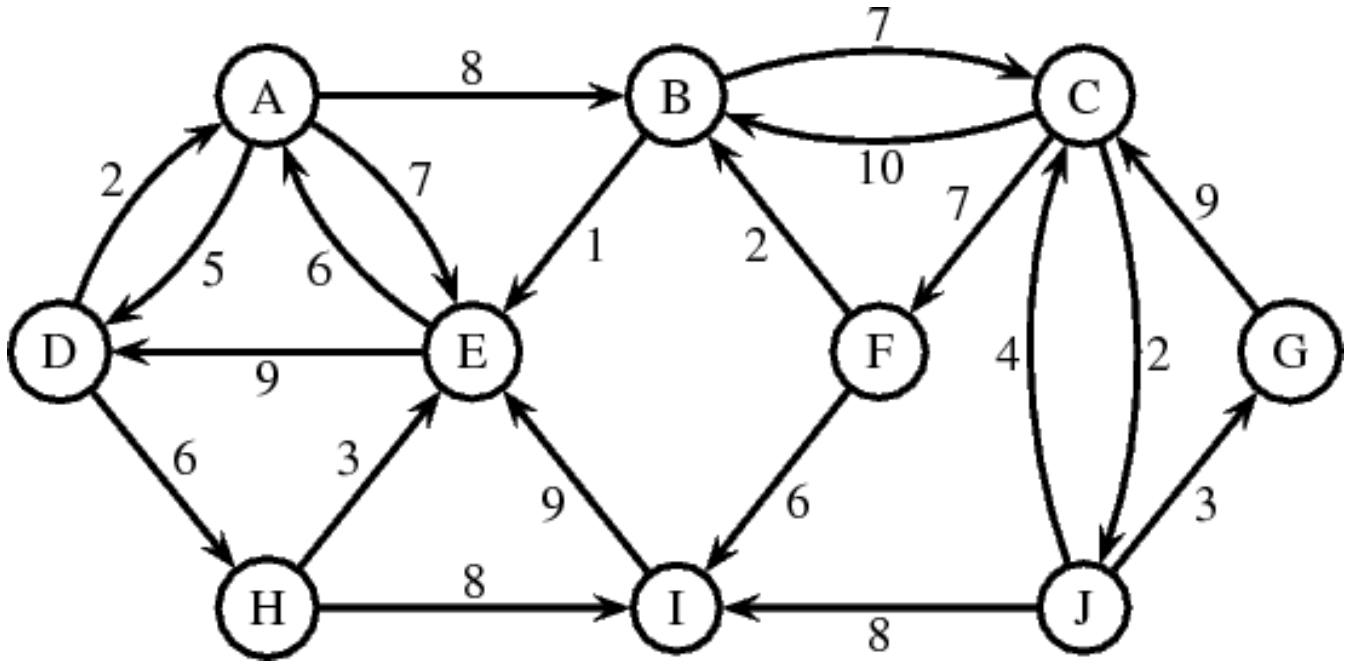
Uniform Cost Search (Problem 04)

Queue	
Path	**Cost**
<E,B,A,D>	11
<I,H,D>	14
<A,E,H,D>	15
<C,B,A,D>	17



Uniform Cost Search (Problem 04)

Queue	
Path	**Cost**
<E,B,A,D>	11
<I,H,D>	14
<A,E,H,D>	15
<C,B,A,D>	17

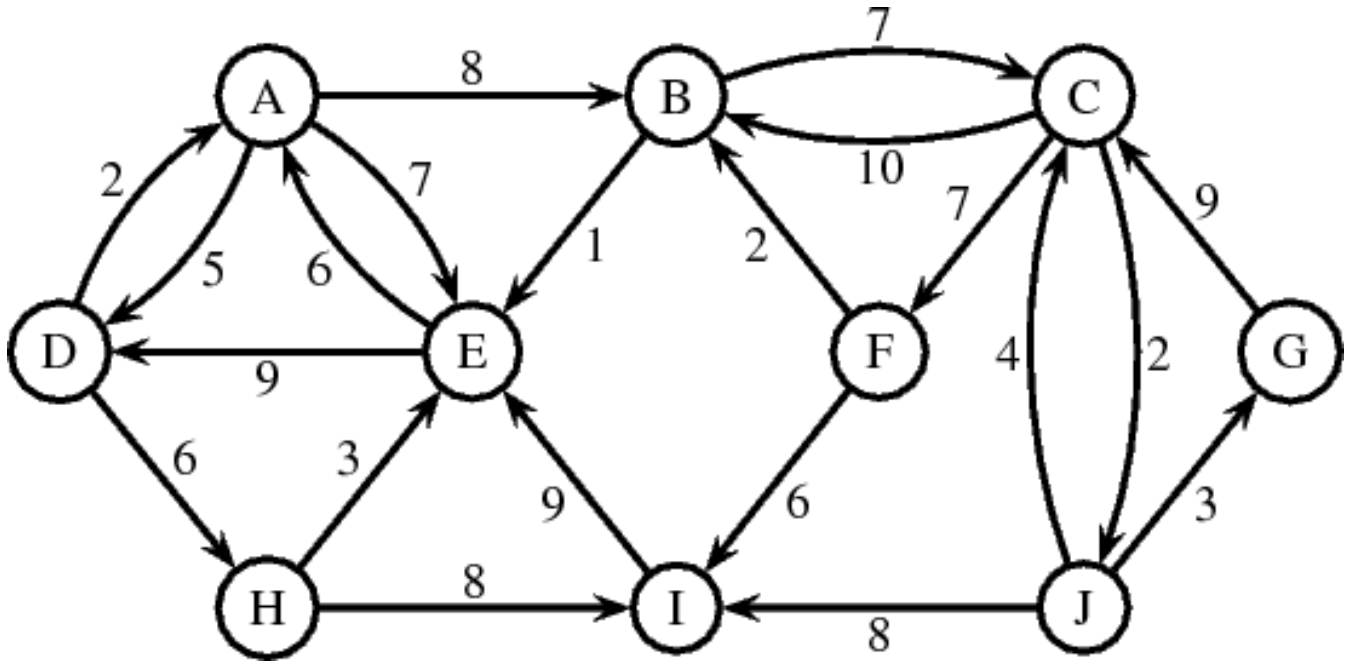


From the path we have:
<E,B,A,D> = 11

<A,E,B,A,D> = 18 (Cycle)
<D,E,B,A,D> = 20 (Cycle)
Avoid Cycles (Discard them)

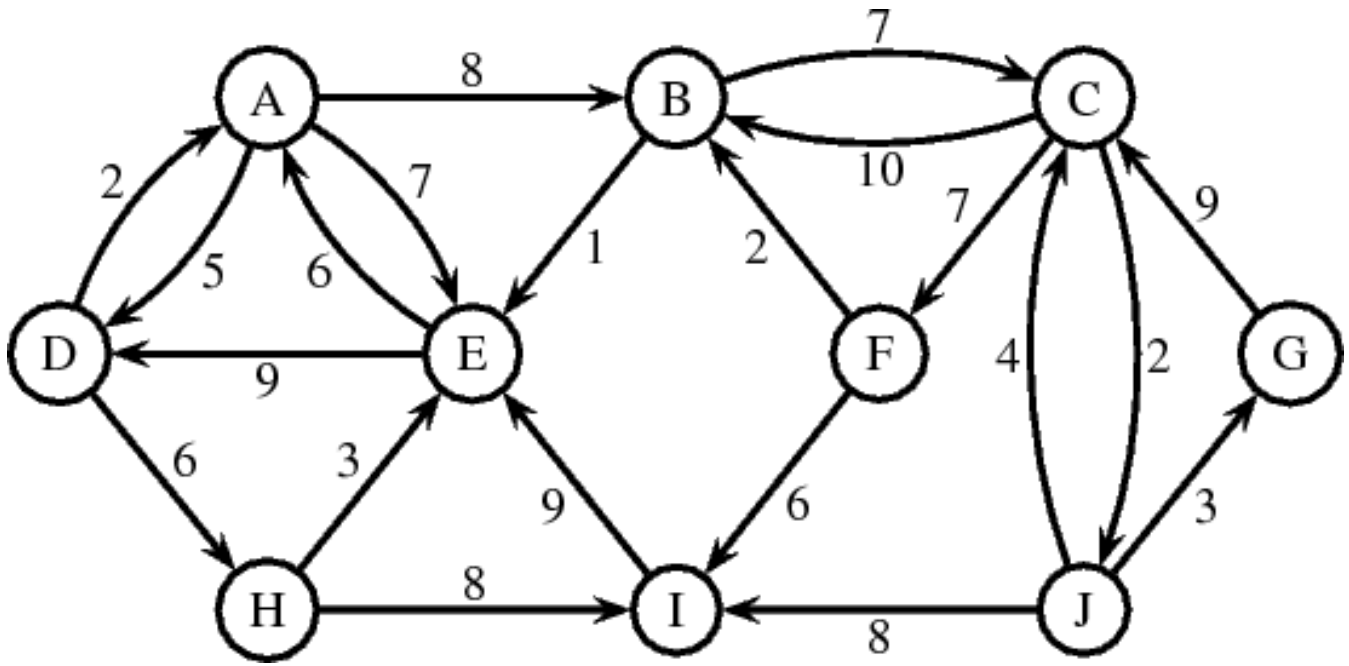
Uniform Cost Search (Problem 04)

Queue	
Path	**Cost**
<I,H,D>	14
<A,E,H,D>	15
<C,B,A,D>	17



Uniform Cost Search (Problem 04)

Queue	
Path	**Cost**
<A,E,H,D>	15
<C,B,A,D>	17
<E,I,H,D>	23

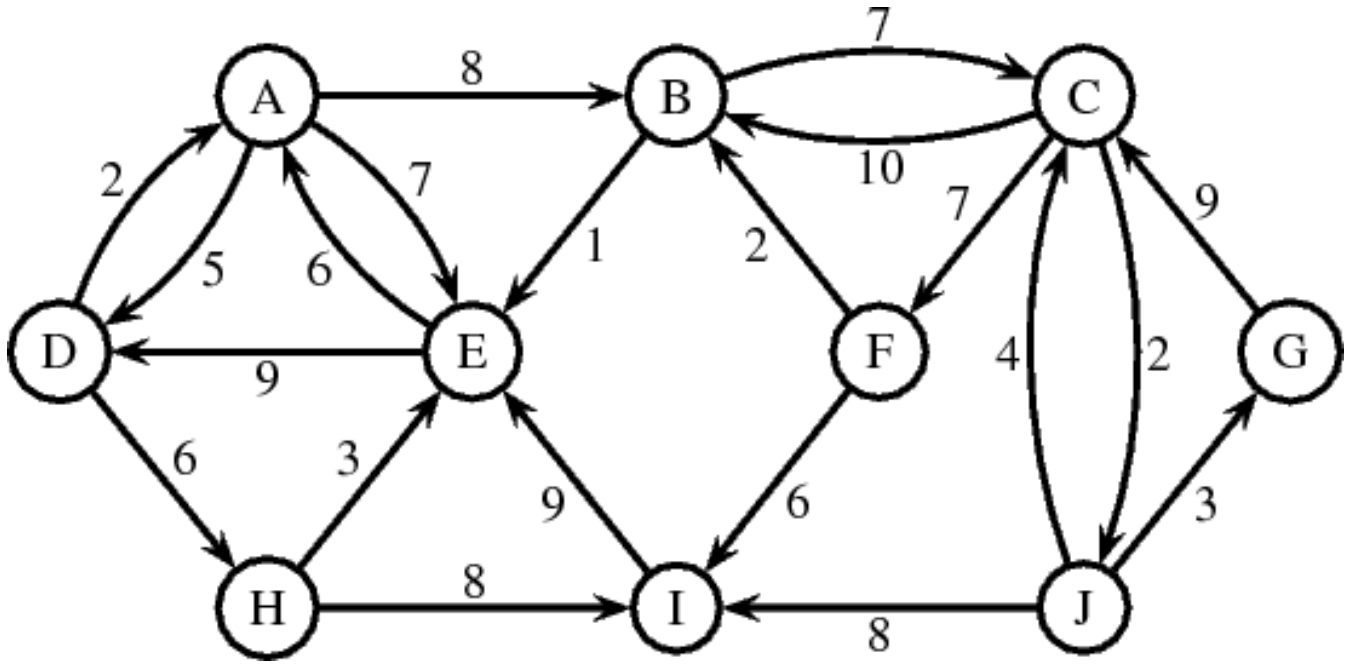


From the path we have:
<A,E,H,D> = 15

<D,A,E,H,D> = 20 (Cycle)
<E,A,E,H,D> = 22 (Cycle)
<B,A,E,H,D> = 23
Avoid Cycles (Discard them)

Uniform Cost Search (Problem 04)

Queue	
Path	**Cost**
<C,B,A,D>	17
<E,I,H,D>	23
<B,A,E,H,D>	23

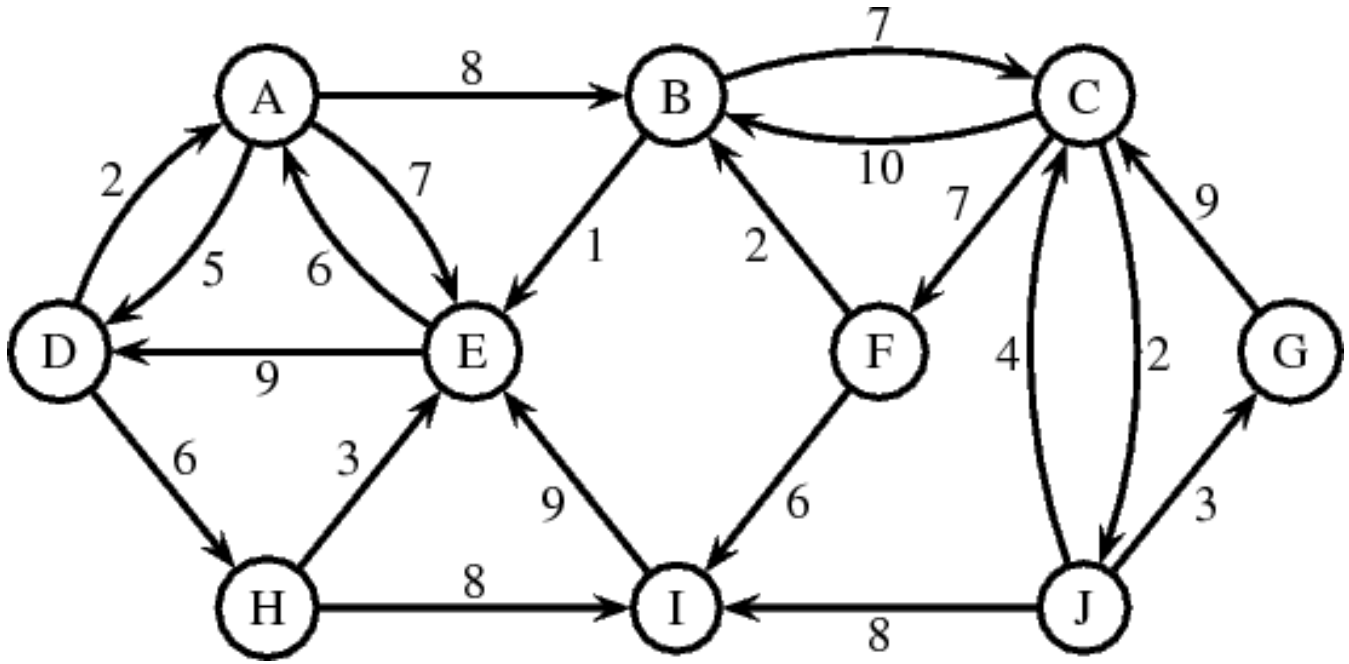


From the path we have:
<C,B,A,D> = 17

<B,C,B,A,D> = 27 (Cycle)
<F,C,B,A,D> = 24
<J,C,B,A,D> = 19
Avoid Cycles (Discard them)

Uniform Cost Search (Problem 04)

Queue	
Path	**Cost**
<J,C,B,A,D>	19
<E,I,H,D>	23
<B,A,E,H,D>	23
<F,C,B,A,D>	24

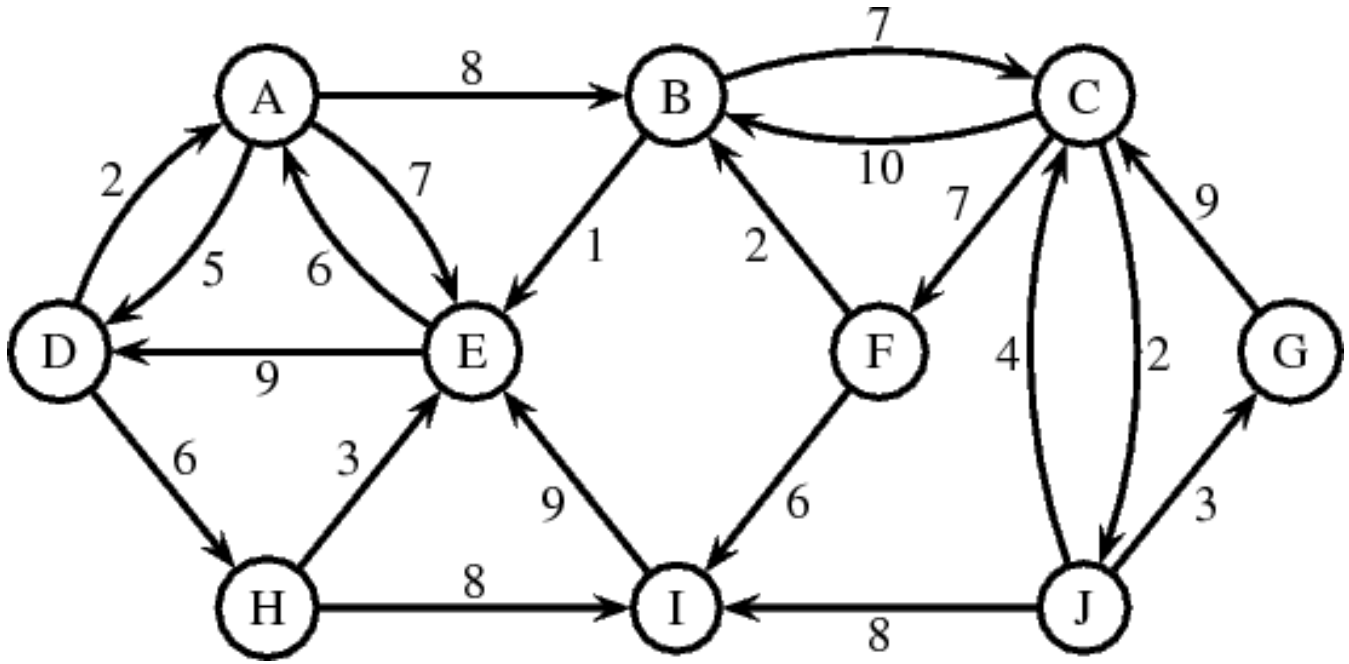


From the path we have:
<J,C,B,A,D> = 19

<I,J,C,B,A,D> = 27
<C,J,C,B,A,D> = 23 (Cycle)
<G,J,C,B,A,D> = 22
Avoid Cycles (Discard them)

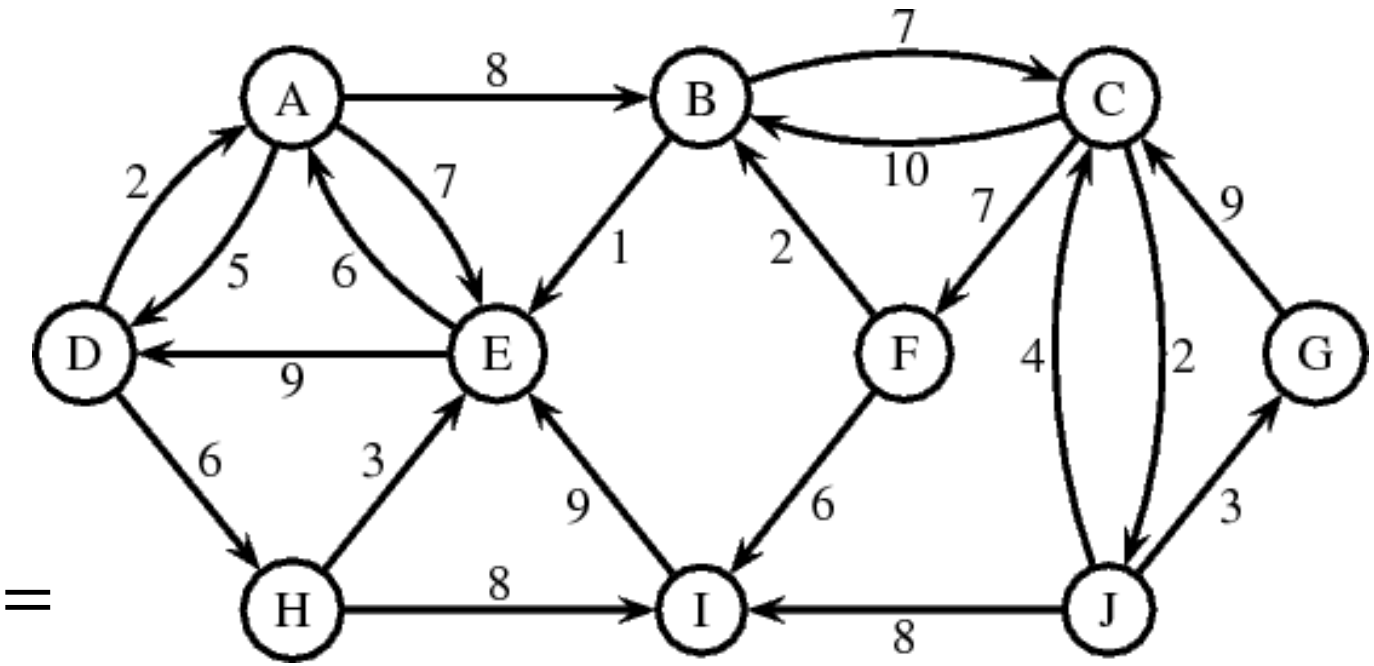
Uniform Cost Search (Problem 04)

Queue	
Path	**Cost**
<G,J,C,B,A,D>	22
<E,I,H,D>	23
<B,A,E,H,D>	23
<F,C,B,A,D>	24
<I,J,C,B,A,D>	27



Uniform Cost Search (Problem 04)

Queue	
Path	**Cost**
<G,J,C,B,A,D>	22
<E,I,H,D>	23
<B,A,E,H,D>	23
<F,C,B,A,D>	24
<I,J,C,B,A,D>	27



Shortest Path = Solution =
D → A → B → C → J → G

As minimum path is the goal path, so we have found a solution.

Properties of Uniform Cost Search

Property	Description
Completeness	UCS is complete, meaning that it is guaranteed to find a solution if one exists, provided that the cost increments are greater than zero.
Optimality	UCS is optimal for graphs with non-negative path costs, ensuring that the lowest-cost solution is found.
Time Complexity	The time complexity of UCS depends on the cost of the optimal solution and may vary based on the problem instance.
Space Complexity	The space complexity of UCS depends on the cost of the optimal solution and may vary based on the problem instance.
Frontier	UCS maintains a priority queue ordered by path cost, ensuring that nodes with the lowest cost are explored first.

Algorithm	Completeness	Optimality	Time Complexity	Space Complexity	Frontier
Breadth-First Search (BFS)	Complete	Optimal	$O(b^d)$	$O(b^d)$	Queue
Backward Breadth-First Search	Complete	Optimal	$O(b^d)$	$O(b^d)$	Queue
Bi-Directional Breadth-First Search	Complete	Optimal	$O(b^{d/2})$	$O(b^{d/2})$	Two Queues (Forward and Backward)
Depth-First Search (DFS)	Incomplete	Suboptimal	$O(b^m)$	$O(bm)$	Stack/Recursion
Depth-Limited Search (DLS)	Incomplete	Suboptimal	$O(b^l)$	$O(bl)$	Stack/Recursion with Depth Limit
Depth-First Iterative Deepening Search (DFID)	Complete	Optimal	$O(b^d)$	$O(bd)$	Stack/Recursion with Iterative Depth Limits
Uniform Cost Search (UCS)	Complete	Optimal	Depends on the cost of optimal solution	Depends on the cost of optimal solution	Priority Queue

Algorithm	Real-Life Example
Breadth-First Search (BFS)	Finding the shortest path in a road network for GPS navigation.
Backward Breadth-First Search	Tracing the source of a network connectivity issue in a computer network.
Bi-Directional Breadth-First Search	Finding the shortest path between two locations in a city map.
Depth-First Search (DFS)	Maze solving in a video game or finding the exit in a labyrinth.
Depth-Limited Search (DLS)	Searching for a lost item in a limited area within a house.
Depth-First Iterative Deepening Search (DFID)	Solving a Rubik's Cube puzzle by trying different depths of moves iteratively.
Uniform Cost Search (UCS)	Planning the shortest route for a delivery truck with varying road costs.

More Real Life Examples of Search Algorithms

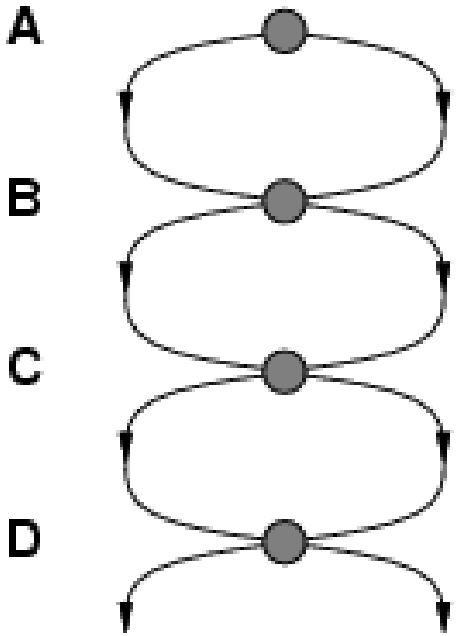
1. **Breadth-First Search (BFS):**
   - **Social Networking:** BFS is used in social networking platforms to find the shortest path between users or to explore connections between users within a certain distance.
   - **Network Routing:** BFS is used in network routing algorithms to find the shortest path between routers or nodes in a computer network.
2. **Backward Breadth-First Search:**
   - **Grammar Checking:** Backward BFS can be used in grammar checking algorithms to analyze the structure of sentences by starting from the end and traversing backward to identify errors.
   - **Network Connectivity:** Backward BFS can be used to identify the source of a network connectivity issue by tracing the path from the destination back to the source.
3. **Bi-Directional Breadth-First Search:**
   - **Shortest Path Finding:** Bi-BFS is used in finding the shortest path between two points in a graph by simultaneously exploring from both the start and end nodes and meeting in the middle.
   - **Friend Recommendations:** Bi-BFS can be used in social networking platforms to suggest mutual connections between users by exploring the network from both users simultaneously.
4. **Depth-First Search (DFS):**
   - **Maze Solving:** DFS is commonly used in solving mazes, where it explores a path until it reaches a dead end, then backtracks and tries a different path.
   - **Web Crawling:** DFS is used in web crawling algorithms to explore the web by following hyperlinks from one page to another in a recursive manner.

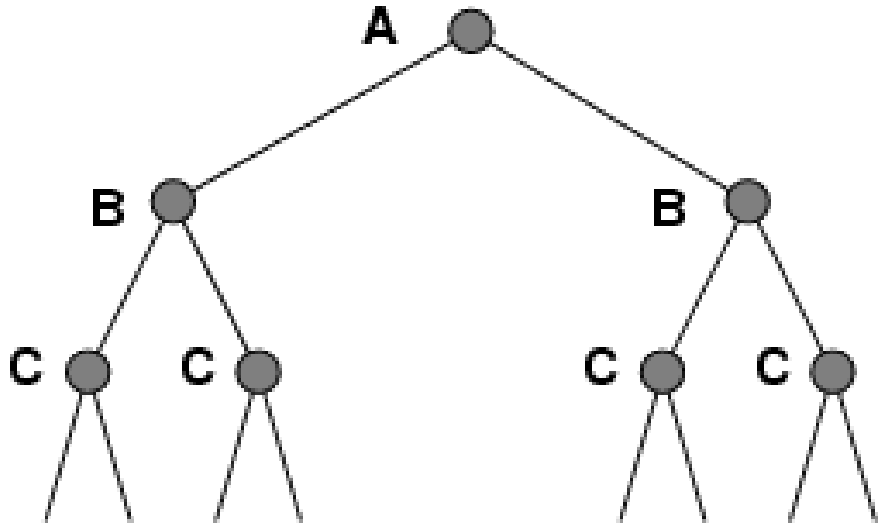
More Real Life Examples of Search Algorithms

5. **Depth-Limited Search (DLS):**
   - **Chess and Other Board Games:** In chess, DLS can be used to limit the depth of the search tree during game tree traversal to avoid infinite exploration.
   - **Robot Navigation:** DLS can be used in robot navigation systems to limit the depth of exploration in complex environments to ensure real-time decision-making.
6. **Depth-First Iterative Deepening Search (DFID):**
   - **Puzzle Solving:** DFID is commonly used in solving puzzles like the Rubik's Cube, where it iteratively deepens the depth limit until a solution is found.
   - **Scheduling:** DFID can be used in scheduling algorithms to find an optimal sequence of tasks with varying depths of exploration.
7. **Uniform Cost Search (UCS):**
   - **GPS Navigation:** UCS is used in GPS navigation systems to find the shortest path between two locations, considering the varying costs of different routes (e.g., traffic conditions, toll roads).
   - **Robot Path Planning:** UCS is used in robotics for path planning, where the cost of traversing different terrain or obstacles varies, and the goal is to find the path with the minimum cost.

Repeated States

- Failure to detect repeated states can turn a linear problem into an exponential one!





How to deal with repeated states?

- Do not return to the state you just came from.
  - *Have the expand function refuse to generate any successor that is the same state as the node's parent.*
- Do not repeat paths with cycles in them.
  - *Have the expand function refuse to generate any successor of a node that is the same as any of the node's ancestors.*
- Do not generate any state that was ever generated before.

Questions

1. What distinguishes an uninformed search algorithm from an informed one?
2. How does an uninformed search explore the search space to find a solution?
3. Explain how breadth-first search explores a search tree or graph.
4. Discuss the advantages and disadvantages of breadth-first search compared to other search algorithms.
5. What is backward breadth-first search, and in what scenarios is it useful?
6. Compare and contrast backward breadth-first search with forward search algorithms.
7. Describe the concept of bi-directional breadth-first search.
8. How does bi-directional breadth-first search improve upon traditional breadth-first search?
9. Explain how depth-first search explores a search tree or graph.
10. Discuss the limitations of depth-first search in terms of finding optimal solutions.
11. Define depth-limited search and discuss its purpose in problem-solving.
12. How does depth-limited search differ from depth-first search?
13. What is depth-first iterative deepening search, and why is it used?
14. Discuss the advantages of DFID over traditional depth-first search and breadth-first search.
15. What criteria are used to evaluate the performance of search algorithms?
16. How can the effectiveness of search algorithms be compared and analyzed?
17. Describe the principles behind uniform cost search.
18. Explain how uniform cost search ensures optimality in finding the least-cost path.
19. What are repeated states in the context of search algorithms?
20. Discuss the challenges posed by repeated states and their impact on search efficiency.
21. What strategies can be employed to handle repeated states in search algorithms?
22. Discuss the importance of cycle detection and state revisitation strategies in overcoming repeated states.

Programming Tasks

1. Implement a program to find a solution path in a maze using an uninformed search algorithm such as depth-first search or breadth-first search.
2. Develop a program to find the shortest path between two nodes in a weighted graph using breadth-first search.
3. Implement a program to find the path from the goal state to the initial state in a problem space using backward breadth-first search.
4. Write a program to find the shortest path between two nodes in a graph using bi-directional breadth-first search, exploring from both the start and goal nodes simultaneously.
5. Create a program to generate all possible configurations of a puzzle (e.g., Sudoku, 8-puzzle) using depth-first search.
6. Implement a program to find a solution path in a large state space while limiting the depth of exploration using depth-limited search.
7. Develop a program to solve the N-Queens problem using depth-first iterative deepening search, gradually increasing the depth limit until a solution is found.
8. Write a benchmarking program to compare the performance of various search algorithms (e.g., BFS, DFS, UCS) on different problem instances, analyzing factors such as time complexity and memory usage.
9. Implement a program to find the cheapest route between two cities in a road network using uniform cost search, considering different costs associated with each road segment.
10. Develop a strategy to handle repeated states efficiently within search algorithms to avoid unnecessary revisitation of states and optimize memory usage.